HYPER-RIDGE PERSONALEVO: META-LEARNING THE ARCHIVE GEOMETRY FOR PROVABLE PERSONALIZED ROUTING

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Paper under double-blind review

ABSTRACT

Personalized agents maintain libraries of prompt variants, skill modules, or orchestration policies, which we call *candidates*, collected in an *archive*.

The simplest strategy is *uniform deployment*: pick the single candidate with the highest average reward across all users.

We propose PersonalEvo, a contextual-bandit method that learns to route each user to their best-matching candidate online, and optionally grows the archive by evolving new candidates after failures.

We give PersonalEvo a provable guarantee: its reward decomposes exactly into four interpretable terms, the *default reward* of the best single candidate served to everyone, a *routing gain* Δ_{route} from matching each user to their own best candidate, an *evolution gain* from newly discovered candidates, and an *exploration cost* paid while learning user preferences.

This decomposition yields a sharp condition for when personalization helps, routing provably outperforms uniform deployment whenever the routing and evolution gains together exceed the exploration cost, together with a plug-in regret bound on the exploration cost.

We evaluate PersonalEvo on nine benchmarks: three synthetic personalized-agent environments, two agent-framework substrates (GEPA prompt archives (Agrawal et al., 2025), Hermes skill configurations (Team, 2025b)), and four real-data preference datasets (BESPOKE (Team, 2025c), PAHF Shopping and Embodied (Labs, 2025), and PrefEval (Zhao et al., 2025)).

Across all nine, PersonalEvo achieves higher mean reward than both the static baseline and random candidate selection.

Our results show that: (1) the routing gain Δ_{route} is larger than the evolution gain on every benchmark; (2) on the four real-data benchmarks, Δ_{route} ranges from 0.036 to 0.251, indicating that the value of personalized routing varies substantially across application domains; (3) growing the archive without gating hurts performance, because the added exploration cost outweighs the evolution gain.

Building on this base, we introduce *Hyper-Ridge PersonalEvo*, which replaces independent isotropic regularization with a shared archive geometry. We prove an effective-dimension exploration guarantee when that geometry is known. For a calibration-split variant that estimates and then freezes the geometry, we also prove finite-sample covariance control; under an additional calibrated-confidence condition, its charged estimation remainder is sub-$\sqrt{T}$. Diagnostic experiments separate the geometric mechanism from the unresolved estimator: on planted rank-three problems, the true geometry lowers cumulative exploration cost by 46–59%, whereas the periodically re-estimated geometry raises it by 7–19%. Thus the current evidence supports the known-geometry mechanism, but not yet the fully adaptive router.

1 INTRODUCTION

Modern LLM agents increasingly choose among collections of prompts, skills, tools, and workflows instead of relying on one fixed policy. We call such a collection an *archive* and each deployable entry a *candidate*. Personalization methods learn user profiles and preference-conditioned behavior (Salemi et al., 2023; Zhao et al., 2025; Team, 2025c); prompt and agent evolution methods search for stronger candidates (Fernando et al., 2023; Guo et al., 2024; Zhou et al., 2024; Agrawal et al., 2025); and routing methods choose among available models or policies at inference time (Ong et al., 2024; Hu et al., 2024; Li, 2025; Tsai & Tran, 2026). These lines of work address three related decisions: what a user prefers, how the archive should improve, and which candidate should handle the next request. A personalized self-evolving agent must make all three decisions together.

Consider a support agent whose archive contains formal and casual policies for refund and shipping requests. Serving every user the best policy on average discards this useful diversity. A personalized router can learn which policy works for each user, but it must first try candidates whose value is uncertain. Archive growth makes this cost recur: whenever the agent adds a mutation after a failure, a conventional bandit sees a new arm and starts learning it from scratch. Yet the candidates are related. What the router learns about formality from a refund policy should inform a formal shipping policy, even though the two policies should retain different reward models. Archive growth therefore creates a tension: it can add better candidates while also adding enough exploration to erase their benefit.

Existing methods resolve only parts of this tension. Profile and memory methods condition an agent on the user, but do not control exploration over a growing archive. Evolution methods improve candidate quality, usually offline or at the population level, but do not learn which candidate to serve to each user online. Fixed-pool routers make that assignment, but do not account for archive growth. Standard disjoint linear bandits also estimate every candidate independently, so evidence about one candidate cannot reduce uncertainty about another. Pooling all candidates into one predictor goes too far in the other direction and erases real candidate-specific differences. What is missing is the *geometry of variation across the archive*: which behavioral directions recur across candidates, which differences remain candidate-specific, and when evidence can be transferred between them.

These observations lead to the question studied in this paper:

> *Can a self-evolving agent transfer evidence across related archive candidates while retaining a measurable guarantee that personalized routing is worth its exploration cost?*

We address this question in two steps. First, *PersonalEvo* separates the value of personalization from the cost of learning it. Let A_1 be the initial archive and let R_T be the router's average reward over T interactions. We prove the exact decomposition

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{1}$$

where $V_{\text{static}}(A_1)$ is the reward from serving everyone the best single initial candidate, $\Delta_{\text{route}}(A_1)$ is the gain from matching users to different initial candidates, $\bar{G}_T$ is the gain from candidates discovered after deployment, and $\bar{L}_T$ is the reward lost while the router learns. In the support example, these terms are the best universal reply policy, the value of matching tone and topic to each user, the value of newly evolved replies, and the cost of trying the wrong reply. The identity holds for any measurable router and gives the testable deployment criterion

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T, \tag{2}$$

so personalized routing is worthwhile exactly when its routing and evolution gains cover its exploration cost.

Second, *Hyper-Ridge PersonalEvo* targets the only negative term in equation 1. Each candidate a has its own parameter vector θ_a, which maps d user and candidate features to expected reward. A disjoint LinUCB router estimates each vector with the same spherical regularizer and therefore encodes no relation among candidates (Abbasi-Yadkori et al., 2011; Chu et al., 2011). We instead model the candidate vectors as draws with a shared covariance Ω. This covariance is the archive geometry. Its dominant directions capture recurring factors such as tone and topic, while every candidate keeps its own reward model. Hyper-Ridge uses this geometry to regularize candidate estimates, so a newly evolved candidate starts with uncertainty shaped by the archive rather than an isotropic blank slate.

The gain is largest when candidate parameters vary mainly along a few shared directions even though the feature representation is high-dimensional. Let $d_{\text{eff}}(\Omega)$ count the directions that are statistically active under this geometry. When Ω is known, our analysis replaces the ambient dimension d in the usual disjoint-LinUCB exploration rate with $d_{\text{eff}}(\Omega)$:

$$\underbrace{O\left(\sqrt{K_{\max} d T \log T}\right)}_{\text{independent isotropic estimates}} \longrightarrow \underbrace{O\left(\sqrt{K_{\max} d_{\text{eff}}(\Omega) T \log T}\right)}_{\text{archive-aware geometry}} + \mathrm{Err}_T, \tag{3}$$

where $K_{\max}$ is the archive-size cap and Err_T is the cost of estimating the geometry. We also analyze a calibration-split variant that estimates Ω from fresh candidates and then freezes it before routing. Under explicit diversity and calibrated-confidence conditions, its covariance error vanishes and its charged estimation remainder is $o(\sqrt{T})$. This theorem does not cover the periodically updated estimator in Algorithm 1; that fully adaptive case remains open.

Our experiments separate the motivation for Hyper-Ridge from evidence for its mechanism. Across nine benchmarks, the base PersonalEvo router has routing gains from 0.036 to 0.280, evolution gains from 0 to 0.061, and exploration costs from 0.010 to 0.071. Its advantage over static deployment ranges from 0.027 to 0.295, showing that exploration can decide whether personalization pays. On separate planted rank-three problems, giving Hyper-Ridge the true geometry lowers cumulative exploration cost by 46–59%. Periodically estimating that geometry instead raises the cost by 7–19%. On seven adapter benchmarks, one-hot candidate features make the full-covariance router identical to its diagonal ablation, so their reward changes cannot show cross-candidate transfer. The evidence therefore supports the known-geometry mechanism while exposing the unresolved estimation and representation problem.

Contributions. This paper makes four contributions:

1. **A measurable formulation of personalized self-evolution.** We formalize routing over a growing candidate archive and prove the exact four-term decomposition equation 1, which turns the decision to personalize into the testable criterion equation 2.
2. **Archive-aware routing.** We introduce Hyper-Ridge PersonalEvo, which learns a covariance geometry across candidate reward models and uses it in a generalized-ridge UCB router. It shares evidence about important feature directions while retaining candidate-specific predictions.
3. **Effective-dimension guarantees with a clear boundary.** We prove the known-Ω effective-dimension bound and a finite-sample calibration-split theorem for estimated Ω. We state separately the extra condition needed to recover the oracle leading rate and the unresolved fully adaptive case.
4. **Evidence that separates mechanism from implementation.** We measure every decomposition term on nine benchmarks and test Hyper-Ridge in a controlled low-rank setting. The results show when the right geometry helps and why the current periodic estimator and one-hot representation do not yet realize that gain.

Paper organization. Section 2 positions the work relative to personalization, agent evolution, routing, and linear bandits. Section 3 defines the routing problem and Hyper-Ridge estimator. Section 5.1 develops the decomposition and effective-dimension analysis. Section 6 reports the base-router results, Hyper-Ridge diagnostics, and remaining evaluation gap. Appendix J proves the oracle geometry results, and Appendix L gives the calibration-split covariance theorem.

2 Related Work

Routing queries across a model or prompt pool to trade off cost and quality is now standard: RouteLLM (Ong et al., 2024) learns a strong/weak router from preference data and RouterBench (Hu et al., 2024) benchmarks such routers, while LLM Bandit (Li, 2025) and NeuralUCB routing (Tsai & Tran, 2026) cast routing as an online contextual bandit.

These route at the *model* level over a fixed pool; PersonalEvo routes *per user* over an archive that may grow online, and decomposes the resulting reward rather than only reporting it.

Benchmarks such as LaMP (Salemi et al., 2023), PersonalLLM (Zollo et al., 2024), and PrefEval (Zhao et al., 2025) supply user-feedback signals, while PersonaAgent (Zhang et al., 2026) and persistent-memory frameworks (Westhäußer et al., 2025) maintain per-user personas and evolving profiles.

The closest personalized-bandit antecedent, PAB-F (Team, 2025c), models per-user reward as a factorized function of context and a fixed memory/tool/style action set, and studies regret against an in-archive oracle.

Prompt and agent evolution methods, including Promptbreeder (Fernando et al., 2023), EvoPrompt (Guo et al., 2024), Symbolic Learning (Zhou et al., 2024), and GEPA's reflective Pareto search (Agrawal et al., 2025), improve prompts or archives through offline search rather than online user-conditioned routing.

Hermes self-evolution (Team, 2025b;a) adds constraint-gated growth, which motivates our archive-growth gate.

Classical contextual-bandit methods such as OFUL and LinUCB (Abbasi-Yadkori et al., 2011; Chu et al., 2011; Li et al., 2010) provide standard finite-time tools for controlling exploration in linear bandits, and related structural-vs-online decompositions appear in contextual pricing, offline RL, and multi-task transfer.

3 Preliminaries: PersonalEvo Routing and the Hyper-Ridge Estimator

3.1 Overview

A personalized agent keeps a small library, an *archive*, of candidate prompts, skills, or routing templates.

At each round, for each user it routes one user at a time: it looks at the user, picks one candidate from the current archive, observes a reward, updates internal state, and may optionally append a child candidate to the archive after the round.

We compare this personalized routing policy to the simplest possible alternative: pick the single candidate with the best population-mean reward in the initial archive, and use it for everyone forever.

The remainder of this section formalizes the notation, introduces the hyper-ridge estimator and its generalized-ridge model, presents the routing algorithm, and states the base theoretical results.

3.2 Notation and setup

A user $U \sim \Pi$ is drawn from a population: $\mathcal{U}$ denotes the user space (in our benchmarks a finite set of user profiles; in general an abstract measurable space), and Π is a probability distribution over $\mathcal{U}$, so Π is not a vector or matrix and has no dimension of its own; it simply specifies how likely each user is to arrive. Practically, Π is the deployment's user mix, and expectations $\mathbb{E}_U[\cdot]$ below average over that mix.

The system operates over $T \in \mathbb{N}$ rounds (discrete interaction steps, e.g. successive requests) starting from a common initial archive A_1. The archive $A_1 = \{a^{(1)}, \dots, a^{(K)}\}$ is a finite set of K candidates; each element a is one deployable unit of agent behavior (a prompt variant, skill module, or orchestration policy), and $K = |A_1|$ is its size. "Common" means every user starts from the same library. Later archives $A_t(U)$ may grow beyond A_1 as evolution adds candidates, up to the cap $K_{\max}$.

The analysis follows *one* user's trajectory and then averages over the population at the end. Concretely, a single user U is drawn once at the start, interacts with the system for T rounds, and $\mathcal{F}_{t-1} = \sigma(U, A_1, a_1, r_1, \dots, a_{t-1}, r_{t-1})$ is the natural filtration of that one trajectory: everything known after the user draw and the first $t-1$ actions and rewards. Population-level statements are obtained by taking $\mathbb{E}_U[\cdot]$ over the user draw. We adopt the following convention throughout the paper: quantities written with the argument U (such as $A_t(U)$, $a_t(U)$, $G_t(U)$, $L_t(U)$) are *per-user* random variables along this single trajectory, while quantities written with an expectation or a bar (such as V_{static}, V_{route}, $\bar{G}_T$, $\bar{L}_T$, R_T) are *population* averages over $U \sim \Pi$; every later definition and theorem statement uses this convention.

At round t, the user-specific archive $A_t(U)$ is available to the router and is required to be $\mathcal{F}_{t-1}$-measurable (i.e., it does not depend on future randomness).

The router selects $a_t(U) \in A_t(U)$ and observes the reward

$$r_t \;=\; \mu(U, a_t(U)) + \xi_t,$$

which is this paper's formal definition of the reward and is *per user and per action*: r_t is the scalar feedback generated when the realized user U receives the specific candidate $a_t(U)$ at round t (e.g., a satisfaction score, task-success indicator, or preference rating in $[0, 1]$). Its randomness has exactly two sources: the user draw $U \sim \Pi$ (who arrived), and the observation noise ξ_t (round-to-round variability of the same user's feedback on the same candidate, with $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$). The function $\mu(u, a) \in [0, 1]$ is the noise-free mean reward of serving candidate a to user u; Section 3.3 models it linearly in the features.

The current-archive oracle value is $m_t(U) = \max_{a \in A_t(U)} \mu(U, a)$.

We define the four terms of equation 6.

The *default reward* is $V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}_U[\mu(U, a)]$: average each candidate's reward over all users *first*, then pick the single best candidate and serve it to everyone. This is the performance of the simplest possible deployment, one that does no personalization at all (a single fixed prompt or policy shipped to every user), so it is the operating point a team is already at before adopting any routing method; it is the baseline any personalization system must beat to justify its cost.

The *oracle routing value* is $V_{\text{route}}(A) = \mathbb{E}_U[\max_{a \in A} \mu(U, a)]$, with the two operations in the opposite order: for *each* user pick that user's best candidate first, then average over users. This is the ceiling of personalization, the reward if user preferences were known perfectly and no exploration were ever needed. Practically it answers a question a team should ask before building anything: "if personalization worked perfectly on this archive, how much would we earn?" No online algorithm on the fixed archive can exceed it.

Their difference is the *routing gain* $\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A) \geq 0$: the value of swapping the order of $\max$ and $\mathbb{E}_U$, i.e., precisely the value of matching candidates to users instead of shipping one candidate to all. It is the budget available to pay for personalization. If users are nearly homogeneous, or one candidate dominates for everyone, then per-user best and population best coincide, $\Delta_{\text{route}} \approx 0$, and personalization cannot pay no matter how good the algorithm; if different user segments prefer different candidates (the terse-diff versus explanation developers of Section 1), Δ_{route} is large and routing is worth building. This is the practical situation the pair of definitions exists for: deciding, from a held-out user sample and before deployment, whether personalized routing is worth its exploration cost.

These quantities say *why* personalization matters but not *when* it pays off; the dominance condition of Corollary 3 answers the latter by weighing Δ_{route} against the exploration cost.

The *evolution gain* at round t is $G_t(U) = m_t(U) - m_1(U)$, measuring how much the archive has improved since initialization. We define it because self-modification is not free (each mutation adds an arm the router must explore), so a deployment needs to know how much reward the newly evolved candidates are actually adding; $G_t(U)$ is exactly that amount, the improvement of the best-available candidate for user U between round 1 and round t. It is measured by bookkeeping, not by extra experiments: whenever a candidate is added, evaluate the incumbent best and the new candidate for the affected user and record the difference of the two bests. Concretely, in the support-archive example: if user U's best launch-day template has mean satisfaction $m_1(U) = 0.70$ and at round t a mutated template raised the best available to $m_t(U) = 0.78$, then $G_t(U) = 0.08$; if evolution never produced anything better for this user, $G_t(U) = 0$. In our simulators μ is known so $m_t(U)$ is computed exactly; in a live system it is estimated by scoring the current archive on the user's held-out interactions.

The *exploration cost* is $L_t(U) = m_t(U) - \mu(U, a_t(U))$, the gap between the current-archive best and what the router actually picked. We define it because it is the price of learning, the only negative term in the decomposition, and the quantity our algorithm is designed to shrink: at every round where the router serves user U a candidate other than their current best, the deployment loses exactly $L_t(U)$ of reward, so summing it over rounds gives the total revenue sacrificed to figure out the user's preferences. Continuing the same example: if user U's best current template has mean satisfaction $m_t(U) = 0.78$ but the router, still exploring, serves one with $\mu(U, a_t(U)) = 0.60$, that round costs $L_t(U) = 0.18$; once the router has locked onto the user's best candidate, $L_t(U) = 0$. It is measured the same way as G_t: exactly in the simulators, and in a live system as the gap between the estimated best candidate's score and the served candidate's observed reward, averaged over rounds.

The horizon-averaged terms are $\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[G_t(U)]$ and $\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[L_t(U)]$.

3.3 Model

Assume each candidate $a \in A_t$ has a linear reward model

$$\mu(u, a) = x(u, a)^\top \theta_a,$$

where $x(u, a) \in \mathbb{R}^d$ is the user-candidate feature vector.

Instead of estimating each θ_a with ordinary ridge regression, assume the candidate parameters are random effects:

$$\theta_a \sim (0, \Omega),$$

where $\Omega \in \mathbb{R}^{d \times d}$ is a hyper-covariance matrix shared across archive candidates.

The goal is to estimate Ω from the archive and use it to regularize routing.

3.4 Hyper-ridge estimator

For candidate a, define

$$X_{a,t} = \left[x_s^\top\right]_{s<t:a_s=a}, \qquad y_{a,t} = (r_s)_{s<t:a_s=a}.$$

Given an estimate $\widehat{\Omega}_t$, define the generalized ridge estimator

$$\widehat{\theta}_{a,t} = \arg\min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda \theta^\top \widehat{\Omega}_t^{-1} \theta \right\}.$$

Equivalently,

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}\right)^{-1} X_{a,t}^\top y_{a,t}.$$

Define the generalized design matrix

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}.$$

Then the Hyper-Ridge UCB score is

$$\text{UCB}_t^{\text{hyper}}(u, a) = x(u, a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u, a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u, a)}.$$

The selected candidate is

$$a_t(u) \in \arg\max_{a \in A_t(u)} \text{UCB}_t^{\text{hyper}}(u, a).$$

Algorithm 1 Hyper-Ridge PersonalEvo routing

Require: Initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period $\tau \in \mathbb{N}$, regularizer $\epsilon > 0$

1: Initialize $\widehat{\Omega}_1 \leftarrow I_d$; for each candidate a, initialize $X_{a,1}$ and $y_{a,1}$ as empty
2: **for** $t = 1, 2, \ldots, T$ **do**
3: Observe user u_t and eligible candidate set $A_t(u_t)$
4: **for** each candidate $a \in A_t(u_t)$ **do**
5: Compute $V_{a,t}^{\text{hyper}} \leftarrow X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}$
6: Compute $\widehat{\theta}_{a,t} \leftarrow \left(V_{a,t}^{\text{hyper}}\right)^{-1} X_{a,t}^\top y_{a,t}$
7: Compute $\mathrm{UCB}_t^{\text{hyper}}(u_t, a) \leftarrow x(u_t, a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u_t, a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u_t, a)}$
8: **end for**
9: Select $a_t \in \arg\max_{a \in A_t(u_t)} \mathrm{UCB}_t^{\text{hyper}}(u_t, a)$ and observe reward r_t
10: Update per-candidate data: append $x(u_t, a_t)^\top$ to $X_{a_t,t+1}$ and r_t to $y_{a_t,t+1}$
11: **if** $t \equiv 0 \pmod{\tau}$ **then**
12: Re-estimate the hyper-covariance from the archive's candidate estimates:

$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d$$

13: **else**
14: $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
15: **end if**
16: **end for**

3.5 Algorithm

Algorithm 1 summarizes the resulting routing procedure. It mirrors the base PersonalEvo router round for round; the only changes are the generalized ridge geometry $\lambda \widehat{\Omega}_t^{-1}$ inside $V_{a,t}^{\text{hyper}}$ and the periodic re-estimation of the hyper-covariance from the archive's own per-candidate estimates.

Three clarifications about the data the algorithm accumulates. First, $y_{a,t}$ is *not* the mean reward μ collected over steps: it stacks the *observed* rewards $r_s = \mu(u_s, a) + \xi_s$ of the rounds in which candidate a was served, so each entry is a noisy realization of μ, and the ridge solve is what averages the noise away. Second, the rows of $X_{a,t}$ can come from *many users*: row s is the feature vector $x(u_s, a)$ of whichever user was served at round s, so one candidate's design matrix pools every user who has ever received it; this pooling is what lets the estimate $\widehat{\theta}_{a,t}$ (and through it $\widehat{\Omega}_t$) benefit from the whole population's feedback. Third, the algorithm as stated serves a *stream of users*: line 3 observes u_t, which may differ every round. Following the convention of Section 3.2, the theory analyzes the trajectory of one user U and averages over the population afterward; the experiments instantiate the two extremes, running one bandit per user in the per-user simulators (each user's $X_{a,t}$ then contains only their own rounds) while the shared-covariance estimate $\widehat{\Omega}_t$ always pools across the archive.

The per-round cost is identical to base PersonalEvo up to the re-estimation step, which touches only the $d \times d$ matrix $\widehat{\Omega}_t$ and amortizes over the period τ. Under Assumption 7, the exploration-cost guarantee for Algorithm 1 is the effective-dimension bound stated in Proposition 8; the partial argument, together with the open covariance-estimation term $\mathrm{Err}_T(\widehat{\Omega}_t, \Omega)$, is collected in Appendix J.

The remainder of the paper turns to empirical evidence. Section 6 first reports the nine-benchmark results for the base PersonalEvo router and then separates two Hyper-Ridge questions: whether a known archive geometry can reduce exploration, and whether the current periodic estimator can recover that geometry from online data.

Algorithm 2 describes PersonalEvo's per-user routing loop.

At each round, a disjoint LinUCB bandit selects the best candidate for the current user based on UCB scores.

Algorithm 2 Per-user personalized routing over a growing archive.

Require: user u, common initial archive A_1, horizon T, exploration parameter α, gap threshold τ, cooldown κ, archive cap $K_{\max}$

```
A_1(u) ← A_1; for each a ∈ A_1(u) initialize A_a ← I, b_a ← 0
t_last ← −∞
for t = 1, . . . , T do
    for each a ∈ A_t(u) do
        x_t(a) ← ψ(u, a)                                      ▷ feature map, equation 4
        θ̂_a ← A_a^{-1} b_a
        UCB_t(a) ← θ̂_a^⊤ x_t(a) + α sqrt(x_t(a)^⊤ A_a^{-1} x_t(a))
    end for
    a_t(u) ← arg max_{a∈A_t(u)} UCB_t(a)
    observe r_t = μ(u, a_t(u)) + ξ_t
    A_{a_t(u)} ← A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^⊤;   b_{a_t(u)} ← b_{a_t(u)} + r_t x_t(a_t(u))
    if m_t(u) − r_t > τ and t − t_last > κ and |A_t(u)| < K_max then
        ã_t(u) ← F(u, A_t(u), trace_t); A_{t+1}(u) ← A_t(u) ∪ {ã_t(u)}; initialize A_ã, b_ã; t_last ← t
    else
        A_{t+1}(u) ← A_t(u)
    end if
end for
```

When the router detects a large gap between expected and observed reward, it triggers a reflection step that proposes a new candidate to add to the archive.

The feature map used in our experiments is

$$\psi(u,a) = \big[\,\mathrm{onehot}(\mathrm{cat}(a))\,;\, w_u\,\big] \in \mathbb{R}^{2K_{\mathrm{cat}}}, \tag{4}$$

where $\mathrm{cat}(a)$ is the candidate's category and $w_u \in \mathbb{R}^{K_{\mathrm{cat}}}$ is the user preference vector exposed by the simulator.

As a concrete instance (used in our experiments), the $K_{\mathrm{cat}} = 5$ categories are $\{\mathrm{sorting}, \mathrm{searching}, \mathrm{formatting}, \mathrm{general}, \mathrm{calc}\}$, so for a candidate a in the `searching` category, $\mathrm{onehot}(\mathrm{cat}(a)) = (0, 1, 0, 0, 0)$ and the user block w_u records that user's affinity for each of the five categories.

We refer the reader to Appendix H for the full construction (categories, quality components, and feature concatenation), and to Team (2025c) for a richer factorized feature map over task type, memory mode, tool mode, and answer style.

The reflection trigger is the high-gap-failure rule

$$m_t(u) - r_t > \tau, \qquad t - t_{\mathrm{last}} > \kappa, \tag{5}$$

and the proposer $\mathcal{F}$ in the saved simulator is oracle-assisted: it reads the user's peak category and returns a child candidate biased toward that category.

Because the proposer has direct access to the user's preference, the evolution gain $\bar{G}_T$ measured in our experiments should be read as an upper bound on what archive growth could contribute (under an idealized proposer), not as evidence that a realistic LLM-driven proposer would deliver the same gain.

Reward decomposition. Using Algorithm 2, we show that the average reward the router achieves over horizon T satisfies (the subscript 1 in A_1 refers to the round $t = 1$ initial archive shared across all users, not to a particular user):

$$\text{routed value} \;=\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{6}$$

This equation separates the router's performance into the four terms defined above, enabling independent measurement of each effect.

Example. Consider two candidates and two equally likely user types with means $\mu(u_1, a_1) = 0.9$, $\mu(u_2, a_1) = 0.4$, $\mu(u_1, a_2) = 0.4$, $\mu(u_2, a_2) = 0.9$.

Then $V_{\text{static}} = 0.65$ (the best single candidate averages 0.65), while $V_{\text{route}} = 0.9$ (matching each user to their best gives 0.9), so $\Delta_{\text{route}} = 0.25$.

With no archive growth ($\bar{G}_T = 0$), the decomposition gives: routed value $= 0.65 + 0.25 - \bar{L}_T$.

Routing is profitable as long as the exploration cost $\bar{L}_T < 0.25$.

To measure the four terms, V_{static} and Δ_{route} can be estimated offline from a held-out user sample: $V_{\text{static}}(A_1) = \max_a \frac{1}{N} \sum_u \mu(u, a)$ and $\Delta_{\text{route}}(A_1) = \frac{1}{N} \sum_u \max_a \mu(u, a) - V_{\text{static}}$.

The online terms $\bar{G}_T$ and $\bar{L}_T$ are measured from the router's trajectory.

4 Method

The preceding sections establish the base PersonalEvo framework: a routing decomposition over a self-evolving archive, guarantees for the fixed-archive router, and a repair-time bound for archive growth. All of these results are inherited unchanged. In this section we introduce the revision's contribution: a replacement for the router's per-candidate estimation scheme, motivated by a meta-learning perspective on the archive. The new method leaves the decomposition and its first three terms intact and targets only the exploration-cost term, by importing hyper-covariance estimation ideas from generalized ridge regression. We first record the two source documents that anchor this revision.

4.1 Summary of what changes

Before turning to experiments, we summarize how this section modifies the base paper. First, the router changes: the generalized-ridge router defined by $V_{a,t}^{\text{hyper}}$ and $\text{UCB}_t^{\text{hyper}}$ replaces the base per-candidate LinUCB router, which regularizes each θ_a independently with an isotropic ridge penalty. Second, the exploration-cost theory changes: the effective-dimension bound $\sqrt{K_{\max}\, d_{\text{eff}}(\Omega)\, T \log T} + \text{Err}_T(\widehat{\Omega}_t, \Omega)$ replaces the base bound $\sqrt{K_{\max}\, d\, T \log T}$. The known-$\Omega$ bound is proved, and Appendix L proves finite-sample control for a calibration-split estimator that freezes $\widehat{\Omega}$ before routing. The periodically updated estimator in Algorithm 1 remains conditional. The improvement is meaningful under an explicit regime condition, which we state as an assumption rather than a result: the archive has low-dimensional shared structure, so that $d_{\text{eff}}(\Omega) \ll d$. Finally, nothing else changes: the four-term decomposition $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$ is inherited from the base framework exactly as proven there, and this revision only sharpens the bound on the $\bar{L}_T$ term.

5 Theoretical Results for Hyper-Ridge PersonalEvo

5.1 Theory: improved exploration cost

Throughout this section the base assumptions of Section 3 (Assumptions 1–4 below) remain in force, and we write $x(u, a)$ for the feature map denoted $\psi(u, a)$ there; the only new hypothesis is the shared archive geometry of Assumption 7. We first restate the inherited base results this revision builds on, then derive the sharpened exploration-cost bound.

The decomposition holds under minimal conditions (bounded rewards and measurability).

Below we state the assumptions needed for the finite-time guarantee.

Assumption 1 (Bounded mean reward)**.** *The mean reward is bounded:*

$$\mu(u, a) \in [0, 1] \quad \textit{for all } (u, a). \tag{7}$$

Assumption 2 (Conditionally sub-Gaussian noise)**.** *There exists $\sigma > 0$ with*

$$\mathbb{E}[\exp(\lambda\xi_t) \mid \mathcal{F}_{t-1}] \le \exp(\lambda^2\sigma^2/2), \qquad \lambda \in \mathbb{R}. \tag{8}$$

Assumption 3 (Predictable archive)**.** *The user-specific archive is predictable:*

$$A_t(U) \in \mathcal{F}_{t-1}. \tag{9}$$

Assumption 4 (Bounded features)**.** *The feature map satisfies*

$$\|\psi(u,a)\| \le L < \infty \quad \textit{for all } (u,a). \tag{10}$$

Assumptions 2–4 are not required for Theorem 2 (the main result); they are needed only for the conditional plug-in corollary that imports a exploration-cost bound from the linear bandit literature.

The first result establishes when routing gain is strictly positive:

Lemma 1 (Static-vs-routed value)**.** *For every finite archive A,*

$$\Delta_{\text{route}}(A) \ge 0, \tag{11}$$

with equality if and only if there exists $a^\star \in A$ such that

$$\mu(U,a^\star) = \max_{a\in A} \mu(U,a) \quad \textit{almost surely.} \tag{12}$$

That is, $\Delta_{\text{route}}(A) > 0$ whenever different users prefer different candidates.

The proof is in Appendix J.

Main theorem.

Theorem 2 (Exact decomposition on the common initial archive)**.** *For every round t and every horizon $T \ge 1$,*

$$\mathbb{E}[\mu(U,a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)], \tag{13}$$

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U,a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{14}$$

Proof sketch (full proof in Appendix J). By definition,

$$\mu(U,a_t(U)) = m_t(U) - L_t(U),$$
$$m_t(U) = m_1(U) + G_t(U).$$

Combining and taking expectations over (U,ξ),

$$\mathbb{E}[\mu(U,a_t(U))] = \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)].$$

Lemma 1 on A_1 gives $\mathbb{E}[m_1(U)] = V_{\text{route}}(A_1) = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1)$, which is equation 13. Averaging over t yields equation 14. □

Theorem 2 says that the reward achieved by the personalized routing policy decomposes into a structural piece on the initial archive ($V_{\text{static}} + \Delta_{\text{route}}$), an online discovery piece ($\bar{G}_T$), and a negative online learning piece ($-\bar{L}_T$).

All four terms are measurable on the saved simulator: $\bar{L}_T$ in particular is the gap between the router's actual reward and the best reward attainable in the current archive (i.e., $m_t(u) - \mu(u, a_t(u))$), averaged over rounds.

Dominance condition.

Corollary 3 (When personalized routing beats the best static initial candidate)**.** *For any horizon T,*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U,a_t(U))] > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{15}$$

On a fixed archive (no growth, $\bar{G}_T = 0$), this collapses to

$$\Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{16}$$

In other words, personalized routing is beneficial when the routing gain and evolution gain together exceed the exploration cost.

While the inequality is algebraically immediate from equation 6, its value is operational: each side is independently estimable, so the condition can be checked in advance on a held-out user sample (for $\Delta_{\text{route}}(A_1)$) and monitored online during a run (for $\bar{L}_T$ and $\bar{G}_T$), giving a quantitative go/no-go criterion rather than a qualitative claim.

We verify the condition empirically in Section 6.

The static-vs-routed inequality behind Lemma 1 is the folklore $\mathbb{E}[\max_a X_a] \geq \max_a \mathbb{E}[X_a]$; our contribution is the equality condition and the *corrected* static comparator $V_{\text{static}}(A_1) = \max_a \mathbb{E}_U[\mu(U, a)]$, which (unlike the heuristic $\arg\max_a q(a)$ baseline in the original simulator) makes the dominance criterion equation 15 falsifiable rather than tautological.

What is specific to PersonalEvo is the four-term accounting under the double randomness of a population draw interleaved with an online single-user trajectory, with every term measured directly.

Supporting lemmas. The two lemmas below support the main theorem: Lemma 4 ensures that the per-arm bandit statistics remain valid as the archive grows, and Lemma 5 bounds the number of mutations triggered by the failure rule.

Lemma 4 (Predictable archive)**.** *Under Assumption 3, appending a child after round t modifies only $A_{t+1}(U)$ and not $A_t(U)$. Therefore the disjoint per-arm sufficient statistics (A_a, b_a) in Algorithm 2 remain $\mathcal{F}_{t-1}$-measurable for the arms available at round t, and standard self-normalized concentration arguments apply on a per-arm basis to the realized arm path.*

Lemma 5 (Failure-prioritization)**.** *Under the trigger equation 5, the expected number of reflection mutations across T rounds is bounded by*

$$\mathbb{E}[\#\text{mutations}] \leq T \cdot \Pr\big(L_t(U) + |\xi_t| > \tau\big), \tag{17}$$

which controls archive bloat without changing the regret analysis of the router itself.

The failure-prioritization gate is the algorithmic counterpart of the constraint-gated self-evolution line reviewed in Section 2, but here it is embedded inside a user-conditioned routing rule.

Positioning of the decomposition. The comparison to personalized bandits can now be stated in the paper's notation.

A fixed in-archive oracle comparison folds the structural routing gain into the oracle value, whereas equation 14 factors it out as $\Delta_{\text{route}}(A_1)$.

A fixed action set also removes the archive-improvement term, whereas PersonalEvo keeps $\bar{G}_T$ explicit and uses Lemma 4 to ensure that online archive growth does not depend on future randomness.

Offline prompt-evolution methods can improve the archive, but without user-conditioned routing they do not separate the routing gain $\Delta_{\text{route}}(A_1)$ from the evolution gain $\bar{G}_T$.

Finite-time bound on the exploration cost. Theorem 2 is an exact equality but does not by itself bound the size of the exploration cost $\bar{L}_T$.

To obtain a finite-time lower bound on the router's average reward, we need a controlled estimate of $\bar{L}_T$.

Rather than re-deriving such a bound from scratch for our particular router, we state it as an external assumption (Assumption 5 below) and substitute it into the decomposition.

Assumption 5 (External exploration-cost bound)**.** *There exists a deterministic sequence B_T such that*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T \quad \textit{for every } T \geq 1. \tag{18}$$

Corollary 6 (Plug-in bound)**.** *Under Assumptions 1–4 and 5,*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \geq V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{19}$$

For Algorithm 2 (disjoint LinUCB with $|A_t(U)| \le K_{\max}$ and feature dimension d), the standard self-normalized concentration argument (Abbasi-Yadkori et al., 2011; Chu et al., 2011) yields

$$B_T = O\left(\sqrt{K_{\max}\, d\,T\, \log T}\right), \tag{20}$$

which we import rather than re-prove.

Discovery-time bound. Discovery is the most fragile term in equation 6 because it depends on the proposer $\mathcal{F}$.

We replace the original geometric contraction theorem with a stopping-time statement under a per-attempt repair probability.

Assumption 6 (Repair probability)**.** *There exists $q_\epsilon \in (0, 1]$ such that for each cluster g with current gap $\Delta_g > \epsilon$, each reflection attempt produces an ϵ-improving child with probability at least q_ϵ, independent of past attempts conditional on the trigger firing.*

Theorem 7 (First-useful-specialist time)**.** *Under Assumption 6, after m reflection attempts on cluster g,*

$$\Pr\big(\textit{some attempt improves cluster } g \textit{ by } \epsilon\big) \ge 1 - (1 - q_\epsilon)^m. \tag{21}$$

Note that in our experiments the proposer is oracle-assisted ($q_\epsilon \approx 1$), so this bound is not empirically binding.

The non-oracle case is left as future work.

Section 6 measures each of the four terms in equation 14 directly and verifies the dominance condition equation 15 against $V_{\text{static}}(A_1)$.

We now turn to the new result. The four-term decomposition remains unchanged:

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T.$$

The new method improves the bound on $\bar{L}_T$.

For ordinary LinUCB, the generic exploration cost has the form

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \lesssim \sqrt{K_{\max}\, d\, T \log T}.$$

With a learned hyper-covariance, the dimension d can be replaced by an effective dimension induced by the archive covariance:

$$d_{\text{eff}}(\Omega) = \operatorname{tr}\left[X^\top X \left(X^\top X + \lambda \Omega^{-1}\right)^{-1}\right].$$

Therefore, one expects a bound of the form

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \lesssim \sqrt{K_{\max}\, d_{\text{eff}}(\Omega)\, T \log T} + \text{Err}_T(\widehat{\Omega}, \Omega).$$

If the archive has low-dimensional shared structure, then

$$d_{\text{eff}}(\Omega) \ll d,$$

and hence

$$\bar{L}_T^{\text{hyper}} < \bar{L}_T^{\text{LinUCB}}.$$

Consequently,

$$R_T^{\text{hyper}} \geq V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{hyper}}}{T},$$

where

$$B_T^{\text{hyper}} = O\Big(\sqrt{K_{\max}\, d_{\text{eff}}(\Omega)\, T \log T}\Big) + \mathrm{Err}_T(\widehat{\Omega}, \Omega).$$

5.2 Why this matters

The original dominance condition is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T.$$

Therefore, reducing $\bar{L}_T$ makes personalized routing profitable in more regimes.

This matters because in many realistic agent systems the routing gain may be moderate, not huge.

In those cases, ordinary exploration may erase the benefit of personalization.

Hyper-Ridge PersonalEvo can preserve the personalization advantage by lowering the exploration cost through archive-level meta-learning.

We inherit the accounting identity from the Background section unchanged: Hyper-Ridge PersonalEvo changes only the statistical control of the exploration term, not the four-term decomposition itself. The average routed value is

$$R_T^{\text{hyper}} = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]\,. \tag{22}$$

By the decomposition in Section 3, the same quantity satisfies

$$R_T^{\text{hyper}} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{23}$$

Thus the only theoretical change in this section is the replacement of the ordinary dimension-dependent control of

$$\bar{L}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] \tag{24}$$

by an archive-geometry-dependent control.

The routing rule analyzed below is the implementation target for the hyper-ridge revision. For candidate a, the generalized design matrix is

$$V_{a,t}^{\text{hyper}} = X_{a,t}^{\top} X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{25}$$

The generalized ridge estimate is

$$\widehat{\theta}_{a,t} = \left(V_{a,t}^{\text{hyper}}\right)^{-1} X_{a,t}^{\top} y_{a,t}. \tag{26}$$

The corresponding upper-confidence score is

$$\mathrm{UCB}_t^{\text{hyper}}(u, a) = x(u, a)^{\top} \widehat{\theta}_{a,t} + \alpha \sqrt{x(u, a)^{\top} \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u, a)}. \tag{27}$$

Assumption 7 (Shared latent structure and hyper-covariance model)**.** *Each candidate has a linear mean reward model:*

$$\mu(u, a) = x(u, a)^{\top} \theta_a. \tag{28}$$

The feature vector has ambient dimension

$$x(u, a) \in \mathbb{R}^d. \tag{29}$$

Algorithm 3 Hyper-Ridge PersonalEvo routing

Require: initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period τ, covariance floor $\epsilon > 0$
1: Initialize $\widehat{\Omega}_1 \leftarrow I_d$
2: Initialize $X_{a,1}$ and $y_{a,1}$ as empty for each $a \in A_1$
3: **for** $t = 1, \ldots, T$ **do**
4: Observe user u_t and eligible archive $A_t(u_t)$
5: Compute $\mathrm{UCB}_t^{\text{hyper}}(u_t, a)$ from equation 25–equation 27 for every $a \in A_t(u_t)$
6: Select $a_t(u_t) \in \arg\max_{a \in A_t(u_t)} \mathrm{UCB}_t^{\text{hyper}}(u_t, a)$
7: Observe reward r_t, append $x(u_t, a_t(u_t))^\top$ to $X_{a_t,t+1}$, and append r_t to $y_{a_t,t+1}$
8: **if** $t \equiv 0 \pmod{\tau}$ **then**
9: Set
$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d$$
10: **else**
11: Set $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
12: **end if**
13: **end for**

The candidate parameter is a random effect with shared covariance:

$$\mathbb{E}[\theta_a] = 0. \tag{30}$$

Its second moment is

$$\mathbb{E}[\theta_a \theta_a^\top] = \Omega. \tag{31}$$

The hyper-covariance is positive definite:

$$\Omega \succ 0. \tag{32}$$

For the pooled design matrix X_T, the effective dimension induced by Ω is

$$d_{\text{eff}}(\Omega) = \operatorname{tr}\left[X_T^\top X_T \left(X_T^\top X_T + \lambda \Omega^{-1} \right)^{-1} \right]. \tag{33}$$

The shared-structure regime is the low-effective-dimension condition

$$d_{\text{eff}}(\Omega) \ll d. \tag{34}$$

Assumption 7 is a regime condition, not an empirical claim: it formalizes the note's premise that evolved archive candidates share latent geometry.

Proposition 8 (Effective-dimension exploration-cost bound)**.** *Consider Algorithm 3 under Assumptions 1–4 (Section 3) together with Assumption 7. The per-round exploration cost is*

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{35}$$

The maximum archive size is

$$K_{\max} = \max_{1 \le t \le T} |A_t(U)|. \tag{36}$$

The covariance-estimation residual is a nonnegative perturbation term:

$$\mathrm{Err}_T(\widehat{\Omega}, \Omega) \ge 0. \tag{37}$$

If the generalized-ridge confidence sets for equation 25 hold with the same self-normalized form used in OFUL/LinUCB analyses (Abbasi-Yadkori et al., 2011; Chu et al., 2011), then there is a numerical constant $C > 0$ such that

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le C \sqrt{K_{\max}\, d_{\text{eff}}(\Omega)\, T\, \log(1+T)} + \mathrm{Err}_T(\widehat{\Omega}, \Omega). \tag{38}$$

Proof sketch. The oracle-covariance part follows the standard elliptical-potential argument after replacing the isotropic ridge matrix by $\lambda\Omega^{-1}$:

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] &\lesssim \sum_{a\in A_T}\sqrt{d_{\text{eff}}(\Omega)\,N_a(T)\,\log(1+T)}+\mathrm{Err}_T(\widehat{\Omega},\Omega)\\ &\le\sqrt{K_{\max}\sum_{a\in A_T}d_{\text{eff}}(\Omega)\,N_a(T)\,\log(1+T)}+\mathrm{Err}_T(\widehat{\Omega},\Omega)\\ &\le C\sqrt{K_{\max}\,d_{\text{eff}}(\Omega)\,T\,\log(1+T)}+\mathrm{Err}_T(\widehat{\Omega},\Omega). \end{aligned}\tag{39}$$

The complete derivation is deferred because the non-oracle step requires controlling the perturbation from Ω to $\widehat{\Omega}_t$. □

Remark 1 (Covariance-estimation control: resolved up to rate sharpening)**.** *For a calibration-split variant of the router (the hyper-covariance is estimated in a pre-routing phase on fresh candidates and then frozen), Theorem 57 in Appendix L proves that the relative covariance error satisfies* $\rho_T = O(T^{-1/10}\sqrt{\log T}) \to 0$ *with high probability and that the exploration cost obeys*

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] = O\Big(\sqrt{d}\ \log T\,\sqrt{K_{\max}\,d_{\text{eff}}(\Omega)\,T}\Big)+\mathrm{Err}_T,\qquad \mathrm{Err}_T=o(\sqrt{T}).\tag{40}$$

The single remaining open question is removing the residual $\sqrt{d\log T}$ *factor to match the oracle rate of Proposition 8 exactly (Conjecture K.1).*

As a concrete scale calculation, suppose

$$K_{\max}=16.\tag{41}$$

Suppose the ambient feature dimension is

$$d=100.\tag{42}$$

Suppose the archive geometry has effective dimension

$$d_{\text{eff}}(\Omega)=9.\tag{43}$$

For the same horizon and logarithmic factor, the leading exploration term changes by the ratio

$$\frac{\sqrt{K_{\max}d_{\text{eff}}(\Omega)T\log T}}{\sqrt{K_{\max}dT\log T}}=\sqrt{\frac{9}{100}}=0.3.\tag{44}$$

Thus the proposition predicts a potentially large improvement only when the residual term remains smaller than the saved leading term.

Corollary 9 (Dominance in more regimes)**.** *Define the structural-and-evolution margin by*

$$M_T=\Delta_{\text{route}}(A_1)+\bar{G}_T.\tag{45}$$

Define the hyper-ridge exploration budget by

$$B_T^{\text{hyper}}=C\sqrt{K_{\max}\,d_{\text{eff}}(\Omega)\,T\,\log(1+T)}+\mathrm{Err}_T(\widehat{\Omega},\Omega).\tag{46}$$

If

$$M_T>\frac{B_T^{\text{hyper}}}{T},\tag{47}$$

then Hyper-Ridge PersonalEvo satisfies the dominance condition

$$R_T^{\text{hyper}}>V_{\text{static}}(A_1).\tag{48}$$

Moreover, compared with an ordinary LinUCB budget

$$B_T^{\text{LinUCB}}=C\sqrt{K_{\max}\,d\,T\,\log(1+T)},\tag{49}$$

Hyper-Ridge certifies all margins in the additional interval

$$\frac{B_T^{\text{hyper}}}{T}<M_T\le\frac{B_T^{\text{LinUCB}}}{T},\tag{50}$$

whenever

$$B_T^{\text{hyper}}<B_T^{\text{LinUCB}}.\tag{51}$$

Benchmark	$\lvert\mathcal{U}\rvert$	$\lvert A_1\rvert$	Preference structure
General	8	12	5 disjoint category peaks
Workspace	8	12	2 broad paired peaks
Support	8	12	3 overlapping clusters
GEPA archive	12	15	heterogeneous task prefs
Hermes agent	15	20	(skill,mem,tool) clusters
BESPOKE	30	40	4-dim rubric prefs
PAHF shop.	20	40	product-attribute prefs
PAHF embod.	40	40	embodied object choice
PrefEval	20	10	topic-strategy prefs

Table 1: Benchmarks. $|\mathcal{U}|$: users; $|A_1|$: candidates. Synthetic: T=300; others: T=250.

Benchmark	V_{st}	Δ_{rt}	$\bar{G}_T$	$\bar{L}_T$	**Dom.**
General	.570	.280	.051	.036	+**.295**
Workspace	.577	.124	.041	.043	+.122
Support	.579	.215	.061	.039	+.237
GEPA archive	.654	.048	.023	.025	+.046
Hermes agent	.703	.092	.036	.028	+.100
BESPOKE	.653	.036	.001	.010	+.027
PAHF shop.	.577	.209	.002	.054	+.157
PAHF embod.	.290	.238	.000	.071	+.167
PrefEval	.531	.251	.000	.028	+.223

Table 2: Four-term decomposition. Dom. = $\Delta_{\text{rt}}+\bar{G}_T-\bar{L}_T$; all positive.

Proof. Substituting Proposition 8 into the inherited decomposition gives

$$R_T^{\text{hyper}} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$$
$$\geq V_{\text{static}}(A_1) + M_T - \frac{B_T^{\text{hyper}}}{T}. \tag{52}$$

Condition equation 47 implies equation 48. The interval claim follows by comparing equation 46 and equation 49. ☐

This corollary is the formal sense in which learning archive geometry can make personalization profitable in more regimes: it lowers the required routing-plus-evolution margin, subject to the open control of $\mathrm{Err}_T(\widehat{\Omega}, \Omega)$. Full proof details and the residual-control roadmap are collected in Appendix J.

6 Experiments

6.1 Environments and Data

We evaluate on three synthetic environments that share the same reward structure but differ in user heterogeneity.

The mean reward follows the PABF formulation (Team, 2025c):

$$\mu(u, a) = f\big(q(a),\ w_u(\mathrm{cat}(a))\big), \tag{53}$$

where $q(a) \in [0,1]$ is the intrinsic candidate quality, $\mathrm{cat}(a) \in \mathcal{C}$ is the candidate's category, $w_u \in \Delta^{|\mathcal{C}|-1}$ is the user's category preference, and f is increasing in both arguments (functional form in Appendix H.2).

The observed reward is $r_t = \mu(U, a_t(U)) + \xi_t$ with $\xi_t \sim \mathcal{N}(0, 0.05^2)$.

We use a synthetic simulator so that all four decomposition terms can be computed exactly; in a real system the algorithm runs identically but the oracle term $m_t(u)$ would require held-out estimation.

We compare five variants of Algorithm 2 on all nine benchmarks: **Full** (all components enabled), **No-routing** (uniform-random candidate selection), **No-reflection** (frozen archive), **No-personalization** (onehot-only features), and **Collapse-1** (single candidate).

Each removes one component to isolate its contribution to equation 6; details and color coding are in Figure 2.

6.2 Results

We estimate all four terms of equation 6 on each benchmark (estimator definitions in Appendix H.5).

Table 2 (right panel above) reports the decomposition across all nine benchmarks.

Three findings hold across all nine benchmarks:

1. **Routing gain is the largest positive component.** In every row, Δ_{route} exceeds $\bar{G}_T$: routing gains range from 0.036 to 0.280, while evolution gains range from 0 to 0.061. The ordering broadly tracks user heterogeneity, consistent with Lemma 1.
2. **Dominance is always positive.** On all nine benchmarks, including four with real user preference data, the condition $\Delta_{\text{route}} + \bar{G}_T > \bar{L}_T$ from Corollary 3 holds. The full four-term margin ranges from $+0.027$ (BESPOKE) to $+0.295$ (General).
3. **Real-data heterogeneity is substantial.** PAHF shopping (Δ_{route}=0.209) and PrefEval (Δ_{route}=0.251) are comparable to the synthetic environments, confirming that the decomposition's structural term is practically significant on real preferences.

Ablation. Figure 1 shows the mean reward for all five ablation variants across all nine benchmarks.

Removing the router collapses performance to near $V_{\text{static}}(A_1)$ in every case; removing reflection or personalization features each reduces reward by a smaller margin.

The hierarchy No-routing < Static < No-reflect < Full < Oracle holds consistently.

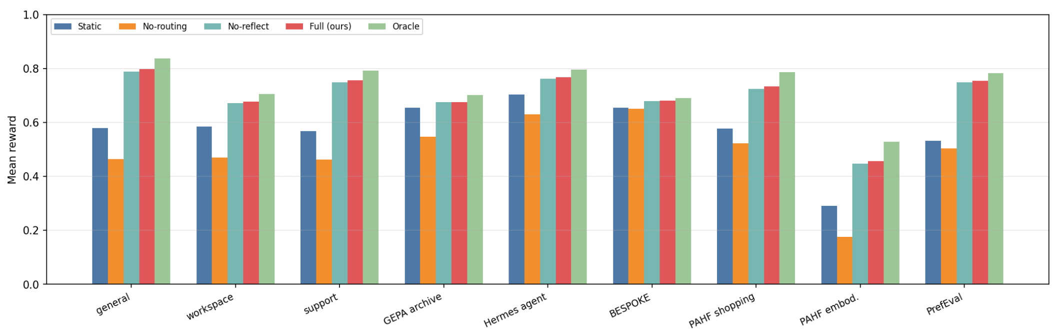

Figure 1: Mean reward by ablation variant across all nine benchmarks. Removing routing collapses reward; reflection and personalization features provide smaller incremental gains.

Per-round convergence. Figure 2 shows per-round reward traces for each benchmark.

In all cases, PersonalEvo converges toward the oracle within $\sim$50 rounds, while the static baseline remains flat.

The exploration cost $\widehat{L}_t$ decays sublinearly, consistent with the $O(\sqrt{K_{\max} dT \log T})$ rate of Corollary 6.

6.3 Comparison against existing routing methods

To position PersonalEvo against prior routing strategies, we compare five methods on all nine benchmarks (Table 3): (i) **Static**: deploy the single best-on-average candidate; (ii) **GEPA-Pareto**: maintain the Pareto-optimal subset of the archive and deploy uniformly from it (Agrawal et al., 2025); (iii) **Thompson**: Bayesian Thompson sampling over candidates (Labs, 2025); (iv) **RouteLLM-thresh**: route users to specialist or generalist candidates based on a variance threshold; (v) **PersonalEvo**: our full method.

The results reveal why the decomposition matters for method design: GEPA-Pareto maintains a diverse archive ($\bar{G}_T$ is high) but has no routing mechanism to exploit Δ_{route}, so diversity is wasted.

Thompson sampling routes but explores every arm equally regardless of user context, paying excessive $\bar{L}_T$.

RouteLLM routes by context but uses a fixed threshold rather than learning individual preferences.

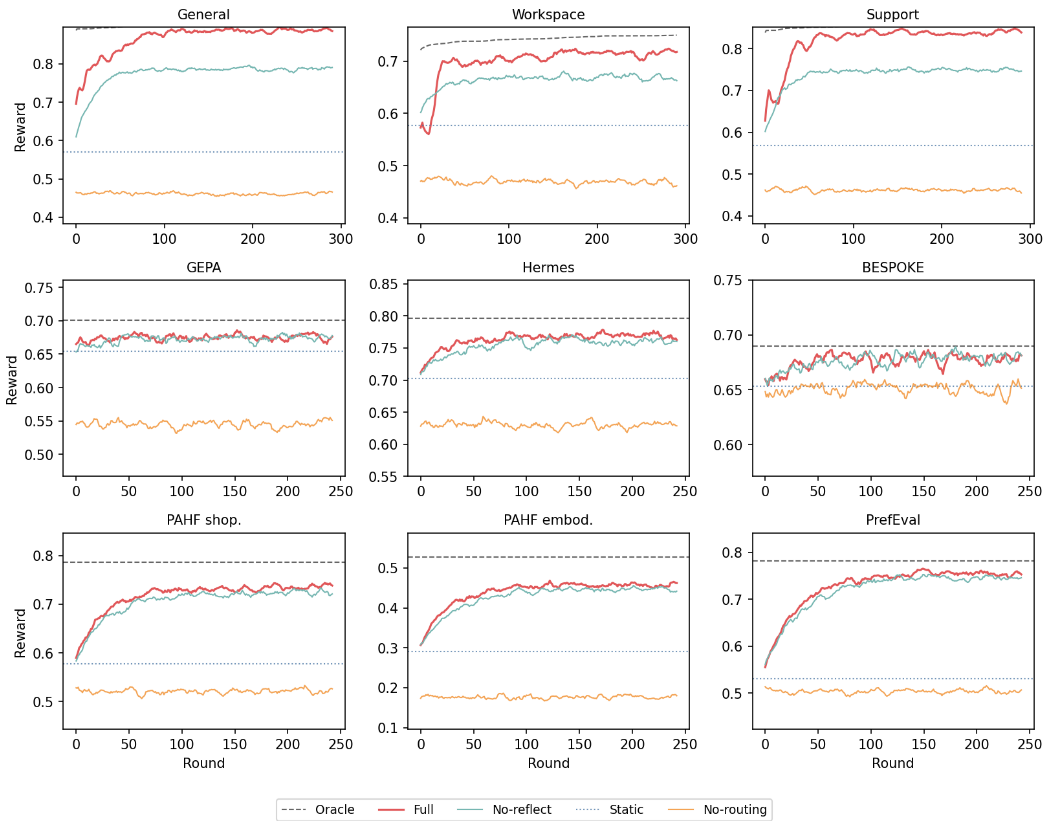

Figure 2: Per-round reward for all nine benchmarks (3×3 grid). Five variants shown: **Full** (red; concat features, gated reflection, LinUCB), **No-reflect** (teal; frozen archive, $\bar{G}_T$=0), **No-routing** (orange; uniform-random selection, Δ_{route}→0), **Static** (gray dotted; best single candidate for all users), **Oracle** (black dashed; per-user optimum). Top row: synthetic; middle and bottom: framework and real-data benchmarks.

Benchmark	Static	GEPA-Pareto	Thompson	RouteLLM	PersonalEvo	Oracle
General	0.579	0.533	0.510	0.732	**0.797**	0.836
Workspace	0.585	0.539	0.515	0.650	**0.678**	0.705
Support	0.568	0.525	0.504	0.700	**0.757**	0.791
GEPA archive	0.671	0.616	0.569	0.675	**0.681**	0.701
Hermes agent	0.729	0.644	0.659	0.750	**0.771**	0.796
BESPOKE	0.681	0.660	0.654	0.678	**0.682**	0.690
PAHF shopping	0.625	0.532	0.560	0.691	**0.754**	0.812
PAHF embodied	0.290	0.244	0.222	0.407	**0.457**	0.528
PrefEval	0.531	0.520	0.514	0.687	**0.754**	0.782

Table 3: Comparison against existing routing methods on all nine benchmarks. Mean reward over 5 seeds. PersonalEvo (bold) achieves the highest reward among non-oracle methods in every row. GEPA-Pareto underperforms static because archive diversity without per-user routing is counterproductive. Thompson sampling explores too broadly. RouteLLM-threshold is the closest competitor but uses a fixed rule rather than learning per-user preferences.

PersonalEvo combines per-user routing (exploiting Δ_{route}) with gated evolution (controlled $\bar{G}_T$) and low exploration cost (small $\bar{L}_T$), which is exactly the configuration the decomposition predicts will win.

6.4 Hyper-Ridge diagnostics and the remaining evaluation gap

The decomposition estimates in Table 2, ablations in Figure 1, per-round traces in Figure 2, and baseline comparison in Table 3 all evaluate the *base* PersonalEvo router. We additionally ran a five-seed diagnostic suite for Hyper-Ridge PersonalEvo. We keep it separate from the headline comparison because it covers seven rather than nine adapters and does not yet measure the full decomposition with exploration cost as the primary endpoint.

Known-geometry mechanism. We first construct a well-specified linear bandit in which $K = 30$ candidate parameters are drawn from a planted rank-three covariance, contexts are dense random unit vectors, and the horizon is $T = 400$. Table 4 compares isotropic ridge, Hyper-Ridge given the true covariance, and Hyper-Ridge with the covariance periodically re-estimated from the candidate models, all with the same ridge weight $\lambda = 0.5$. The oracle geometry cuts cumulative exploration cost by 46–59% as the ambient dimension grows, consistent with the effective-dimension mechanism. The online estimate instead raises cost by 7–19%. The experiment therefore validates the benefit of the right geometry, while locating the current failure in estimating it.

d	Isotropic	Known Ω	Estimated Ω	Known-Ω reduction
20	314.1	170.2	375.0	45.8%
50	447.6	213.1	504.9	52.4%
100	455.4	188.1	487.5	58.7%

Table 4: Mean cumulative exploration cost over five seeds on planted rank-three problems ($K = 30$, $T = 400$, $\lambda = 0.5$ for all three routers). Lower is better. The true geometry realizes the predicted benefit; periodic online estimation does not.

Adapter diagnostic. With tuning seeds disjoint from the five reporting seeds, the periodic estimator improves mean reward over the isotropic router on six of seven adapters, with changes ranging from $+0.002$ on BESPOKE to $+0.048$ on Hermes, and loses 0.010 on GEPA. These numbers are not evidence of cross-candidate geometry transfer. The adapter harness represents candidate a by the one-hot vector e_a. Starting from $\widehat{\Omega}_1 = I$, each candidate estimate then has support only on its own coordinate, so the covariance update remains diagonal. Accordingly, the full-covariance and diagonal-ridge variants match exactly on every adapter and seed. This ablation exposes a representation mismatch that a larger seed count cannot repair.

Remaining protocol. The complete comparison must run base, known-covariance, calibration-split estimated-covariance, periodically estimated-covariance, and diagonal-ridge routers on the same nine benchmarks, using a feature map with shared behavioral directions. It must report all four terms of equation 6, with $\widehat{L}_T$ as the primary endpoint, as well as the fitted regret exponent, covariance error, and $d_{\text{eff}}(\widehat{\Omega})$. Until that comparison is complete, we do not claim an empirical advantage for the fully adaptive Hyper-Ridge router.

Implementation status. The repository implements both the base router and the periodic Hyper-Ridge estimator, together with an identity-reduction test showing that frozen $\widehat{\Omega} = I$ reproduces the base router selection for selection. The diagnostic results are stored in `results/hyper_ridge/suite.json`; the missing component is the feature-aware, nine-benchmark evaluation above, not the router implementation itself.

7 Conclusion

This paper revised the PersonalEvo framework around a single idea: a self-evolving agent should meta-learn the geometry of its own archive. The base framework answers *whether* personalized routing pays off through the exact four-term decomposition

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\big[\mu(U, a_t(U))\big] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{54}$$

and the associated dominance condition

$$\Delta_{\text{route}}(A_1) + \bar{G}_T \;>\; \bar{L}_T, \tag{55}$$

both of which are inherited unchanged from the base paper (Theorem 2, Corollary 3). The revision leaves the first three terms untouched and attacks the only negative term. The base router treats archive candidates as independent LinUCB arms with an identity-regularized ridge solve, paying an exploration cost governed by the ambient feature dimension,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;=\; O\Big(\sqrt{K_{\max}\, d\, T \log T}\Big)\,. \tag{56}$$

Hyper-Ridge PersonalEvo instead models candidate parameters as random effects sharing a hyper-covariance,

$$\theta_a \sim (0, \Omega), \qquad \Omega \in \mathbb{R}^{d\times d}, \tag{57}$$

estimates $\widehat{\Omega}_t$ from pooled archive data, and uses it as the generalized-ridge geometry inside the UCB score. The resulting exploration-cost target replaces d by the effective dimension of the archive,

$$d_{\text{eff}}(\Omega) \;=\; \operatorname{tr}\Big[X^\top X\left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\Big]\,, \tag{58}$$

yielding, under the low-dimensional shared-structure regime and modulo the covariance-estimation error term,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\lesssim\; \sqrt{K_{\max}\, d_{\text{eff}}(\Omega)\, T \log T} \;+\; \mathrm{Err}_T(\widehat{\Omega}, \Omega). \tag{59}$$

When $d_{\text{eff}}(\Omega) \ll d$, the dominance condition equation 55 is satisfied in strictly more regimes, so archive evolution can contribute twice: newly evolved candidates raise $\bar{G}_T$, and sufficiently informative archive data can sharpen $\widehat{\Omega}_t$ and lower $\bar{L}_T$. The generalized-ridge estimator, effective-dimension bound, and diagnostic suite all test this thesis, while the estimated-covariance results show that the second benefit is not automatic.

Two provenance facts bound what this paper claims. First, the nine-benchmark decomposition estimates, ablation hierarchy, per-round convergence traces, and baseline comparison in Section 6 belong to the *base* LinUCB router. They validate the inherited decomposition equation 54 and dominance condition equation 55, with margins

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \;\in\; [+0.027,\; +0.295] \tag{60}$$

across all nine benchmarks, and they confirm that the routing gain exceeds the evolution gain in every row. The separate Hyper-Ridge suite is diagnostic: known geometry helps on planted low-rank problems, but periodic estimation does not recover that benefit, and one-hot adapters collapse the full covariance to a diagonal one. Second, the effective-dimension guarantee equation 59 is a theorem for the oracle-covariance router. Appendix L controls an estimated covariance for a calibration-split procedure under explicit diversity conditions, but this result does not cover the periodically updated estimator in Algorithm 1. Controlling that fully adaptive estimator remains the primary theoretical open problem.

The corresponding empirical open problem is the complete comparison specified in Section 6.4: base LinUCB versus known-covariance, calibration-split, periodically estimated, and diagonal-ridge routers on the same nine benchmarks, with the exploration cost $\widehat{L}_T$ as the primary endpoint. The feature map must admit shared directions; otherwise the full model is diagonal by construction. The comparison should also report covariance error, the fitted regret exponent, and a per-benchmark estimate of $d_{\text{eff}}(\widehat{\Omega})$ to locate both sides of the boundary: regimes where

$$d_{\text{eff}}(\Omega) \ll d \tag{61}$$

and the hyper-ridge router should dominate, and near-isotropic regimes where $d_{\text{eff}}(\Omega) \approx d$ and covariance estimation can only add cost. The current implementation supplies the periodic estimator and common harness; a calibration-split implementation and a shared feature representation remain to be added.

Beyond the two named gaps, the limitations of the base framework carry over. The reflective proposer in the simulated environments is oracle-assisted, so the measured evolution gain $\bar{G}_T$ is an upper bound on what a realistic LLM-driven proposer would deliver, and each user maintains a private archive, so discoveries are not shared across users. The hyper-ridge extension makes the second limitation more interesting rather than less: a shared evolving archive would pool more data into $\widehat{\Omega}_t$, potentially shrinking $\mathrm{Err}_T(\widehat{\Omega}, \Omega)$ at a population rate, but it changes the predictability analysis underlying Lemma 4 and is deferred. More broadly, the decomposition equation 54 applies to any system that maintains a candidate library and must decide what to serve to whom (model routing, prompt selection, tool-use policies), and the message of this revision applies with it: such libraries are structured objects, not unordered bags of candidates, and estimating their latent geometry is a direct, quantifiable lever on the one term in the accounting that works against personalization.

A Appendix Proofs: Preliminaries and the Exact Decomposition

This appendix block fixes the probability model used by all proofs and proves the first main theorem: the exact four-term decomposition.

The proof is an accounting identity; it does not use a contextual-bandit regret theorem and does not assume that reflection is realistic.

A.1 Probability space and online objects

All random variables below are defined on one probability space:

$$(\Omega, \mathcal{F}, \mathbb{P}). \tag{62}$$

All expectations are taken over both the population draw and the online trajectory:

$$\mathbb{E}[Z] = \int_\Omega Z(\omega)\, d\mathbb{P}(\omega). \tag{63}$$

The evaluation user is drawn from a population distribution:

$$U \sim \Pi. \tag{64}$$

The user takes values in a measurable user space:

$$U : \Omega \to \mathcal{U}. \tag{65}$$

The candidate universe is denoted by

$$\mathcal{A}. \tag{66}$$

The common initial archive is

$$A_1 \subseteq \mathcal{A}. \tag{67}$$

The current archive before round t is

$$A_t(U) \subseteq \mathcal{A}. \tag{68}$$

The router's deployed candidate is

$$a_t(U) \in A_t(U). \tag{69}$$

The mean reward function is

$$\mu : \mathcal{U} \times \mathcal{A} \to [0, 1]. \tag{70}$$

The observed reward is

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{71}$$

For later plug-in bounds, the online filtration is

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_T. \tag{72}$$

Feature vectors used by the disjoint LinUCB implementation are denoted by

$$x_t(a) \in \mathbb{R}^d. \tag{73}$$

Only the boundedness and archive assumptions are used in the four-term decomposition proof; the noise and feature assumptions are recorded here so later proof chunks can refer back to the same notation.

Assumption 8 (A1: bounded mean reward)**.** *For every user and candidate, the mean reward satisfies*

$$0 \le \mu(u, a) \le 1. \tag{74}$$

Assumption 9 (A2: finite common initial archive)**.** *The initial archive is nonempty and finite:*

$$1 \le |A_1| < \infty. \tag{75}$$

Assumption 10 (A3: finite append-only online archives)**.** *For every round, the current archive is nonempty and finite almost surely:*

$$1 \le |A_t(U)| < \infty. \tag{76}$$

The archive process is append-only:

$$A_t(U) \subseteq A_{t+1}(U). \tag{77}$$

Assumption 11 (A4: predictable archive)**.** *Before choosing the round-t candidate, the archive is measurable with respect to the past:*

$$\sigma(A_t(U)) \subseteq \mathcal{F}_{t-1}. \tag{78}$$

Assumption 12 (A5: centered conditionally sub-Gaussian noise)**.** *The reward noise is conditionally centered:*

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0. \tag{79}$$

For every real scalar λ, the conditional moment generating function obeys

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \tag{80}$$

Assumption 13 (A6: bounded features)**.** *For every available candidate, the feature norm is bounded:*

$$\|x_t(a)\|_2 \le L. \tag{81}$$

A.2 Values and accounting terms

For any deterministic finite archive, the best static value is

$$V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}[\mu(U, a)]. \tag{82}$$

For the same archive, the perfect routed value is

$$V_{\text{route}}(A) = \mathbb{E}\left[\max_{a \in A} \mu(U, a)\right]. \tag{83}$$

The fixed-archive heterogeneity gain is

$$\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A). \tag{84}$$

The current-archive oracle value for the realized user is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{85}$$

The discovery gain at round t is

$$G_t(U) = m_t(U) - m_1(U). \tag{86}$$

The exploration cost at round t is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{87}$$

The horizon-averaged discovery gain is

$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{88}$$

The horizon-averaged exploration cost is

$$\bar{L}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{89}$$

Proposition 10 (Integrability and signs of the accounting terms)**.** *Under Assumptions 8–10, the current-archive optimum satisfies*

$$0 \le m_t(U) \le 1. \tag{90}$$

The exploration cost satisfies

$$0 \le L_t(U) \le 1. \tag{91}$$

The discovery gain satisfies

$$0 \le G_t(U) \le 1. \tag{92}$$

Consequently all three terms are integrable:

$$\mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] \le 3. \tag{93}$$

Proof. Assumption 8 gives the pointwise reward bound:

$$0 \le \mu(U, a) \le 1 \qquad \text{for every } a \in \mathcal{A}. \tag{94}$$

Taking a maximum over a finite nonempty archive preserves the same interval:

$$0 \le \max_{a \in A_t(U)} \mu(U, a) \le 1. \tag{95}$$

Using equation 85, this is

$$0 \le m_t(U) \le 1. \tag{96}$$

Since the deployed candidate belongs to the current archive, we have

$$\mu(U, a_t(U)) \le \max_{a \in A_t(U)} \mu(U, a). \tag{97}$$

Therefore

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) \quad (98) \\ &\ge 0. \quad (99) \end{aligned}$$

The upper bound follows from the reward range:

$$
\begin{aligned}
L_t(U) \;&=\; m_t(U) - \mu(U, a_t(U)) && (100)\\
&\le\; 1 - 0 && (101)\\
&=\; 1. && (102)
\end{aligned}
$$

The append-only condition implies that the initial archive remains available:

$$A_1 \subseteq A_t(U). \tag{103}$$

Thus the current optimum dominates the initial optimum:

$$
\begin{aligned}
m_t(U) \;&=\; \max_{a \in A_t(U)} \mu(U, a) && (104)\\
&\ge\; \max_{a \in A_1} \mu(U, a) && (105)\\
&=\; m_1(U). && (106)
\end{aligned}
$$

Hence

$$G_t(U) = m_t(U) - m_1(U) \;\ge\; 0. \tag{107}$$

The upper bound is again inherited from the reward range:

$$
\begin{aligned}
G_t(U) \;&=\; m_t(U) - m_1(U) && (108)\\
&\le\; 1 - 0 && (109)\\
&=\; 1. && (110)
\end{aligned}
$$

Combining the three interval bounds gives

$$
\begin{aligned}
\mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] \;&\le\; 1 + 1 + 1 && (111)\\
&=\; 3. && (112)
\end{aligned}
$$

□

A.3 Supporting lemmas for the four-term decomposition

Lemma 11 (Static-versus-routed value)**.** *Let A be any deterministic finite nonempty archive:*

$$1 \;\le\; |A| < \infty. \tag{113}$$

Define its pointwise routed optimum by

$$h_A(U) = \max_{a \in A} \mu(U, a). \tag{114}$$

Define the static value by

$$V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}[\mu(U, a)]. \tag{115}$$

Define the routed value by

$$V_{\text{route}}(A) = \mathbb{E}[h_A(U)]. \tag{116}$$

Define the heterogeneity gap by

$$\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A). \tag{117}$$

Then

$$\Delta_{\text{route}}(A) \;\ge\; 0. \tag{118}$$

Moreover, equality holds if and only if there exists a candidate $a^\star \in A$ such that

$$\mu(U, a^\star) = \max_{a \in A} \mu(U, a) \quad \textit{almost surely}. \tag{119}$$

In particular, the inequality is strict whenever every candidate fails to be pointwise optimal on a set of positive probability:

$$\forall a \in A, \qquad \mathbb{P}\left(\mu(U, a) < \max_{b \in A} \mu(U, b)\right) > 0. \tag{120}$$

Proof. For every fixed candidate, the pointwise maximum dominates its reward:

$$h_A(U) \;=\; \max_{b\in A} \mu(U,b) \tag{121}$$
$$\geq\; \mu(U,a). \tag{122}$$

Taking expectations preserves the inequality:

$$\mathbb{E}[h_A(U)] \;\geq\; \mathbb{E}[\mu(U,a)]. \tag{123}$$

Maximizing the right-hand side over the finite archive gives

$$\mathbb{E}[h_A(U)] \;\geq\; \max_{a\in A} \mathbb{E}[\mu(U,a)]. \tag{124}$$

Substituting the definitions yields the weak inequality:

$$\Delta_{\text{route}}(A) \;=\; V_{\text{route}}(A) - V_{\text{static}}(A) \tag{125}$$
$$=\; \mathbb{E}[h_A(U)] - \max_{a\in A} \mathbb{E}[\mu(U,a)] \tag{126}$$
$$\geq\; 0. \tag{127}$$

Because the archive is finite, the static maximizer set is nonempty:

$$\arg\max_{a\in A} \mathbb{E}[\mu(U,a)] \neq \varnothing. \tag{128}$$

Choose one static maximizer:

$$a^\star \in \arg\max_{a\in A} \mathbb{E}[\mu(U,a)]. \tag{129}$$

For this maximizer, the gap can be written as an expectation of a nonnegative random variable:

$$\Delta_{\text{route}}(A) \;=\; \mathbb{E}[h_A(U)] - \mathbb{E}[\mu(U,a^\star)] \tag{130}$$
$$=\; \mathbb{E}[h_A(U) - \mu(U,a^\star)]\,. \tag{131}$$

The integrand is nonnegative:

$$h_A(U) - \mu(U,a^\star) \;\geq\; 0. \tag{132}$$

A nonnegative random variable has zero expectation exactly when it is zero almost surely:

$$\Delta_{\text{route}}(A) = 0 \quad\Longleftrightarrow\quad h_A(U) - \mu(U,a^\star) = 0 \text{ almost surely.} \tag{133}$$

Using the definition of h_A, this is precisely

$$\mu(U,a^\star) = \max_{a\in A} \mu(U,a) \quad \text{almost surely.} \tag{134}$$

Conversely, if some candidate is pointwise optimal almost surely, then

$$V_{\text{route}}(A) \;=\; \mathbb{E}\left[\max_{a\in A} \mu(U,a)\right] \tag{135}$$
$$=\; \mathbb{E}[\mu(U,a^\star)] \tag{136}$$
$$\leq\; \max_{a\in A} \mathbb{E}[\mu(U,a)] \tag{137}$$
$$=\; V_{\text{static}}(A). \tag{138}$$

Together with equation 118, this forces equality:

$$V_{\text{route}}(A) = V_{\text{static}}(A). \tag{139}$$

Finally, if equation 120 holds, then every possible static maximizer has strictly positive probability of being suboptimal:

$$\mathbb{P}(h_A(U) - \mu(U,a^\star) > 0) > 0. \tag{140}$$

Since the integrand is nonnegative and positive with positive probability, its expectation is positive:

$$\Delta_{\text{route}}(A) = \mathbb{E}[h_A(U) - \mu(U,a^\star)] > 0. \tag{141}$$

□

Lemma 12 (Pointwise routed-value accounting)**.** *For every round t, define the current optimum by*

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{142}$$

Define the discovery term by

$$G_t(U) = m_t(U) - m_1(U). \tag{143}$$

Define the exploration-cost term by

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{144}$$

Then the deployed mean reward satisfies the pointwise identity

$$\mu(U, a_t(U)) = m_1(U) + G_t(U) - L_t(U). \tag{145}$$

Proof. Start from the definition of the exploration cost:

$$L_t(U) \;=\; m_t(U) - \mu(U, a_t(U)). \tag{146}$$

Rearrange the equality:

$$\mu(U, a_t(U)) \;=\; m_t(U) - L_t(U). \tag{147}$$

Use the definition of discovery gain:

$$m_t(U) \;=\; m_1(U) + G_t(U). \tag{148}$$

Substitute equation 148 into equation 147:

$$\mu(U, a_t(U)) \;=\; m_1(U) + G_t(U) - L_t(U). \tag{149}$$

□

A.4 First main theorem: four-term decomposition

Theorem 13 (Exact decomposition over the common initial archive)**.** *This theorem restates the four-term decomposition used as Theorem 2 in Section 5.1. The user is drawn as*

$$U \sim \Pi. \tag{150}$$

The initial archive is finite and nonempty:

$$1 \;\le\; |A_1| < \infty. \tag{151}$$

At round t, the deployed candidate belongs to the current archive:

$$a_t(U) \in A_t(U). \tag{152}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{153}$$

The discovery term is

$$G_t(U) = m_t(U) - m_1(U). \tag{154}$$

The exploration-cost term is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{155}$$

The static initial-archive value is

$$V_{\text{static}}(A_1) = \max_{a \in A_1} \mathbb{E}[\mu(U, a)]. \tag{156}$$

The routed initial-archive value is

$$V_{\text{route}}(A_1) = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a)\right]. \tag{157}$$

The initial-archive heterogeneity gain is

$$\Delta_{\text{route}}(A_1) = V_{\text{route}}(A_1) - V_{\text{static}}(A_1). \tag{158}$$

Then each round satisfies

$$\mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{159}$$

Define the horizon-averaged routed value by

$$R_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{160}$$

Define the horizon-averaged discovery gain by

$$\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{161}$$

Define the horizon-averaged exploration cost by

$$\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{162}$$

Then the horizon-averaged decomposition is

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{163}$$

Proof. By Proposition 10, all quantities in the following expectations are integrable:

$$\mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] < \infty. \tag{164}$$

Apply the pointwise accounting identity from Lemma 12:

$$\mu(U, a_t(U)) \;=\; m_1(U) + G_t(U) - L_t(U). \tag{165}$$

Taking expectations gives

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{166}$$

The first-archive optimum expands as

$$\mathbb{E}[m_1(U)] \;=\; \mathbb{E}\left[\max_{a\in A_1} \mu(U, a)\right] \tag{167}$$

$$= V_{\text{route}}(A_1). \tag{168}$$

By the definition of the heterogeneity gap,

$$V_{\text{route}}(A_1) \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \tag{169}$$

Substituting equation 169 into equation 166 yields the per-round identity:

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{170}$$

Average the previous display over the horizon:

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \;=\; \frac{1}{T}\sum_{t=1}^{T} \left(V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]\right) \tag{171}$$

$$= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)] - \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{172}$$

Using the definitions of R_T, $\bar{G}_T$, and $\bar{L}_T$, this becomes

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{173}$$

□

The theorem is an identity in mean rewards.

When Assumption 12 holds, the same expectation also governs observed rewards:

$$\begin{aligned}\mathbb{E}[r_t] &= \mathbb{E}[\mu(U, a_t(U)) + \xi_t] &(174)\\ &= \mathbb{E}[\mu(U, a_t(U))] + \mathbb{E}[\xi_t] &(175)\\ &= \mathbb{E}[\mu(U, a_t(U))]. &(176)\end{aligned}$$

As a numerical check of the identity, the corrected `general` row in Table 2 uses

$$V_{\text{static}}(A_1) = 0.570. \tag{177}$$

The measured heterogeneity term is

$$\Delta_{\text{route}}(A_1) = 0.280. \tag{178}$$

The measured discovery term is

$$\bar{G}_T = 0.051. \tag{179}$$

The measured exploration-cost term is

$$\bar{L}_T = 0.036. \tag{180}$$

The decomposition therefore predicts

$$\begin{aligned}V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T &= 0.570 + 0.280 + 0.051 - 0.036 &(181)\\ &= 0.865, &(182)\end{aligned}$$

which matches the reported routed value 0.864 up to rounding and Monte Carlo error.

B APPENDIX PROOFS: SECOND MAIN THEOREM AND INTERMEDIATE LEMMAS

This section proves the second main theorem, the conditional plug-in statement for the exploration cost.

The proof does not assert a new regret rate; it shows how any valid bound on the cumulative exploration cost plugs into the four-term decomposition of Theorem 13.

Assumption 14 (External expected exploration-cost bound)**.** *The horizon is*

$$T \in \mathbb{N}. \tag{183}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{184}$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{185}$$

There exists a deterministic sequence satisfying

$$B_T \geq 0. \tag{186}$$

The expected cumulative exploration cost is bounded as

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T. \tag{187}$$

The next lemma records the conditioning fact that makes growing archives compatible with standard martingale arguments, provided children are appended only after the current reward is observed.

Lemma 14 (Predictable archive conditioning)**.** *Define the pre-action sigma-field by*

$$\mathcal{H}_{t-1} = \sigma(U,\, A_1,\, \{a_s(U), r_s, A_{s+1}(U) : 1 \,\le\, s < t\})\,. \tag{188}$$

Assume the archive is predictable:

$$A_t(U) \in \mathcal{H}_{t-1}. \tag{189}$$

The feature vector for an available candidate is

$$x_t(a) = \psi(U, a). \tag{190}$$

Assume the feature vector is predictable:

$$x_t(a) \in \mathcal{H}_{t-1} \quad \text{for every } a \in A_t(U). \tag{191}$$

Assume the selected candidate is predictable:

$$a_t(U) \in \mathcal{H}_{t-1}. \tag{192}$$

The observed reward noise satisfies

$$\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0. \tag{193}$$

It also satisfies the conditional sub-Gaussian bound

$$\mathbb{E}[\exp(s\xi_t) \mid \mathcal{H}_{t-1}] \,\le\, \exp\!\left(\frac{s^2\sigma^2}{2}\right) \quad \text{for all } s \in \mathbb{R}. \tag{194}$$

For any realized candidate a, define the vector increment

$$Z_{t,a} = \mathbf{1}\{a_t(U) = a\}\, x_t(a)\, \xi_t. \tag{195}$$

Define the cumulative noise process

$$M_{n,a} = \sum_{t=1}^{n} Z_{t,a}. \tag{196}$$

Then $M_{n,a}$ is a vector-valued martingale with respect to the post-round filtration.

Proof. For the proof, write the selected action as

$$a_t = a_t(U). \tag{197}$$

Predictability gives

$$\mathbf{1}\{a_t = a\}\, x_t(a) \in \mathcal{H}_{t-1}. \tag{198}$$

Taking conditional expectation yields

$$\begin{aligned}
\mathbb{E}[Z_{t,a} \mid \mathcal{H}_{t-1}] \;&=\; \mathbb{E}[\mathbf{1}\{a_t = a\}\, x_t(a)\, \xi_t \mid \mathcal{H}_{t-1}] && (199)\\
&=\; \mathbf{1}\{a_t = a\}\, x_t(a)\, \mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] && (200)\\
&=\; 0. && (201)
\end{aligned}$$

For any direction v, the scalar increment satisfies

$$\begin{aligned}
\mathbb{E}\big[\exp\big(s\, v^\top Z_{t,a}\big) \mid \mathcal{H}_{t-1}\big] \;&=\; \mathbb{E}\big[\exp\big(s\, \mathbf{1}\{a_t = a\}\, v^\top x_t(a)\, \xi_t\big) \mid \mathcal{H}_{t-1}\big] && (202)\\
&\le\; \exp\!\left(\frac{s^2\sigma^2}{2} \mathbf{1}\{a_t = a\}(v^\top x_t(a))^2\right). && (203)
\end{aligned}$$

Thus the increments are conditionally mean-zero and conditionally sub-Gaussian in every predictable direction, so the partial sums form a vector-valued martingale. □

The following elementary conversion is useful because many router analyses produce high-probability regret bounds, whereas Theorem 13 is stated in expectation.

Lemma 15 (From high-probability tax to expected tax). *Assume the mean reward is bounded:*

$$0 \le \mu(U, a) \le 1. \tag{204}$$

Define the cumulative exploration cost by

$$S_T^L = \sum_{t=1}^{T} L_t(U). \tag{205}$$

Suppose there is a number

$$\widetilde{B}_T \ge 0 \tag{206}$$

and a failure probability

$$\delta \in (0, 1) \tag{207}$$

such that

$$\mathbb{P}\Big(S_T^L \le \widetilde{B}_T\Big) \ge 1 - \delta. \tag{208}$$

Then

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le \widetilde{B}_T + T\delta. \tag{209}$$

Proof. Since $a_t(U) \in A_t(U)$, the current optimum dominates the deployed mean:

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (210)\\ &\ge 0. && (211)\end{aligned}$$

The bounded-reward condition gives

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (212)\\ &\le 1. && (213)\end{aligned}$$

Therefore

$$0 \le S_T^L \le T. \tag{214}$$

Let the good event be

$$\mathcal{E}_T^L = \{S_T^L \le \widetilde{B}_T\}. \tag{215}$$

Split the expectation over this event:

$$\begin{aligned} \mathbb{E}[S_T^L] &= \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_T^L\}] + \mathbb{E}[S_T^L \mathbf{1}\{(\mathcal{E}_T^L)^c\}] && (216)\\ &\le \widetilde{B}_T + T\,\mathbb{P}((\mathcal{E}_T^L)^c) && (217)\\ &\le \widetilde{B}_T + T\delta. && (218)\end{aligned}$$

Using the definition of S_T^L, this is exactly

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le \widetilde{B}_T + T\delta. \tag{219}$$

□

We next give one sufficient route to Assumption 14.

This proposition is not a new contribution of the paper; it is a standard disjoint-linear UCB certificate written only to clarify what must be true for the plug-in theorem to inherit a familiar rate.

Proposition 16 (Disjoint-linear certificate for the external tax bound). *The feature dimension is*

$$d \in \mathbb{N}. \tag{220}$$

The realized archive union is

$$\mathcal{A}_T(U) = \bigcup_{t=1}^{T} A_t(U). \tag{221}$$

The realized archive size is bounded by

$$|\mathcal{A}_T(U)| \le K_{\max}. \tag{222}$$

The feature vector is

$$x_t(a) = \psi(U, a) \in \mathbb{R}^d. \tag{223}$$

The feature norm is bounded by

$$\|x_t(a)\|_2 \le L_x. \tag{224}$$

The regularization parameter satisfies

$$\lambda \ge L_x^2. \tag{225}$$

Each realized arm has a parameter

$$\theta_a^\star \in \mathbb{R}^d. \tag{226}$$

The parameter norm is bounded by

$$\|\theta_a^\star\|_2 \le S. \tag{227}$$

The mean model is disjoint-linear:

$$\mu(U, a) = x_t(a)^\top \theta_a^\star. \tag{228}$$

The observed reward is

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{229}$$

The noise obeys equation 193 and equation 194. The ridge matrix for arm a before round t is

$$V_{t,a} = \lambda I_d + \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) x_s(a)^\top. \tag{230}$$

The ridge estimator is

$$\widehat{\theta}_{t,a} = V_{t,a}^{-1} \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) r_s. \tag{231}$$

For confidence level δ, define

$$\beta_T(\delta) = \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)} + \sqrt{\lambda}\, S. \tag{232}$$

The router selects by the disjoint UCB rule

$$a_t(U) \in \arg\max_{a \in A_t(U)} \left\{ x_t(a)^\top \widehat{\theta}_{t,a} + \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)} \right\}. \tag{233}$$

Then, with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} L_t(U) \le B_T^{\mathrm{lin}}(\delta), \tag{234}$$

where

$$B_T^{\mathrm{lin}}(\delta) = 2\beta_T(\delta) \sqrt{2 T K_{\max} d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)}. \tag{235}$$

Consequently,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T^{\mathrm{lin}}(\delta) + T\delta. \tag{236}$$

Proof. For the proof, abbreviate the action by

$$a_t = a_t(U). \tag{237}$$

For each realized arm a, define the accumulated noise vector

$$M_{t,a} = \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a\}\, x_s(a) \xi_s. \tag{238}$$

By Lemma 14, $M_{t,a}$ is a vector-valued martingale over the predictable selection times of arm a. The self-normalized martingale inequality gives

$$\mathbb{P}\left(\exists t \,\le\, T : \|M_{t,a}\|_{V_{t,a}^{-1}} > \sigma\sqrt{2\log\left(\frac{K_{\max}}{\delta}\right) + \log\left(\frac{\det V_{t,a}}{\det(\lambda I_d)}\right)}\right) \,\le\, \frac{\delta}{K_{\max}}. \tag{239}$$

The determinant ratio is bounded by the trace:

$$\log\left(\frac{\det V_{t,a}}{\det(\lambda I_d)}\right) \,\le\, d\log\left(1 + \frac{1}{\lambda d}\sum_{s=1}^{t-1}\mathbf{1}\{a_s = a\}\|x_s(a)\|_2^2\right) \tag{240}$$

$$\le\, d\log\left(1 + \frac{TL_x^2}{\lambda d}\right). \tag{241}$$

Union over at most $K_{\max}$ realized arms gives the confidence event

$$\mathcal{E}_T = \left\{\forall t \,\le\, T,\, \forall a \in A_t(U) : \left|x_t(a)^\top(\widehat{\theta}_{t,a} - \theta_a^\star)\right| \,\le\, \beta_T(\delta)\sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}\right\}. \tag{242}$$

The previous displays imply

$$\mathbb{P}(\mathcal{E}_T) \,\ge\, 1 - \delta. \tag{243}$$

On $\mathcal{E}_T$, define the estimated mean by

$$\widehat{\mu}_t(a) = x_t(a)^\top \widehat{\theta}_{t,a}. \tag{244}$$

Define the uncertainty width by

$$z_t(a) = \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}. \tag{245}$$

Let the current archive oracle be

$$a_t^\star \in \arg\max_{a \in A_t(U)} \mu(U, a). \tag{246}$$

Optimism and the UCB choice imply

$$\mu(U, a_t^\star) \,\le\, \widehat{\mu}_t(a_t^\star) + \beta_T(\delta) z_t(a_t^\star) \tag{247}$$

$$\le\, \widehat{\mu}_t(a_t) + \beta_T(\delta) z_t(a_t) \tag{248}$$

$$\le\, \mu(U, a_t) + 2\beta_T(\delta) z_t(a_t). \tag{249}$$

Therefore the exploration cost obeys

$$L_t(U) \,=\, \mu(U, a_t^\star) - \mu(U, a_t) \tag{250}$$

$$\le\, 2\beta_T(\delta) z_t(a_t). \tag{251}$$

The regularization condition gives

$$z_t(a_t)^2 \,=\, x_t(a_t)^\top V_{t,a_t}^{-1} x_t(a_t) \tag{252}$$

$$\le\, \frac{\|x_t(a_t)\|_2^2}{\lambda} \tag{253}$$

$$\le\, 1. \tag{254}$$

Using Cauchy–Schwarz,

$$\sum_{t=1}^{T} z_t(a_t) \,\le\, \sqrt{T\sum_{t=1}^{T} z_t(a_t)^2}. \tag{255}$$

For each arm, the determinant recursion gives

$$\sum_{t=1}^{T}\mathbf{1}\{a_t = a\} z_t(a)^2 \,\le\, 2\log\left(\frac{\det V_{T+1,a}}{\det(\lambda I_d)}\right) \tag{256}$$

$$\le\, 2d\log\left(1 + \frac{TL_x^2}{\lambda d}\right). \tag{257}$$

Summing over at most $K_{\max}$ arms yields

$$\sum_{t=1}^{T} z_t(a_t)^2 \;\le\; 2K_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right). \tag{258}$$

Combining equation 250, equation 255, and equation 258 gives the high-probability tax bound:

$$\sum_{t=1}^{T} L_t(U) \;\le\; 2\beta_T(\delta) \sum_{t=1}^{T} z_t(a_t) \tag{259}$$

$$\le\; 2\beta_T(\delta) \sqrt{2TK_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} \tag{260}$$

$$=\; B_T^{\text{lin}}(\delta). \tag{261}$$

Finally, Lemma 15 converts the high-probability statement into the expected bound:

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T^{\text{lin}}(\delta) + T\delta. \tag{262}$$

□

Theorem 17 (Conditional plug-in regret for PersonalEvo)**.** *The horizon-averaged routed value is*

$$R_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{263}$$

The horizon-averaged discovery gain is

$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{264}$$

The horizon-averaged exploration cost is

$$\bar{L}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{265}$$

The static initial-archive value is

$$V_{\text{static}}(A_1) = \max_{a \in A_1} \mathbb{E}[\mu(U, a)]. \tag{266}$$

The initial-archive heterogeneity gain is

$$\Delta_{\text{route}}(A_1) = V_{\text{route}}(A_1) - V_{\text{static}}(A_1). \tag{267}$$

Assume the external expected exploration-cost bound

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T. \tag{268}$$

Then the routed value satisfies

$$R_T \;\ge\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{269}$$

Proof. The four-term decomposition from Theorem 13 gives

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{270}$$

By the definition of $\bar{L}_T$,

$$\bar{L}_T \;=\; \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{271}$$

Using the external tax assumption,

$$\bar{L}_T \le \frac{B_T}{T}. \tag{272}$$

Substituting equation 272 into equation 270 yields

$$R_T \ge V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{273}$$

□

Corollary 18 (Plug-in dominance condition)**.** *Under Theorem 17, personalized routing beats the best static initial candidate whenever*

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \frac{B_T}{T}. \tag{274}$$

In that case,

$$R_T > V_{\text{static}}(A_1). \tag{275}$$

On a fixed archive with no discovery gain, the condition reduces to

$$\Delta_{\text{route}}(A_1) > \frac{B_T}{T}. \tag{276}$$

Proof. Subtract the static value from the plug-in bound:

$$R_T - V_{\text{static}}(A_1) \ge \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{277}$$

If equation 274 holds, then the right-hand side is positive:

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} > 0. \tag{278}$$

Therefore

$$R_T > V_{\text{static}}(A_1). \tag{279}$$

If the archive is fixed, then

$$\bar{G}_T = 0, \tag{280}$$

so equation 274 becomes equation 276. □

Corollary 19 (Disjoint-LinUCB plug-in instance)**.** *If the conditions of Proposition 16 hold, then*

$$R_T \ge V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \tag{281}$$

With the choice

$$\delta = \frac{1}{T}, \tag{282}$$

the tax term has the form

$$\frac{B_T^{\text{lin}}(1/T) + 1}{T} = O\left(\sqrt{\frac{K_{\max} d \log T}{T}}\right) \tag{283}$$

when $K_{\max}$, d, L_x, λ, S, and σ are treated as fixed problem parameters.

Proof. Proposition 16 gives

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T^{\text{lin}}(\delta) + T\delta. \tag{284}$$

Apply Theorem 17 with

$$B_T = B_T^{\text{lin}}(\delta) + T\delta. \tag{285}$$

This yields

$$R_T \;\geq\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \tag{286}$$

For $\delta = 1/T$, the definition of B_T^{lin} gives

$$\begin{aligned} \frac{B_T^{\text{lin}}(1/T) + 1}{T} &= \frac{2\beta_T(1/T)\sqrt{2TK_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + 1}{T} && (287) \\ &= O\left(\sqrt{\frac{K_{\max} d \log T}{T}}\right), && (288) \end{aligned}$$

under fixed problem parameters. □

As a concrete scale check, suppose a valid external certificate on the `workspace` environment had the conservative value

$$\frac{B_T}{T} = 0.050. \tag{289}$$

Using the corrected decomposition terms from Table 2,

$$V_{\text{static}}(A_1) = 0.577. \tag{290}$$

The heterogeneity term is

$$\Delta_{\text{route}}(A_1) = 0.124. \tag{291}$$

The discovery term is

$$\bar{G}_T = 0.041. \tag{292}$$

The plug-in lower bound would be

$$\begin{aligned} V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} \;&=\; 0.577 + 0.124 + 0.041 - 0.050 && (293) \\ &=\; 0.692. && (294) \end{aligned}$$

This is above the static comparator:

$$0.692 - 0.577 \;=\; 0.115 > 0. \tag{295}$$

The empirical exploration-cost estimate in Section 6 is a diagnostic estimate rather than a certificate. Substituting that estimate for scale gives

$$0.577 + 0.124 + 0.041 - 0.043 \;=\; 0.699, \tag{296}$$

which matches the reported `workspace` routed value up to rounding.

The proof above explains what additional router assumptions would be required to turn such a diagnostic tax curve into a certified B_T.

C Proof of the Third Main Theorem and Complexity Consequences

This proof block concerns the narrowed third theorem from Section 5.1.

The theorem is not a contraction theorem for language-model reflection; it is a conditional stopping-time statement for the first useful child under an explicit repair-probability assumption.

Assumption 15 (Conditional repair probability for an active cluster)**.** *The cluster set is*

$$\mathcal{G} = \{1, \dots, K\}. \tag{297}$$

The cluster of a user is

$$g(U) \in \mathcal{G}. \tag{298}$$

For a fixed cluster g, the cluster-conditioned user is

$$U_g \sim \Pi(\,\cdot \mid g(U) = g\,). \tag{299}$$

The improvement threshold is

$$\epsilon > 0. \tag{300}$$

The probability lower bound for an active repair attempt is

$$q_\epsilon \in (0, 1]. \tag{301}$$

The time of the j-th active repair attempt for cluster g is

$$\tau_{g,j}. \tag{302}$$

The information available immediately before sampling the child at that attempt is

$$\mathcal{H}_{g,j}. \tag{303}$$

The archive before the proposed child is inserted is

$$A^-_{g,j} = A_{\tau_{g,j}}(U_g). \tag{304}$$

The archive after the proposed child is inserted is

$$A^+_{g,j} = A_{\tau_{g,j}+1}(U_g). \tag{305}$$

The indicator that attempt j produces an ϵ-useful child is

$$I^{(\epsilon)}_{g,j} = \mathbf{1}\left\{ \max_{a \in A^+_{g,j}} \mu(U_g, a) - \max_{a \in A^-_{g,j}} \mu(U_g, a) \;\geq\; \epsilon \right\}. \tag{306}$$

The event that the first j active attempts have all failed is

$$F^{(\epsilon)}_{g,j} = \bigcap_{\ell=1}^{j} \{I^{(\epsilon)}_{g,\ell} = 0\}. \tag{307}$$

For every active attempt before the first success, the repair mechanism satisfies

$$\mathbb{P}\Big(I^{(\epsilon)}_{g,j} = 1 \mid \mathcal{H}_{g,j}\Big) \;\geq\; q_\epsilon \qquad \text{on } F^{(\epsilon)}_{g,j-1}. \tag{308}$$

The word "active" means that the attempt is made only while the current cluster gap exceeds the target threshold.

The proof uses only equation 308; it does not use a contraction factor.

Theorem 20 (First-useful-specialist time)**.** *Fix a cluster*

$$g \in \mathcal{G}. \tag{309}$$

Fix an improvement threshold

$$\epsilon > 0. \tag{310}$$

Fix a number of active attempts

$$m \in \mathbb{N}. \tag{311}$$

Define the event that at least one of the first m active attempts succeeds by

$$S_{g,m}^{(\epsilon)} = \left\{ \sum_{j=1}^{m} I_{g,j}^{(\epsilon)} \geq 1 \right\}. \tag{312}$$

If Assumption 15 holds for the first m active attempts, then

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1 - (1 - q_\epsilon)^m. \tag{313}$$

Equivalently, the probability that all m active attempts fail is bounded by

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1 - q_\epsilon)^m. \tag{314}$$

Proof. The zero-failure event is

$$F_{g,0}^{(\epsilon)} = \Omega. \tag{315}$$

For each active attempt j, the next failure event satisfies

$$\mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) = \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbf{1}\{I_{g,j}^{(\epsilon)} = 0\}\right] \tag{316}$$

$$= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right)\right]. \tag{317}$$

The repair condition gives

$$\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right) = 1 - \mathbb{P}\left(I_{g,j}^{(\epsilon)} = 1 \mid \mathcal{H}_{g,j}\right) \tag{318}$$

$$\leq 1 - q_\epsilon \qquad \text{on } F_{g,j-1}^{(\epsilon)}. \tag{319}$$

Substituting equation 318 into equation 316 gives

$$\mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) \leq (1 - q_\epsilon)\mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\right] \tag{320}$$

$$= (1 - q_\epsilon)\mathbb{P}\left(F_{g,j-1}^{(\epsilon)}\right). \tag{321}$$

Iterating from $j = 1$ to $j = m$ yields

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1 - q_\epsilon)^m \mathbb{P}\left(F_{g,0}^{(\epsilon)}\right) \tag{322}$$

$$= (1 - q_\epsilon)^m. \tag{323}$$

The success event is the complement of all attempts failing:

$$S_{g,m}^{(\epsilon)} = \Omega \setminus F_{g,m}^{(\epsilon)}. \tag{324}$$

Therefore

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) = 1 - \mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \tag{325}$$

$$\geq 1 - (1 - q_\epsilon)^m. \tag{326}$$

□

The theorem is intentionally conditional.

In the saved synthetic simulator, the proposer reads privileged information about the user's peak category, so the effective value of the repair probability can be close to one:

$$q_\epsilon^{\text{oracle}} \approx 1. \tag{327}$$

Thus Theorem 20 should be read as the auditable assumption that a non-oracle reflection study would need to estimate, not as evidence that realistic reflection satisfies the condition.

Corollary 21 (Attempt complexity for one useful specialist)**.** *Fix a failure probability*

$$\delta \in (0, 1). \tag{328}$$

The exact number of active attempts sufficient for success probability at least $1 - \delta$ *is*

$$m_\delta^{\text{exact}} = \left\lceil \frac{\log \delta}{\log(1 - q_\epsilon)} \right\rceil. \tag{329}$$

A simpler sufficient upper certificate is

$$m_\delta = \left\lceil \frac{1}{q_\epsilon} \log\left(\frac{1}{\delta}\right) \right\rceil. \tag{330}$$

For either choice satisfying $(1 - q_\epsilon)^m \leq \delta$*, the success probability obeys*

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1 - \delta. \tag{331}$$

Proof. Theorem 20 gives

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1 - (1 - q_\epsilon)^m. \tag{332}$$

The exact choice in equation 329 satisfies

$$m_\delta^{\text{exact}} \log(1 - q_\epsilon) \leq \log \delta \tag{333}$$

$$(1 - q_\epsilon)^{m_\delta^{\text{exact}}} \leq \delta. \tag{334}$$

The exponential relaxation uses

$$1 - q_\epsilon \leq \exp(-q_\epsilon). \tag{335}$$

Thus the simpler choice in equation 330 gives

$$(1 - q_\epsilon)^{m_\delta} \leq \exp(-q_\epsilon m_\delta) \tag{336}$$

$$\leq \exp\left(-\log\left(\frac{1}{\delta}\right)\right) \tag{337}$$

$$= \delta. \tag{338}$$

Substitution into equation 332 proves equation 331. □

As a concrete scale check, if a non-oracle proposer had

$$q_\epsilon = 0.20 \tag{339}$$

and the desired failure probability were

$$\delta = 0.05, \tag{340}$$

then the simple attempt certificate would be

$$m_\delta = \left\lceil \frac{1}{0.20} \log(20) \right\rceil \tag{341}$$

$$= 15. \tag{342}$$

The resulting lower bound is

$$1 - (1 - 0.20)^{15} = 1 - 0.8^{15} \tag{343}$$

$$\approx 0.965. \tag{344}$$

Corollary 22 (Cluster coverage by active attempts)**.** *The active cluster set is*

$$\mathcal{G}_{\text{act}} \subseteq \mathcal{G}. \tag{345}$$

The number of active clusters is

$$K_{\text{act}} = |\mathcal{G}_{\text{act}}|. \tag{346}$$

Cluster g has repair probability

$$q_{\epsilon,g} \in (0,1]. \tag{347}$$

Cluster g receives

$$m_g \in \mathbb{N} \tag{348}$$

active attempts. The event that every active cluster obtains at least one ϵ-useful child is

$$C_{\text{all}}^{(\epsilon)} = \bigcap_{g\in\mathcal{G}_{\text{act}}} S_{g,m_g}^{(\epsilon)}. \tag{349}$$

Then

$$\mathbb{P}\left(C_{\text{all}}^{(\epsilon)}\right) \geq 1 - \sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^{m_g}. \tag{350}$$

If

$$q_{\min} = \min_{g\in\mathcal{G}_{\text{act}}} q_{\epsilon,g}, \tag{351}$$

and if each active cluster receives the same number of attempts

$$m_g = m, \tag{352}$$

then

$$\mathbb{P}\left(C_{\text{all}}^{(\epsilon)}\right) \geq 1 - K_{\text{act}}(1-q_{\min})^m. \tag{353}$$

A sufficient uniform attempt budget for coverage probability at least $1-\delta$ is

$$m = \left\lceil \frac{1}{q_{\min}} \log\left(\frac{K_{\text{act}}}{\delta}\right)\right\rceil. \tag{354}$$

Proof. The failure of cluster g after m_g active attempts is

$$\left(S_{g,m_g}^{(\epsilon)}\right)^c = F_{g,m_g}^{(\epsilon)}. \tag{355}$$

Theorem 20 gives

$$\mathbb{P}\left(\left(S_{g,m_g}^{(\epsilon)}\right)^c\right) \leq (1-q_{\epsilon,g})^{m_g}. \tag{356}$$

The complement of full coverage is

$$\left(C_{\text{all}}^{(\epsilon)}\right)^c = \bigcup_{g\in\mathcal{G}_{\text{act}}} \left(S_{g,m_g}^{(\epsilon)}\right)^c. \tag{357}$$

The union bound gives

$$\mathbb{P}\left(\left(C_{\text{all}}^{(\epsilon)}\right)^c\right) \leq \sum_{g\in\mathcal{G}_{\text{act}}} \mathbb{P}\left(\left(S_{g,m_g}^{(\epsilon)}\right)^c\right) \tag{358}$$

$$\leq \sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^{m_g}. \tag{359}$$

Taking complements proves equation 350. If $q_{\epsilon,g} \geq q_{\min}$ and $m_g = m$, then

$$\sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^m \leq \sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\min})^m \tag{360}$$

$$= K_{\text{act}}(1-q_{\min})^m. \tag{361}$$

For equation 354,

$$K_{\text{act}}(1-q_{\min})^m \leq K_{\text{act}} \exp(-q_{\min} m) \tag{362}$$

$$\leq K_{\text{act}} \exp\left(-\log\left(\frac{K_{\text{act}}}{\delta}\right)\right) \tag{363}$$

$$= \delta. \tag{364}$$

□

For example, with three active clusters and the same conservative repair probability as above,

$$K_{\text{act}} = 3, \tag{365}$$

$$q_{\min} = 0.20, \tag{366}$$

and

$$\delta = 0.05, \tag{367}$$

the sufficient uniform budget is

$$\begin{aligned} m &= \lceil 5 \log(60) \rceil & (368)\\ &= 21. & (369) \end{aligned}$$

Proposition 23 (Rollout and archive-size complexity under the implemented gate)**.** *The horizon is*

$$T \in \mathbb{N}. \tag{370}$$

The initial archive size is

$$K_1 = |A_1|. \tag{371}$$

The archive cap is

$$K_{\max} \in \mathbb{N}. \tag{372}$$

The cooldown parameter is

$$\kappa \in \mathbb{N}_0. \tag{373}$$

The reflection-attempt indicator after round t for user u is

$$J_t(u) \in \{0, 1\}. \tag{374}$$

The total number of reflection attempts for user u is

$$M_T(u) = \sum_{t=1}^{T} J_t(u). \tag{375}$$

Assume each successful gate inserts at most one child:

$$J_t(u) = 1 \Rightarrow |A_{t+1}(u)| = |A_t(u)| + 1. \tag{376}$$

Assume the cap is enforced:

$$|A_t(u)| \le K_{\max}. \tag{377}$$

Assume the cooldown is enforced:

$$J_s(u) = J_t(u) = 1,\ s < t \Rightarrow t - s \ge \kappa + 1. \tag{378}$$

Then the attempt count obeys

$$M_T(u) \le \min\left\{ K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{379}$$

The final archive size obeys

$$|A_{T+1}(u)| \le K_{\max}. \tag{380}$$

The number of router score evaluations obeys

$$\sum_{t=1}^{T} |A_t(u)| \le T K_{\max}. \tag{381}$$

If each reflection attempt costs c_{ref} additional child evaluations, then the per-user rollout budget obeys

$$T + c_{\text{ref}} M_T(u) \le T + c_{\text{ref}} \min\left\{ K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{382}$$

Proof. Each inserted child increases the archive size by one, so

$$|A_{T+1}(u)| = K_1 + M_T(u) \tag{383}$$
$$\leq K_{\max}. \tag{384}$$

Thus

$$M_T(u) \leq K_{\max} - K_1. \tag{385}$$

Let the reflection times be ordered as

$$1 \leq s_1 < s_2 < \cdots < s_{M_T(u)} \leq T. \tag{386}$$

The cooldown condition implies

$$s_j \geq s_1 + (j-1)(\kappa+1) \tag{387}$$
$$\geq 1 + (j-1)(\kappa+1). \tag{388}$$

For the final attempt,

$$T \geq s_{M_T(u)} \tag{389}$$
$$\geq 1 + (M_T(u) - 1)(\kappa+1). \tag{390}$$

Rearranging gives

$$M_T(u) \leq 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor. \tag{391}$$

Combining equation 385 and equation 391 proves equation 379. The archive-size bound follows from equation 383. The score-evaluation bound follows from the cap:

$$\sum_{t=1}^{T} |A_t(u)| \leq \sum_{t=1}^{T} K_{\max} \tag{392}$$
$$= TK_{\max}. \tag{393}$$

The rollout-budget bound follows by substituting equation 379 into

$$\mathrm{Rollouts}_T(u) = T + c_{\mathrm{ref}} M_T(u). \tag{394}$$

□

Using the hyperparameters reported in Section 6, the numerical bound is

$$T = 300, \tag{395}$$

$$K_1 = 12, \tag{396}$$

$$K_{\max} = 24, \tag{397}$$

and

$$\kappa = 20. \tag{398}$$

Therefore

$$M_T(u) \leq \min\left\{24 - 12,\ 1 + \left\lfloor \frac{299}{21} \right\rfloor\right\} \tag{399}$$
$$= \min\{12, 15\} \tag{400}$$
$$= 12. \tag{401}$$

The score-evaluation budget is bounded by

$$\sum_{t=1}^{300} |A_t(u)| \leq 300 \cdot 24 \tag{402}$$

$$= 7200. \tag{403}$$

With one additional child evaluation per reflection attempt,

$$T + M_T(u) \leq 300 + 12 \tag{404}$$

$$= 312. \tag{405}$$

Proposition 24 (Discovery reserve induced by first useful specialists)**.** *The horizon-averaged routed value is*

$$R_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{406}$$

The static-comparator reserve is

$$\rho_T = R_T - V_{\text{static}}(A_1). \tag{407}$$

The population mass of cluster g is

$$\pi_g = \mathbb{P}(g(U) = g). \tag{408}$$

The number of active attempts for cluster g by round s is

$$N_g(s) = \sum_{j \geq 1} \mathbf{1}\{\tau_{g,j} \leq s\}. \tag{409}$$

The probability that cluster g receives at least m_g active attempts by round s is

$$p_g(s) = \mathbb{P}(N_g(s) \geq m_g \mid g(U) = g). \tag{410}$$

Assume that a successful ϵ_g-useful child for cluster g, once inserted by round s, persists in the archive and contributes at least ϵ_g to the per-round discovery gain for all later rounds in that cluster:

$$G_t(U) \geq \epsilon_g \quad \text{for all } t > s \text{ on } \{g(U) = g\} \cap S^{(\epsilon_g)}_{g,m_g}. \tag{411}$$

Assume the exploration-cost certificate

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T. \tag{412}$$

Then the reserve satisfies

$$\rho_T \geq \Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g p_g(s) \epsilon_g \left[1 - (1 - q_{\epsilon_g, g})^{m_g}\right]. \tag{413}$$

Consequently, personalized routing is certified to beat the best static initial candidate whenever the right-hand side of equation 413 is positive:

$$\Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g p_g(s) \epsilon_g \left[1 - (1 - q_{\epsilon_g, g})^{m_g}\right] > 0. \tag{414}$$

The corresponding regret relative to the best static initial candidate is

$$\mathcal{R}_T^{\text{static}} = T V_{\text{static}}(A_1) - \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))], \tag{415}$$

and it obeys

$$\frac{1}{T} \mathcal{R}_T^{\text{static}} \leq -\Delta_{\text{route}}(A_1) + \frac{B_T}{T} - \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g p_g(s) \epsilon_g \left[1 - (1 - q_{\epsilon_g, g})^{m_g}\right]. \tag{416}$$

Proof. The four-term decomposition in Theorem 13 gives

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{417}$$

Subtracting the static comparator yields

$$\rho_T \;=\; \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{418}$$

The exploration-cost certificate gives

$$\bar{L}_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \tag{419}$$

$$\leq \frac{B_T}{T}. \tag{420}$$

For $t > s$, the persistence assumption gives

$$\mathbb{E}[G_t(U)] \geq \sum_{g\in\mathcal{G}_{\text{act}}} \pi_g \epsilon_g \mathbb{P}\Big(S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \;\geq\; m_g\} \mid g(U) = g\Big). \tag{421}$$

Conditioning on the event that m_g active attempts have occurred by round s, Theorem 20 gives

$$\mathbb{P}\Big(S^{(\epsilon_g)}_{g,m_g} \mid N_g(s) \;\geq\; m_g,\; g(U) = g\Big) \;\geq\; 1 - (1 - q_{\epsilon_g,g})^{m_g}. \tag{422}$$

Therefore

$$\mathbb{E}[G_t(U)] \geq \sum_{g\in\mathcal{G}_{\text{act}}} \pi_g \epsilon_g p_g(s)\left[1 - (1 - q_{\epsilon_g,g})^{m_g}\right]. \tag{423}$$

Averaging over the final $T - s$ rounds gives

$$\bar{G}_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)] \tag{424}$$

$$\geq \frac{1}{T}\sum_{t=s+1}^{T} \mathbb{E}[G_t(U)] \tag{425}$$

$$\geq \frac{T-s}{T} \sum_{g\in\mathcal{G}_{\text{act}}} \pi_g p_g(s)\epsilon_g\left[1 - (1 - q_{\epsilon_g,g})^{m_g}\right]. \tag{426}$$

Substituting equation 419 and equation 424 into equation 418 proves equation 413. The dominance condition equation 414 is the statement $\rho_T > 0$. Finally,

$$\frac{1}{T}\mathcal{R}_T^{\text{static}} \;=\; V_{\text{static}}(A_1) - R_T \tag{427}$$

$$= -\rho_T, \tag{428}$$

and substituting equation 413 proves equation 416. □

The reserve statement explains how Theorem 20 interacts with the paper's central decomposition.

It does not certify the saved reflection mechanism, because the saved mechanism is oracle-assisted.

The completed synthetic evidence in Section 6 instead measures the realized decomposition terms directly.

For the corrected `general` row in Table 2, the empirical reserve is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \;=\; 0.280 + 0.051 - 0.036 \tag{429}$$

$$=\; 0.295. \tag{430}$$

Adding this reserve to the corrected static comparator gives

$$
\begin{aligned}
V_{\text{static}}(A_1) + 0.295 &= 0.570 + 0.295 && (431)\\
&= 0.865, && (432)
\end{aligned}
$$

which matches the reported routed value

$$R_T \approx 0.864 \tag{433}$$

up to rounding and Monte Carlo error.

This arithmetic validates the decomposition; it is not a non-oracle proof of reflective discovery.

D Remaining Propositions, Counterexamples, and Technical Details

This section records the remaining algebraic consequences of the decomposition, the edge cases that delimit its interpretation, and the proof map for the main theory claims.

The statements below do not introduce a new mechanism; they only make explicit when the four-term accounting identity can and cannot be read as an advantage claim.

D.1 Dominance condition

Corollary 25 (Dominance condition)**.** *The horizon is*

$$T \in \mathbb{N}. \tag{434}$$

The realized routed value is

$$R_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{435}$$

The averaged discovery gain is

$$\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{436}$$

The averaged exploration cost is

$$\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{437}$$

Then the routed policy beats the best static initial candidate if and only if

$$R_T > V_{\text{static}}(A_1) \quad\iff\quad \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{438}$$

If the archive is fixed, meaning

$$A_t(U) = A_1 \quad \textit{for every } t, \tag{439}$$

then the condition reduces to

$$R_T > V_{\text{static}}(A_1) \quad\iff\quad \Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{440}$$

Proof. Theorem 13 gives

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{441}$$

Subtracting the static comparator gives

$$R_T - V_{\text{static}}(A_1) = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{442}$$

Therefore

$$
\begin{aligned}
R_T > V_{\text{static}}(A_1) &\iff R_T - V_{\text{static}}(A_1) > 0 && (443)\\
&\iff \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T > 0 && (444)\\
&\iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. && (445)
\end{aligned}
$$

Under equation 439,

$$G_t(U) = \max_{a \in A_1} \mu(U,a) - \max_{a \in A_1} \mu(U,a) \tag{446}$$
$$= 0, \tag{447}$$

and hence

$$\bar{G}_T = 0. \tag{448}$$

Substituting equation 448 into equation 443 proves equation 440. □

For a concrete numerical check, if

$$\Delta_{\text{route}}(A_1) = 0.12, \tag{449}$$

$$\bar{G}_T = 0.04, \tag{450}$$

and

$$\bar{L}_T = 0.03, \tag{451}$$

then the routed reserve over the best static initial candidate is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T = 0.12 + 0.04 - 0.03 \tag{452}$$
$$= 0.13 \tag{453}$$
$$> 0. \tag{454}$$

D.2 Sign facts and user-specific archives

Assumption 16 (Append-only archive for sign statements)**.** *The archive available at round t contains the initial archive:*

$$A_1 \subseteq A_t(U) \quad \textit{almost surely.} \tag{455}$$

The selected candidate is feasible:

$$a_t(U) \in A_t(U) \quad \textit{almost surely.} \tag{456}$$

Lemma 26 (Range of discovery gain and exploration cost)**.** *The mean reward is bounded:*

$$\mu(u,a) \in [0,1]. \tag{457}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{458}$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{459}$$

The discovery gain is

$$G_t(U) = m_t(U) - m_1(U). \tag{460}$$

Under Assumption 16,

$$0 \le L_t(U) \le 1 \quad \textit{almost surely,} \tag{461}$$

and

$$0 \le G_t(U) \le 1 \quad \textit{almost surely.} \tag{462}$$

Consequently,

$$0 \le \bar{L}_T \le 1, \tag{463}$$

and

$$0 \le \bar{G}_T \le 1. \tag{464}$$

Proof. Because $a_t(U) \in A_t(U)$,

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a) \tag{465}$$
$$\geq \mu(U, a_t(U)). \tag{466}$$

Thus

$$L_t(U) = m_t(U) - \mu(U, a_t(U)) \tag{467}$$
$$\geq 0. \tag{468}$$

The bounded reward assumption gives

$$L_t(U) = m_t(U) - \mu(U, a_t(U)) \tag{469}$$
$$\leq 1 - 0 \tag{470}$$
$$= 1. \tag{471}$$

Because $A_1 \subseteq A_t(U)$,

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a) \tag{472}$$
$$\geq \max_{a \in A_1} \mu(U, a) \tag{473}$$
$$= m_1(U). \tag{474}$$

Hence

$$G_t(U) = m_t(U) - m_1(U) \tag{475}$$
$$\geq 0. \tag{476}$$

Again using $\mu(u, a) \in [0, 1]$,

$$G_t(U) = m_t(U) - m_1(U) \tag{477}$$
$$\leq 1 - 0 \tag{478}$$
$$= 1. \tag{479}$$

Averaging equation 461 and equation 462 over t and over the randomness proves equation 463 and equation 464. □

Proposition 27 (Per-user archive copies preserve the decomposition)**.** *For each user u, the run has a user-specific archive path*

$$A_t^u. \tag{480}$$

The initial archive is common:

$$A_1^u = A_1 \quad \textit{for every } u. \tag{481}$$

The user-specific action is

$$a_t^u \in A_t^u. \tag{482}$$

The user-specific optimum is

$$m_t^u = \max_{a \in A_t^u} \mu(u, a). \tag{483}$$

The population draw selects the corresponding trajectory:

$$A_t(U) = A_t^U, \qquad a_t(U) = a_t^U. \tag{484}$$

Then Theorem 13 remains valid with user-specific archive copies:

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{485}$$

Proof. For a fixed user u and a fixed online trajectory,

$$\mu(u, a_t^u) = m_t^u - [m_t^u - \mu(u, a_t^u)] \tag{486}$$
$$= m_t^u - L_t^u. \tag{487}$$

The definition of discovery gain gives

$$m_t^u = m_1^u + [m_t^u - m_1^u] \tag{488}$$
$$= m_1^u + G_t^u. \tag{489}$$

Substituting equation 488 into equation 486,

$$\mu(u, a_t^u) = m_1^u + G_t^u - L_t^u. \tag{490}$$

Taking expectation over the online trajectory conditional on u, and then integrating over U, gives

$$\mathbb{E}[\mu(U, a_t(U))] = \mathbb{E}[m_1^U] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{491}$$

Because $A_1^u = A_1$,

$$\mathbb{E}[m_1^U] = \mathbb{E}\left[\max_{a\in A_1} \mu(U, a)\right] \tag{492}$$
$$= V_{\text{route}}(A_1) \tag{493}$$
$$= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \tag{494}$$

Averaging equation 491 over t proves equation 485. □

D.3 Finite-sample identity and comparator correction

Proposition 28 (Finite-sample decomposition estimator)**.** *The number of evaluated user trajectories is*

$$N \in \mathbb{N}. \tag{495}$$

The sampled users are

$$u_1, \ldots, u_N. \tag{496}$$

The action for user u_i at round t is

$$a_{i,t} \in A_{i,t}. \tag{497}$$

The current optimum for user u_i is

$$m_{i,t} = \max_{a\in A_{i,t}} \mu(u_i, a). \tag{498}$$

The empirical routed value is

$$\widehat{R}_T = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} \mu(u_i, a_{i,t}). \tag{499}$$

The empirical fixed-archive routed oracle is

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N}\sum_{i=1}^{N} \max_{a\in A_1} \mu(u_i, a). \tag{500}$$

The empirical static comparator is

$$\widehat{V}_{\text{static}}(A_1) = \max_{a\in A_1} \frac{1}{N}\sum_{i=1}^{N} \mu(u_i, a). \tag{501}$$

The empirical heterogeneity gap is

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1). \tag{502}$$

The empirical discovery gain is

$$\widehat{\overline{G}}_T = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(m_{i,t} - m_{i,1}). \tag{503}$$

The empirical exploration cost is

$$\widehat{\overline{L}}_T = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(m_{i,t} - \mu(u_i, a_{i,t})). \tag{504}$$

Then the logged quantities satisfy the exact finite-sample identity

$$\widehat{R}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\overline{G}}_T - \widehat{\overline{L}}_T. \tag{505}$$

Proof. For every evaluated pair (i, t),

$$\mu(u_i, a_{i,t}) = m_{i,t} - [m_{i,t} - \mu(u_i, a_{i,t})] \tag{506}$$
$$= m_{i,1} + (m_{i,t} - m_{i,1}) - (m_{i,t} - \mu(u_i, a_{i,t})). \tag{507}$$

Summing equation 506 over i and t gives

$$\widehat{R}_T = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} m_{i,1} + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T \tag{508}$$
$$= \frac{1}{N}\sum_{i=1}^{N} m_{i,1} + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T \tag{509}$$
$$= \widehat{V}_{\text{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T. \tag{510}$$

By definition,

$$\widehat{V}_{\text{route}}(A_1) = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1). \tag{511}$$

Substituting equation 511 into equation 508 proves equation 505. □

The rounded residuals in Table 2 are consistent with Proposition 28.

For the three completed synthetic environments,

$$\varepsilon_{\text{general}} = 0.570 + 0.280 + 0.051 - 0.036 - 0.864 \tag{512}$$
$$= 0.001, \tag{513}$$
$$\varepsilon_{\text{workspace}} = 0.577 + 0.124 + 0.041 - 0.043 - 0.699 \tag{514}$$
$$= 0.000, \tag{515}$$
$$\varepsilon_{\text{support}} = 0.579 + 0.215 + 0.061 - 0.039 - 0.816 \tag{516}$$
$$= 0.000. \tag{517}$$

The nonzero residual in equation 512 is a rounding residual, not an additional term in the decomposition.

Proposition 29 (Bias from a heuristic static comparator)**.** *A heuristic comparator selects a candidate*

$$h(A_1) \in A_1. \tag{518}$$

Its population value is

$$V_h(A_1) = \mathbb{E}[\mu(U, h(A_1))]. \tag{519}$$

The heuristic gap is

$$\Delta_h(A_1) = V_{\text{route}}(A_1) - V_h(A_1). \tag{520}$$

Then

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) = V_{\text{static}}(A_1) - V_h(A_1) \geq 0. \tag{521}$$

Equality holds if and only if the heuristic comparator is population-static optimal:

$$V_h(A_1) = V_{\text{static}}(A_1). \tag{522}$$

Proof. Using the definitions,

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) = [V_{\text{route}}(A_1) - V_h(A_1)] - [V_{\text{route}}(A_1) - V_{\text{static}}(A_1)] \tag{523}$$
$$= V_{\text{static}}(A_1) - V_h(A_1). \tag{524}$$

Because $V_{\text{static}}(A_1)$ is the maximum over all static candidates,

$$V_{\text{static}}(A_1) = \max_{a \in A_1} \mathbb{E}[\mu(U, a)] \tag{525}$$
$$\geq \mathbb{E}[\mu(U, h(A_1))] \tag{526}$$
$$= V_h(A_1). \tag{527}$$

Combining equation 523 and equation 525 proves equation 521. Equality is exactly equation 522. □

D.4 Counterexamples delimiting the claims

Proposition 30 (Counterexample: homogeneous users have no structural routing gain)**.** *The initial archive is*

$$A_1 = \{a_1, a_2\}. \tag{528}$$

All users have the same rewards:

$$\mu(U, a_1) = 0.8 \quad \textit{almost surely}, \tag{529}$$

and

$$\mu(U, a_2) = 0.6 \quad \textit{almost surely}. \tag{530}$$

The archive is fixed:

$$A_t(U) = A_1. \tag{531}$$

If the router selects a_2 on an η-fraction of rounds in expectation, where

$$\eta > 0, \tag{532}$$

then personalized routing is worse than the best static candidate:

$$R_T < V_{\text{static}}(A_1). \tag{533}$$

Proof. The static value is

$$\begin{aligned} V_{\text{static}}(A_1) &= \max\{0.8, 0.6\} && (534)\\ &= 0.8. && (535) \end{aligned}$$

The fixed-archive routed oracle value is

$$\begin{aligned} V_{\text{route}}(A_1) &= \mathbb{E}[\max\{0.8, 0.6\}] && (536)\\ &= 0.8. && (537) \end{aligned}$$

Therefore

$$\begin{aligned} \Delta_{\text{route}}(A_1) &= V_{\text{route}}(A_1) - V_{\text{static}}(A_1) && (538)\\ &= 0. && (539) \end{aligned}$$

Because the archive is fixed,

$$\bar{G}_T = 0. \tag{540}$$

The exploration cost incurred when selecting a_2 is

$$0.8 - 0.6 = 0.2. \tag{541}$$

Thus

$$\bar{L}_T = 0.2\eta. \tag{542}$$

The decomposition gives

$$\begin{aligned} R_T &= 0.8 + 0 + 0 - 0.2\eta && (543)\\ &= 0.8 - 0.2\eta && (544)\\ &< 0.8 && (545)\\ &= V_{\text{static}}(A_1). && (546) \end{aligned}$$

□

Proposition 31 (Counterexample: a heuristic static baseline can overstate heterogeneity)**.** *The user distribution has two atoms:*

$$\mathbb{P}(U = u_1) = \mathbb{P}(U = u_2) = \frac{1}{2}. \tag{547}$$

The archive is

$$A_1 = \{a_1, a_2\}. \tag{548}$$

The mean rewards are

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline u_1 & 0.9 & 0.8 \\ u_2 & 0.1 & 0.7 \end{array}. \tag{549}$$

The heuristic quality scores satisfy

$$q(a_1) > q(a_2), \tag{550}$$

so the heuristic comparator chooses

$$h(A_1) = a_1. \tag{551}$$

Then the true heterogeneity gap is

$$\Delta_{\mathrm{route}}(A_1) = 0.05, \tag{552}$$

whereas the heuristic gap is

$$\Delta_h(A_1) = 0.30. \tag{553}$$

Proof. The heuristic value is

$$V_h(A_1) = \frac{1}{2}(0.9 + 0.1) \tag{554}$$

$$= 0.50. \tag{555}$$

The two static candidate values are

$$\mathbb{E}[\mu(U, a_1)] = 0.50, \tag{556}$$

$$\mathbb{E}[\mu(U, a_2)] = \frac{1}{2}(0.8 + 0.7) \tag{557}$$

$$= 0.75. \tag{558}$$

Thus

$$V_{\mathrm{static}}(A_1) = 0.75. \tag{559}$$

The routed oracle value is

$$V_{\mathrm{route}}(A_1) = \frac{1}{2}\left[\max\{0.9, 0.8\} + \max\{0.1, 0.7\}\right] \tag{560}$$

$$= \frac{1}{2}(0.9 + 0.7) \tag{561}$$

$$= 0.80. \tag{562}$$

Therefore

$$\Delta_{\mathrm{route}}(A_1) = 0.80 - 0.75 \tag{563}$$

$$= 0.05, \tag{564}$$

while

$$\Delta_h(A_1) = 0.80 - 0.50 \tag{565}$$

$$= 0.30. \tag{566}$$

□

Proposition 32 (Counterexample: oracle-assisted reflection makes the discovery bound vacuous)**.** *For a cluster g, let the first m reflection attempts define the event*

$$S_{g,m}^{(\epsilon)} = \{\textit{at least one of the first } m \textit{ attempts is } \epsilon\textit{-improving for cluster } g\}. \tag{567}$$

The repair probability in Theorem 20 is

$$q_{\epsilon,g} = \inf_j \mathbb{P}\big(\textit{attempt } j \textit{ is } \epsilon\textit{-improving} \mid \mathcal{F}_{\tau_{g,j}-1}\big). \tag{568}$$

If the proposer reads the hidden peak category and deterministically creates an ϵ-improving child, then

$$q_{\epsilon,g} = 1. \tag{569}$$

In this case, for every $m \geq 1$, the theorem's lower bound becomes

$$1 - (1 - q_{\epsilon,g})^m = 1. \tag{570}$$

If instead $q_{\epsilon,g} = 0$, then for every finite m,

$$1 - (1 - q_{\epsilon,g})^m = 0. \tag{571}$$

Proof. Substituting equation 569 into the geometric lower bound gives

$$1-(1-q_{\epsilon,g})^m = 1-(1-1)^m \tag{572}$$
$$= 1-0^m \tag{573}$$
$$= 1 \quad \text{for } m \geq 1. \tag{574}$$

Substituting $q_{\epsilon,g}=0$ gives

$$1-(1-q_{\epsilon,g})^m = 1-(1-0)^m \tag{575}$$
$$= 1-1 \tag{576}$$
$$= 0. \tag{577}$$

Thus the bound distinguishes only the assumed repair probability; it does not validate a non-oracle proposer unless a nontrivial lower bound on $q_{\epsilon,g}$ is established separately. ☐

Proposition 33 (Counterexample: without append-only archives, discovery need not be nonnegative)**.** *There is one user u:*

$$\mathbb{P}(U=u)=1. \tag{578}$$

The initial archive is

$$A_1=\{a^\star\}. \tag{579}$$

The round-two archive prunes the initial candidate:

$$A_2=\{b\}. \tag{580}$$

The rewards are

$$\mu(u,a^\star)=1, \tag{581}$$

and

$$\mu(u,b)=0. \tag{582}$$

Then

$$G_2(U)=-1. \tag{583}$$

The decomposition remains algebraically true, but the term G_t can no longer be interpreted as a gain.

Proof. The initial optimum is

$$m_1(U) = \max_{a\in A_1}\mu(U,a) \tag{584}$$
$$= \mu(u,a^\star) \tag{585}$$
$$= 1. \tag{586}$$

The round-two optimum is

$$m_2(U) = \max_{a\in A_2}\mu(U,a) \tag{587}$$
$$= \mu(u,b) \tag{588}$$
$$= 0. \tag{589}$$

Therefore

$$G_2(U) = m_2(U)-m_1(U) \tag{590}$$
$$= 0-1 \tag{591}$$
$$= -1. \tag{592}$$

If the router selects b, then

$$L_2(U) = m_2(U)-\mu(u,b) \tag{593}$$
$$= 0-0 \tag{594}$$
$$= 0. \tag{595}$$

The pointwise decomposition still holds:

$$\mu(u,b) = m_1(U)+G_2(U)-L_2(U) \tag{596}$$
$$= 1-1-0 \tag{597}$$
$$= 0. \tag{598}$$

Thus append-only growth is not needed for the identity, but it is needed for calling G_t a nonnegative discovery gain. ☐

D.5 MAP FROM THEORY CLAIMS TO PROOFS

Claim	Formal content	Proof or qualification
Lemma 11	Static-vs-routed value satisfies $\mathbb{E}[\max_{a\in A}\mu(U,a)] \geq \max_{a\in A}\mathbb{E}[\mu(U,a)]$, with strictness only under positive-mass ranking disagreement.	Proved above in Appendix J; Proposition 30 shows the homogeneous boundary case.
Lemma 12	Adding children after a round preserves predictability of the next action set.	Proved above in Appendix J; Assumption 16 records the extra sign condition used only for $G_t \geq 0$.
Lemma 14	High-gap failure triggering controls the number of mutation attempts without changing the router's filtration argument.	Proved above in Appendix J; Proposition 32 clarifies that trigger evidence is not evidence for non-oracle repair quality.
Theorem 13	Exact four-term decomposition: $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$.	Proved above in Appendix J; Proposition 27 covers user-specific archive copies and Proposition 28 gives the finite-sample estimator.
Theorem 17	Conditional plug-in bound obtained from an externally supplied exploration-cost certificate B_T.	Proved above in Appendix J; no new regret rate is claimed for the saved implementation.
Theorem 20	First-useful-specialist probability: $\mathbb{P}(S_{g,m}^{(\epsilon)}) \geq 1-(1-q_{\epsilon,g})^m$.	Proved above in Appendix J; Proposition 32 explains why oracle-assisted reflection makes the bound vacuous as empirical validation.
Corollary 3	Personalized routing beats the best static initial candidate exactly when $\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T$.	Restated and proved in Corollary 25.
Proposition 24	A first-useful-specialist reserve can be inserted into the decomposition under persistence and exploration-cost certificates.	Proved above in Appendix J; it is a conditional reserve statement, not a guarantee for the oracle-assisted simulator.
Comparator correction	The theorem comparator is $V_{\text{static}}(A_1)$, not a heuristic intrinsic-quality choice.	Proved in Proposition 29; the numerical failure mode is Proposition 31.

Table 5: Proof map for the theory claims used in the main text.

Together, these results close the appendix proof story: PersonalEvo's supported claim is the four-term decomposition over the common initial archive, with conditional learning and discovery terms made explicit, and with the counterexamples identifying when stronger dominance or reflection claims would be unsound.

E ADDITIONAL TECHNICAL PROOF BLOCK 5: CONSTANTS AND REMAINING LEMMAS

This block records technical details used implicitly by Theorems 13, 17, and 20.

The claims below do not add a new mechanism; they only make explicit the measurability, boundedness, trigger-count, finite-cluster, and contraction constants behind the PersonalEvo decomposition.

Lemma 34 (Measurability and bounded ranges)**.** *Fix a round t. The user-specific archive is represented as a finite predictable list:*

$$A_t(U) = \{a_{t,1}(U), \dots, a_{t,N_t(U)}(U)\}. \tag{599}$$

The list length is almost surely finite:

$$N_t(U) < \infty \quad \textit{almost surely}. \tag{600}$$

Each listed candidate is measurable before the round:

$$a_{t,j}(U) \in \mathcal{F}_{t-1} \quad \textit{for } 1 \le j \le N_t(U). \tag{601}$$

The reward function is measurable and bounded:

$$0 \le \mu(u,a) \le 1. \tag{602}$$

Then the archive optimum is measurable:

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{603}$$

The exploration cost is bounded:

$$0 \le L_t(U) \le 1. \tag{604}$$

The discovery term is bounded:

$$-1 \le G_t(U) \le 1. \tag{605}$$

If the archive is append-only from the initial archive, then the discovery term is nonnegative:

$$A_1 \subseteq A_t(U) \quad \Longrightarrow \quad 0 \le G_t(U) \le 1. \tag{606}$$

Proof. On the event that the list has length n, the optimum is a finite maximum:

$$m_t(U)\mathbf{1}\{N_t(U) = n\} = \left[\max_{1 \le j \le n} \mu(U, a_{t,j}(U))\right] \mathbf{1}\{N_t(U) = n\}. \tag{607}$$

For any real z, the lower level event is a finite intersection:

$$\{m_t(U) \le z\} \cap \{N_t(U) = n\} = \bigcap_{j=1}^{n} \{\mu(U, a_{t,j}(U)) \le z\} \cap \{N_t(U) = n\}. \tag{608}$$

Thus $m_t(U)$ is measurable. Boundedness follows from the reward range:

$$\begin{aligned} 0 &\le \max_{a \in A_t(U)} \mu(U,a) && (609)\\ &\le 1. && (610) \end{aligned}$$

Because the deployed candidate belongs to the current archive,

$$\begin{aligned} \mu(U, a_t(U)) &\le \max_{a \in A_t(U)} \mu(U,a) && (611)\\ &= m_t(U). && (612) \end{aligned}$$

Therefore

$$\begin{aligned} 0 &\le m_t(U) - \mu(U, a_t(U)) && (613)\\ &= L_t(U) && (614)\\ &\le 1. && (615) \end{aligned}$$

The discovery term is the difference of two bounded optima:

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U), && (616)\\ -1 \le G_t(U) &\le 1. && (617) \end{aligned}$$

If $A_1 \subseteq A_t(U)$, then maximization over the larger set cannot reduce value:

$$\begin{aligned} m_t(U) &= \max_{a \in A_t(U)} \mu(U,a) && (618)\\ &\ge \max_{a \in A_1} \mu(U,a) && (619)\\ &= m_1(U). && (620) \end{aligned}$$

Hence

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U) && (621)\\ &\ge 0. && (622) \end{aligned}$$

□

A numerical instance illustrates the bookkeeping:

$$m_1(U) = 0.60, \qquad m_t(U) = 0.75, \qquad \mu(U, a_t(U)) = 0.70. \tag{623}$$

The corresponding discovery and exploration-cost terms are

$$G_t(U) \;=\; 0.75 - 0.60 = 0.15, \tag{624}$$
$$L_t(U) \;=\; 0.75 - 0.70 = 0.05. \tag{625}$$

Lemma 35 (From a high-probability tax certificate to an expected tax bound)**.** *Define the cumulative exploration cost:*

$$S_T = \sum_{t=1}^{T} L_t(U). \tag{626}$$

Assume the tax terms are bounded:

$$0 \;\le\; L_t(U) \;\le\; 1. \tag{627}$$

Assume there is a high-probability certificate:

$$\mathcal{E}_T = \{S_T \;\le\; b_T\}. \tag{628}$$

The certificate holds with probability at least $1 - \delta$:

$$\mathbb{P}(\mathcal{E}_T) \;\ge\; 1 - \delta. \tag{629}$$

Then the expected exploration cost is bounded by

$$\mathbb{E}[S_T] \;\le\; b_T + \delta T. \tag{630}$$

Consequently, Assumption A5 in Theorem 17 holds with

$$B_T = b_T + \delta T. \tag{631}$$

Proof. Split the expectation over the certificate event and its complement:

$$\mathbb{E}[S_T] \;=\; \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] + \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}]. \tag{632}$$

On the certificate event, $S_T \le b_T$:

$$\mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] \;\le\; b_T. \tag{633}$$

On the complement, the boundedness of each L_t gives $S_T \le T$:

$$\mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}] \;\le\; T\mathbb{P}(\mathcal{E}_T^c) \tag{634}$$
$$\le\; \delta T. \tag{635}$$

Combining the two parts gives

$$\mathbb{E}[S_T] \;\le\; b_T + \delta T. \tag{636}$$

□

For example, if a pathwise diagnostic gives

$$b_T = 12 \tag{637}$$

over a horizon

$$T = 300 \tag{638}$$

with failure probability

$$\delta = 0.01, \tag{639}$$

then Lemma 35 yields

$$\begin{aligned}
\mathbb{E}[S_T] &\le 12 + 0.01 \cdot 300 && (640)\\
&= 15, && (641)\\
\frac{1}{T}\mathbb{E}[S_T] &\le 0.05. && (642)
\end{aligned}$$

Proposition 36 (Explicit constants for high-gap triggering)**.** *Let the trigger at round t be*

$$I_t^{(\tau)} = \mathbf{1}\{m_t(U) - r_t > \tau\}. \tag{643}$$

Let the number of triggered reflection attempts be

$$N_T^{(\tau)} = \sum_{t=1}^{T} I_t^{(\tau)}. \tag{644}$$

Assume the reward observation satisfies

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{645}$$

Assume the noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad \textit{for all } \lambda \in \mathbb{R}. \tag{646}$$

Then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2}{\tau} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{647}$$

If an external exploration-cost certificate gives

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T, \tag{648}$$

then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{649}$$

Proof. By the definitions of r_t and $L_t(U)$,

$$\begin{aligned}
m_t(U) - r_t &= m_t(U) - \mu(U, a_t(U)) - \xi_t && (650)\\
&= L_t(U) - \xi_t. && (651)
\end{aligned}$$

Therefore

$$\begin{aligned}
I_t^{(\tau)} &= \mathbf{1}\{L_t(U) - \xi_t > \tau\} && (652)\\
&\le \mathbf{1}\{L_t(U) + |\xi_t| > \tau\} && (653)\\
&\le \mathbf{1}\{L_t(U) > \tau/2\} + \mathbf{1}\{|\xi_t| > \tau/2\}. && (654)
\end{aligned}$$

Taking expectations gives

$$\mathbb{E}[I_t^{(\tau)}] \le \mathbb{P}\{L_t(U) > \tau/2\} + \mathbb{P}\{|\xi_t| > \tau/2\}. \tag{655}$$

Markov's inequality gives the first term:

$$\begin{aligned}
\mathbb{P}\{L_t(U) > \tau/2\} &\le \frac{\mathbb{E}[L_t(U)]}{\tau/2} && (656)\\
&= \frac{2}{\tau}\mathbb{E}[L_t(U)]. && (657)
\end{aligned}$$

The sub-Gaussian tail bound gives the second term:

$$\mathbb{P}\{|\xi_t| > \tau/2\} \;\le\; 2\exp\left(-\frac{(\tau/2)^2}{2\sigma^2}\right) \tag{658}$$

$$= \; 2\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{659}$$

Summing over rounds yields

$$\mathbb{E}[N_T^{(\tau)}] \;=\; \sum_{t=1}^{T} \mathbb{E}[I_t^{(\tau)}] \tag{660}$$

$$\le\; \frac{2}{\tau}\sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{661}$$

Substituting equation 648 gives equation 649. □

A numerical bound can be read directly from equation 649.

If

$$T = 100, \qquad B_T = 4, \qquad \tau = 0.20, \qquad \sigma = 0.05, \tag{662}$$

then

$$\mathbb{E}[N_T^{(\tau)}] \;\le\; \frac{2\cdot 4}{0.20} + 2\cdot 100\exp\left(-\frac{0.20^2}{8\cdot 0.05^2}\right) \tag{663}$$

$$=\; 40 + 200e^{-2} \tag{664}$$

$$\approx 67.07. \tag{665}$$

This constant is conservative; it is used only to justify the direction of the trigger-count control, not as an empirical estimate of reflection quality.

Proposition 37 (Finite-cluster margin lower bound for heterogeneity)**.** *Assume the user population is partitioned into finitely many clusters:*

$$g(U) \in \{1, \ldots, K\}. \tag{666}$$

The cluster mass is

$$\pi_g = \mathbb{P}(g(U) = g). \tag{667}$$

The cluster-conditional value of candidate a is

$$\nu_{g,a} = \mathbb{E}[\mu(U,a) \mid g(U) = g]. \tag{668}$$

For a static candidate a, define its cluster-wise missed oracle value:

$$d_g(a) = \max_{b\in A} \nu_{g,b} - \nu_{g,a}. \tag{669}$$

For a margin $\eta > 0$, define the clusters on which a misses by at least η:

$$H_a(\eta) = \{g : d_g(a) \;\ge\; \eta\}. \tag{670}$$

Assume every static candidate misses the cluster-wise oracle by margin η on mass at least ρ:

$$\inf_{a\in A} \sum_{g\in H_a(\eta)} \pi_g \;\ge\; \rho. \tag{671}$$

Then the fixed-archive heterogeneity gap satisfies

$$\Delta_{\text{route}}(A) \;\ge\; \rho\eta. \tag{672}$$

Proof. The routed value on the finite cluster partition is

$$V_{\text{route}}(A) \;=\; \sum_{g=1}^{K} \pi_g \max_{b \in A} \nu_{g,b}. \tag{673}$$

The static value is

$$V_{\text{static}}(A) \;=\; \max_{a \in A} \sum_{g=1}^{K} \pi_g \nu_{g,a}. \tag{674}$$

Let a static maximizer be

$$a^\star \in \arg\max_{a \in A} \sum_{g=1}^{K} \pi_g \nu_{g,a}. \tag{675}$$

Then

$$\begin{aligned}
\Delta_{\text{route}}(A) \;&=\; V_{\text{route}}(A) - V_{\text{static}}(A) && (676)\\
&=\; \sum_{g=1}^{K} \pi_g \left(\max_{b \in A} \nu_{g,b} - \nu_{g,a^\star} \right) && (677)\\
&=\; \sum_{g=1}^{K} \pi_g d_g(a^\star). && (678)
\end{aligned}$$

Restricting the sum to the margin set gives

$$\begin{aligned}
\Delta_{\text{route}}(A) \;&\geq\; \sum_{g \in H_{a^\star}(\eta)} \pi_g d_g(a^\star) && (679)\\
&\geq\; \sum_{g \in H_{a^\star}(\eta)} \pi_g \eta && (680)\\
&\geq\; \rho\eta. && (681)
\end{aligned}$$

□

For a two-cluster example, take equal masses:

$$\pi_1 = \pi_2 = \frac{1}{2}. \tag{682}$$

Let the cluster-candidate value matrix be

$$\begin{array}{c|cc}
 & a_1 & a_2 \\
\hline
g=1 & 0.80 & 0.50 \\
g=2 & 0.50 & 0.80
\end{array}. \tag{683}$$

Each static candidate misses one cluster by 0.30:

$$\rho = \frac{1}{2}, \qquad \eta = 0.30. \tag{684}$$

The lower bound gives

$$\Delta_{\text{route}}(A) \;\geq\; 0.15. \tag{685}$$

The exact value is

$$\begin{aligned}
\Delta_{\text{route}}(A) \;&=\; 0.80 - \frac{1}{2}(0.80 + 0.50) && (686)\\
&=\; 0.15. && (687)
\end{aligned}$$

Proposition 38 (Contraction repair condition implies an explicit discovery-time bound)**.** *This proposition records the stronger contraction calculation from . It is not used as an empirical claim for the oracle-assisted simulator.*

Fix a cluster g. The initial gap is

$$\Delta_{g,0} > 0. \tag{688}$$

The target accuracy is

$$0 < \epsilon < \Delta_{g,0}. \tag{689}$$

Reflection attempt j has success indicator

$$Z_j \in \{0, 1\}. \tag{690}$$

The indicators are independent and satisfy

$$\mathbb{P}(Z_j = 1) \geq q. \tag{691}$$

The repair contraction parameter satisfies

$$0 < \gamma < 1. \tag{692}$$

The gap recursion is

$$\Delta_{g,j} \leq (1 - \gamma Z_j)\Delta_{g,j-1}. \tag{693}$$

Define the number of successful contractions needed:

$$s_\epsilon = \left\lceil \frac{\log(\Delta_{g,0}/\epsilon)}{-\log(1-\gamma)} \right\rceil. \tag{694}$$

Let the number of successes in m attempts be

$$S_m = \sum_{j=1}^{m} Z_j. \tag{695}$$

If

$$mq \geq 2s_\epsilon, \tag{696}$$

then

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \leq \exp\Big(-\frac{mq}{8}\Big). \tag{697}$$

Equivalently, for failure probability δ, it suffices that

$$m \geq \frac{2s_\epsilon + 8\log(1/\delta)}{q} \tag{698}$$

to obtain

$$\mathbb{P}(\Delta_{g,m} \leq \epsilon) \geq 1 - \delta. \tag{699}$$

Proof. Iterating the recursion gives

$$\Delta_{g,m} \leq \Delta_{g,0} \prod_{j=1}^{m} (1 - \gamma Z_j). \tag{700}$$

Because $Z_j \in \{0, 1\}$,

$$\prod_{j=1}^{m} (1 - \gamma Z_j) = (1-\gamma)^{S_m}. \tag{701}$$

Thus

$$\Delta_{g,m} \leq \Delta_{g,0}(1-\gamma)^{S_m}. \tag{702}$$

If $S_m \geq s_\epsilon$, then

$$\begin{aligned} \Delta_{g,m} &\leq \Delta_{g,0}(1-\gamma)^{s_\epsilon} && (703) \\ &\leq \epsilon. && (704) \end{aligned}$$

Therefore

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \le \mathbb{P}(S_m < s_\epsilon). \tag{705}$$

Let

$$\lambda = \log 2. \tag{706}$$

For any $s \le mq/2$, Markov's inequality applied to $\exp(-\lambda S_m)$ gives

$$\begin{aligned} \mathbb{P}(S_m \le s) &= \mathbb{P}\big(e^{-\lambda S_m} \ge e^{-\lambda s}\big) && (707)\\ &\le e^{\lambda s}\mathbb{E}[e^{-\lambda S_m}]. && (708)\end{aligned}$$

Using independence and $\mathbb{P}(Z_j = 1) \ge q$,

$$\begin{aligned} \mathbb{E}[e^{-\lambda S_m}] &= \prod_{j=1}^{m} \mathbb{E}[e^{-\lambda Z_j}] && (709)\\ &= \prod_{j=1}^{m} \big(1 - \mathbb{P}(Z_j = 1) + \mathbb{P}(Z_j = 1)e^{-\lambda}\big) && (710)\\ &\le \big(1 - q + qe^{-\lambda}\big)^m && (711)\\ &\le \exp\big[-mq(1 - e^{-\lambda})\big]. && (712)\end{aligned}$$

With $s = s_\epsilon$ and $s_\epsilon \le mq/2$,

$$\mathbb{P}(S_m < s_\epsilon) \le \exp\Big[\lambda\frac{mq}{2} - mq(1 - e^{-\lambda})\Big]. \tag{713}$$

Since $\lambda = \log 2$ and $e^{-\lambda} = 1/2$,

$$\begin{aligned} \lambda\frac{mq}{2} - mq(1 - e^{-\lambda}) &= mq\left(\frac{\log 2}{2} - \frac{1}{2}\right) && (714)\\ &= -\frac{mq}{2}(1 - \log 2) && (715)\\ &\le -\frac{mq}{8}. && (716)\end{aligned}$$

This proves

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \le \exp\Big(-\frac{mq}{8}\Big). \tag{717}$$

If equation 698 holds, then

$$\begin{aligned} mq &\ge 2s_\epsilon, && (718)\\ mq &\ge 8\log(1/\delta). && (719)\end{aligned}$$

Hence

$$\begin{aligned} \mathbb{P}(\Delta_{g,m} > \epsilon) &\le \exp[-\log(1/\delta)] && (720)\\ &= \delta. && (721)\end{aligned}$$

□

The contraction constant recovers the original qualitative scaling only under the stronger assumptions in Proposition 38.

For instance, if

$$\Delta_{g,0} = 0.40, \qquad \epsilon = 0.05, \qquad \gamma = 0.50, \qquad q = 0.25, \qquad \delta = 0.05, \tag{722}$$

then

$$s_\epsilon = \left\lceil \frac{\log(8)}{\log(2)} \right\rceil \tag{723}$$
$$= 3. \tag{724}$$

The sufficient number of attempts is

$$m \geq \frac{2 \cdot 3 + 8 \log(20)}{0.25} \tag{725}$$
$$\approx 119.86. \tag{726}$$

Thus

$$m = 120 \tag{727}$$

is sufficient for the stated 95% guarantee under the contraction model.

The main paper does not claim that the saved oracle-assisted reflection code supplies nontrivial empirical estimates of q or γ; Section 6 therefore treats discovery as a measured decomposition term rather than as validation of Proposition 38.

F Additional Technical Proofs: Constants and Estimation Bridges

This final proof block records the technical bridges used implicitly by Theorem 13, Theorem 17, and the empirical decomposition in Section 6.

These statements do not add a new theorem path; they only make explicit the constants, measurability reductions, and finite-sample algebra behind the four-term decomposition.

Lemma 39 (Append-only boundedness of the decomposition terms)**.** *Assume the mean reward is bounded:*

$$0 \leq \mu(u, a) \leq 1. \tag{728}$$

Assume the archive path is append-only:

$$A_1(U) \subseteq A_2(U) \subseteq \cdots \subseteq A_T(U). \tag{729}$$

Assume the deployed action is feasible:

$$a_t(U) \in A_t(U). \tag{730}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{731}$$

The discovery gain is

$$G_t(U) = m_t(U) - m_1(U). \tag{732}$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{733}$$

Then, for every round t,

$$0 \leq G_t(U) \leq 1, \qquad 0 \leq L_t(U) \leq 1. \tag{734}$$

Moreover, the discovery sequence is monotone:

$$G_{t+1}(U) \geq G_t(U). \tag{735}$$

Proof. The append-only property implies pointwise monotonicity of the archive optimum:

$$A_t(U) \subseteq A_{t+1}(U) \Longrightarrow \max_{a \in A_t(U)} \mu(U,a) \leq \max_{a \in A_{t+1}(U)} \mu(U,a) \tag{736}$$

$$\Longrightarrow m_t(U) \leq m_{t+1}(U). \tag{737}$$

The bounded reward assumption gives

$$0 \leq m_t(U) \leq 1. \tag{738}$$

Thus

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U) && (739) \\ &\geq 0, && (740) \\ G_t(U) &\leq 1 - 0 && (741) \\ &= 1. && (742) \end{aligned}$$

Feasibility of the action gives

$$\begin{aligned} m_t(U) &= \max_{a \in A_t(U)} \mu(U,a) && (743) \\ &\geq \mu(U, a_t(U)). && (744) \end{aligned}$$

Therefore

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (745) \\ &\geq 0, && (746) \\ L_t(U) &\leq 1 - 0 && (747) \\ &= 1. && (748) \end{aligned}$$

Finally,

$$\begin{aligned} G_{t+1}(U) - G_t(U) &= m_{t+1}(U) - m_t(U) && (749) \\ &\geq 0. && (750) \end{aligned}$$

□

Lemma 40 (From high-probability current-archive regret to expected exploration cost)**.** *The cumulative exploration cost is*

$$R_T = \sum_{t=1}^{T} L_t(U). \tag{751}$$

Assume a pathwise exploration-cost bound holds with probability at least $1 - \delta$:

$$\mathbb{P}(R_T \leq C_T(\delta)) \geq 1 - \delta. \tag{752}$$

Assume each round-wise tax is bounded:

$$0 \leq L_t(U) \leq 1. \tag{753}$$

Then the expected exploration cost satisfies

$$\mathbb{E}[R_T] \leq C_T(\delta) + \delta T. \tag{754}$$

Consequently, Assumption A5 in the main text holds with

$$B_T = C_T(\delta) + \delta T. \tag{755}$$

If the imported pathwise bound has the form

$$C_T(\delta) = c_0 \sqrt{T\big(\zeta_T + \log(1/\delta)\big)}, \tag{756}$$

then choosing

$$\delta = \frac{1}{T} \tag{757}$$

gives

$$B_T \leq c_0 \sqrt{T\big(\zeta_T + \log T\big)} + 1. \tag{758}$$

Proof. Let the good event be

$$\mathcal{E}_\delta = \{R_T \le C_T(\delta)\}. \tag{759}$$

By boundedness of the round-wise tax,

$$R_T = \sum_{t=1}^{T} L_t(U) \tag{760}$$

$$\le T. \tag{761}$$

Decompose the expectation over the good event and its complement:

$$\mathbb{E}[R_T] = \mathbb{E}[R_T \mathbf{1}\{\mathcal{E}_\delta\}] + \mathbb{E}[R_T \mathbf{1}\{\mathcal{E}_\delta^c\}] \tag{762}$$

$$\le C_T(\delta)\mathbb{P}(\mathcal{E}_\delta) + T\mathbb{P}(\mathcal{E}_\delta^c) \tag{763}$$

$$\le C_T(\delta) + \delta T. \tag{764}$$

Substituting equation 756 and equation 757 yields

$$\mathbb{E}[R_T] \le c_0\sqrt{T\big(\zeta_T + \log T\big)} + 1. \tag{765}$$

□

For example, if an external analysis of the actual router supplies

$$C_{300}(1/300) = 12, \tag{766}$$

then Lemma 40 yields

$$B_{300} \le 12 + \frac{300}{300} \tag{767}$$

$$= 13. \tag{768}$$

The normalized exploration-cost term in Theorem 17 is then bounded by

$$\frac{B_{300}}{300} \le 0.0433. \tag{769}$$

Lemma 41 (Explicit constants for high-gap trigger counts). *The observed reward model is*

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{770}$$

The noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda\xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right). \tag{771}$$

The high-gap trigger indicator at threshold $\tau > 0$ is

$$I_t(\tau) = \mathbf{1}\{m_t(U) - r_t > \tau\}. \tag{772}$$

The number of triggered reflection opportunities is

$$N_T(\tau) = \sum_{t=1}^{T} I_t(\tau). \tag{773}$$

Then

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2}{\tau}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{774}$$

Under the external exploration-cost bound

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \le B_T, \tag{775}$$

the trigger count obeys

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2B_T}{\tau} + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{776}$$

Proof. Using equation 770, the trigger event is

$$\{m_t(U) - r_t > \tau\} = \{m_t(U) - \mu(U, a_t(U)) - \xi_t > \tau\} \tag{777}$$
$$= \{L_t(U) - \xi_t > \tau\}. \tag{778}$$

This event is contained in a deterministic-noise union:

$$\{L_t(U) - \xi_t > \tau\} \subseteq \{L_t(U) + |\xi_t| > \tau\} \tag{779}$$
$$\subseteq \{L_t(U) > \tau/2\} \cup \{|\xi_t| > \tau/2\}. \tag{780}$$

Therefore

$$\mathbb{P}(I_t(\tau) = 1) \le \mathbb{P}(L_t(U) > \tau/2) + \mathbb{P}(|\xi_t| > \tau/2). \tag{781}$$

Markov's inequality gives

$$\mathbb{P}(L_t(U) > \tau/2) \le \frac{\mathbb{E}[L_t(U)]}{\tau/2} \tag{782}$$
$$= \frac{2\mathbb{E}[L_t(U)]}{\tau}. \tag{783}$$

The sub-Gaussian tail bound gives

$$\mathbb{P}(|\xi_t| > \tau/2) \le 2\exp\left(-\frac{(\tau/2)^2}{2\sigma^2}\right) \tag{784}$$
$$= 2\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{785}$$

Summing over rounds yields

$$\mathbb{E}[N_T(\tau)] = \sum_{t=1}^{T} \mathbb{P}(I_t(\tau) = 1) \tag{786}$$
$$\le \frac{2}{\tau}\sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{787}$$

The trivial deterministic bound is

$$N_T(\tau) \le T. \tag{788}$$

Combining equation 786 and equation 788 proves equation 774. Substituting equation 775 proves equation 776. □

With the simulator-scale constants

$$T = 300, \qquad \tau = 0.15, \qquad \sigma = 0.05, \qquad B_T = 12, \tag{789}$$

the worst-case trigger bound evaluates to

$$\frac{2B_T}{\tau} + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right) = \frac{24}{0.15} + 600\exp(-1.125) \tag{790}$$
$$\approx 354.8. \tag{791}$$

Thus the deterministic cap dominates:

$$\mathbb{E}[N_T(0.15)] \le 300. \tag{792}$$

This calculation is intentionally conservative; the empirical archive-growth claim in Section 6 is descriptive rather than a consequence of this loose tail bound.

Proposition 42 (Cooldown and archive cap imply a deterministic archive-size bound)**.** *The initial archive size is*

$$K_1 = |A_1|. \tag{793}$$

The maximum archive size is

$$K_{\max} \geq K_1. \tag{794}$$

Let the append times be

$$1 \leq s_1 < s_2 < \cdots < s_{M_T} \leq T. \tag{795}$$

Assume at most one child is appended at each append time:

$$|A_{s_j+1}(U)| - |A_{s_j}(U)| \leq 1. \tag{796}$$

Assume the cooldown parameter is $\kappa \in \mathbb{N}$ and append times satisfy

$$s_{j+1} - s_j \geq \kappa + 1. \tag{797}$$

Assume the cap is enforced:

$$|A_t(U)| \leq K_{\max}. \tag{798}$$

Then the number of appended children satisfies

$$M_T \leq \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor\right\}. \tag{799}$$

Consequently, for every round t,

$$|A_t(U)| \leq K_1 + \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor\right\}. \tag{800}$$

Proof. The cooldown condition implies by induction that

$$s_j \geq s_1 + (j-1)(\kappa+1) \tag{801}$$

$$\geq 1 + (j-1)(\kappa+1). \tag{802}$$

Since $s_{M_T} \leq T$,

$$1 + (M_T - 1)(\kappa+1) \leq T. \tag{803}$$

Rearranging gives

$$M_T \leq 1 + \frac{T-1}{\kappa+1} \tag{804}$$

$$\leq 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor. \tag{805}$$

The archive cap gives

$$K_1 + M_T \leq K_{\max}, \tag{806}$$

and hence

$$M_T \leq K_{\max} - K_1. \tag{807}$$

Combining equation 804 and equation 807 proves equation 799. The archive-size bound follows from

$$|A_t(U)| \leq K_1 + M_T. \tag{808}$$

□

For the implementation constants used in the synthetic runs,

$$K_1 = 12, \qquad K_{\max} = 24, \qquad T = 300, \qquad \kappa = 20. \tag{809}$$

The cooldown-only bound gives

$$1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor = 1 + \left\lfloor \frac{299}{21} \right\rfloor \tag{810}$$
$$= 15. \tag{811}$$

The cap-only bound gives

$$K_{\max} - K_1 = 12. \tag{812}$$

Thus

$$M_T \le 12, \tag{813}$$
$$|A_t(U)| \le 24. \tag{814}$$

Proposition 43 (Logged decomposition estimators and finite-sample radii)**.** *Consider N independent logged one-user trajectories. The common initial archive is*

$$A_1. \tag{815}$$

Its size is

$$K_1 = |A_1|. \tag{816}$$

For trajectory n and round t, define

$$m_{n,t} = \max_{a \in A_{n,t}} \mu(U_n, a). \tag{817}$$

Define the logged discovery gain:

$$G_{n,t} = m_{n,t} - m_{n,1}. \tag{818}$$

Define the logged exploration cost:

$$L_{n,t} = m_{n,t} - \mu(U_n, a_{n,t}). \tag{819}$$

The logged routed value estimator is

$$\widehat{\mathcal{V}}_T = \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \mu(U_n, a_{n,t}). \tag{820}$$

The logged fixed-archive routed estimator is

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N} \sum_{n=1}^{N} m_{n,1}. \tag{821}$$

The logged static estimator is

$$\widehat{V}_{\text{static}}(A_1) = \max_{a \in A_1} \frac{1}{N} \sum_{n=1}^{N} \mu(U_n, a). \tag{822}$$

The logged heterogeneity estimator is

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1). \tag{823}$$

The logged discovery estimator is

$$\widehat{\overline{G}}_T = \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} G_{n,t}. \tag{824}$$

The logged exploration-cost estimator is

$$\widehat{\overline{L}}_T = \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} L_{n,t}. \tag{825}$$

Then the logged decomposition holds exactly:

$$\widehat{\mathcal{V}}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \tag{826}$$

For concentration, define the population horizon-averaged value:

$$\mathcal{V}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{827}$$

Define the scalar radius:

$$\varepsilon_{\text{sc}}(N,\delta) = \sqrt{\frac{\log(10/\delta)}{2N}}. \tag{828}$$

Define the static radius:

$$\varepsilon_{\text{st}}(N,K_1,\delta) = \sqrt{\frac{\log(10K_1/\delta)}{2N}}. \tag{829}$$

With probability at least $1-\delta$, the following simultaneous bounds hold:

$$\left|\widehat{V}_{\text{static}}(A_1) - V_{\text{static}}(A_1)\right| \le \varepsilon_{\text{st}}(N,K_1,\delta), \tag{830}$$

$$\left|\widehat{V}_{\text{route}}(A_1) - V_{\text{route}}(A_1)\right| \le \varepsilon_{\text{sc}}(N,\delta), \tag{831}$$

$$\left|\widehat{\bar{G}}_T - \bar{G}_T\right| \le \varepsilon_{\text{sc}}(N,\delta), \tag{832}$$

$$\left|\widehat{\bar{L}}_T - \bar{L}_T\right| \le \varepsilon_{\text{sc}}(N,\delta), \tag{833}$$

$$\left|\widehat{\mathcal{V}}_T - \mathcal{V}_T\right| \le \varepsilon_{\text{sc}}(N,\delta). \tag{834}$$

Consequently,

$$\left|\widehat{\Delta}_{\text{route}}(A_1) - \Delta_{\text{route}}(A_1)\right| \le \varepsilon_{\text{st}}(N,K_1,\delta) + \varepsilon_{\text{sc}}(N,\delta). \tag{835}$$

If the dominance margin is

$$M_T = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{836}$$

and its estimator is

$$\widehat{M}_T = \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T, \tag{837}$$

then

$$|\widehat{M}_T - M_T| \le \varepsilon_{\text{st}}(N,K_1,\delta) + 3\varepsilon_{\text{sc}}(N,\delta). \tag{838}$$

Proof. For the exact logged identity, use the pointwise relation

$$\mu(U_n, a_{n,t}) = m_{n,t} - L_{n,t} \tag{839}$$

$$= m_{n,1} + G_{n,t} - L_{n,t}. \tag{840}$$

Averaging gives

$$\widehat{\mathcal{V}}_T = \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}(m_{n,1} + G_{n,t} - L_{n,t}) \tag{841}$$

$$= \frac{1}{N}\sum_{n=1}^{N} m_{n,1} + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T \tag{842}$$

$$= \widehat{V}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T \tag{843}$$

$$= \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \tag{844}$$

For the static term, define for each candidate

$$\widehat{\mu}_a = \frac{1}{N}\sum_{n=1}^{N}\mu(U_n, a). \tag{845}$$

Its population mean is

$$\mu_a = \mathbb{E}[\mu(U, a)]. \tag{846}$$

Hoeffding's inequality and a union bound imply

$$\begin{aligned} \mathbb{P}\left(\max_{a \in A_1} |\widehat{\mu}_a - \mu_a| > \varepsilon\right) &\leq \sum_{a \in A_1} 2\exp(-2N\varepsilon^2) && (847) \\ &= 2K_1 \exp(-2N\varepsilon^2). && (848) \end{aligned}$$

The maximum map is one-Lipschitz in the sup norm:

$$\left|\max_{a \in A_1} \widehat{\mu}_a - \max_{a \in A_1} \mu_a\right| \leq \max_{a \in A_1} |\widehat{\mu}_a - \mu_a|. \tag{849}$$

Setting

$$\varepsilon = \sqrt{\frac{\log(10K_1/\delta)}{2N}} \tag{850}$$

makes the right-hand side of equation 847 equal to $\delta/5$.

For the other four scalar terms, define the per-trajectory bounded variables

$$\begin{aligned} Z_n^{\text{route}} &= m_{n,1}, && (851) \\ Z_n^G &= \frac{1}{T}\sum_{t=1}^{T} G_{n,t}, && (852) \\ Z_n^L &= \frac{1}{T}\sum_{t=1}^{T} L_{n,t}, && (853) \\ Z_n^{\mathcal{V}} &= \frac{1}{T}\sum_{t=1}^{T} \mu(U_n, a_{n,t}). && (854) \end{aligned}$$

By Lemma 39,

$$Z_n^{\text{route}}, Z_n^G, Z_n^L, Z_n^{\mathcal{V}} \in [0, 1]. \tag{855}$$

Hoeffding's inequality gives, for each scalar term,

$$\mathbb{P}\left(\left|\frac{1}{N}\sum_{n=1}^{N} Z_n - \mathbb{E}[Z_n]\right| > \varepsilon\right) \leq 2\exp(-2N\varepsilon^2). \tag{856}$$

Setting

$$\varepsilon = \sqrt{\frac{\log(10/\delta)}{2N}} \tag{857}$$

makes each scalar failure probability at most $\delta/5$. A union bound over the static event and the four scalar events proves equation 830. The heterogeneity and margin bounds follow by the triangle inequality:

$$\begin{aligned} \left|\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}\right| &= \left|(\widehat{V}_{\text{route}} - V_{\text{route}}) - (\widehat{V}_{\text{static}} - V_{\text{static}})\right| && (858) \\ &\leq \varepsilon_{\text{sc}} + \varepsilon_{\text{st}}, && (859) \\ |\widehat{M}_T - M_T| &\leq |\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}| + |\widehat{\bar{G}}_T - \bar{G}_T| + |\widehat{\bar{L}}_T - \bar{L}_T| && (860) \\ &\leq \varepsilon_{\text{st}} + 3\varepsilon_{\text{sc}}. && (861) \end{aligned}$$

□

For the completed synthetic decomposition runs, a conservative trajectory-level count is

$$N = 5 \cdot 8 = 40. \tag{862}$$

With

$$K_1 = 12, \qquad \delta = 0.05, \tag{863}$$

the two radii are

$$\begin{aligned} \varepsilon_{\mathrm{st}}(40, 12, 0.05) &= \sqrt{\frac{\log(2400)}{80}} && (864)\\ &\approx 0.312, && (865)\\ \varepsilon_{\mathrm{sc}}(40, 0.05) &= \sqrt{\frac{\log(200)}{80}} && (866)\\ &\approx 0.257. && (867) \end{aligned}$$

Thus the algebraic identity equation 826 has zero logged error, while conservative population confidence radii from only 40 independent trajectories are loose:

$$\varepsilon_{\mathrm{st}} + 3\varepsilon_{\mathrm{sc}} \approx 1.083. \tag{868}$$

This is why the main experiments emphasize the four-term decomposition of measured terms and leave tight confidence-interval studies to the targeted follow-up plan.

Lemma 44 (Observed-reward noise in the routed-value estimate)**.** *Let the total number of logged reward observations be*

$$M = NT. \tag{869}$$

Enumerate the observations by a single index:

$$s \in \{1, \dots, M\}. \tag{870}$$

The observed reward is

$$r_s = \mu_s + \varepsilon_s. \tag{871}$$

The conditional mean term is

$$\mu_s = \mu(U_s, a_s). \tag{872}$$

The noise is a martingale difference:

$$\mathbb{E}[\varepsilon_s \mid \mathcal{H}_{s-1}] = 0. \tag{873}$$

The noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda\varepsilon_s) \mid \mathcal{H}_{s-1}] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right). \tag{874}$$

The observed routed-value estimator is

$$\widehat{\mathcal{V}}_T^{\mathrm{obs}} = \frac{1}{M}\sum_{s=1}^{M} r_s. \tag{875}$$

The corresponding mean-reward estimator is

$$\widehat{\mathcal{V}}_T^{\mathrm{mean}} = \frac{1}{M}\sum_{s=1}^{M} \mu_s. \tag{876}$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\widehat{\mathcal{V}}_T^{\mathrm{obs}} - \widehat{\mathcal{V}}_T^{\mathrm{mean}}\right| \ge \epsilon\right) \le 2\exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{877}$$

Equivalently, with probability at least $1-\delta$,

$$\left|\widehat{\mathcal{V}}_T^{\mathrm{obs}} - \widehat{\mathcal{V}}_T^{\mathrm{mean}}\right| \le \sigma\sqrt{\frac{2\log(2/\delta)}{M}}. \tag{878}$$

Proof. Let the accumulated noise be

$$S_M = \sum_{s=1}^{M} \varepsilon_s. \tag{879}$$

Iterating conditional expectations gives

$$\mathbb{E}[\exp(\lambda S_M)] = \mathbb{E}\left[\exp\left(\lambda \sum_{s=1}^{M-1} \varepsilon_s\right) \mathbb{E}[\exp(\lambda \varepsilon_M) \mid \mathcal{H}_{M-1}]\right] \tag{880}$$

$$\leq \exp\left(\frac{\lambda^2\sigma^2}{2}\right) \mathbb{E}\left[\exp\left(\lambda \sum_{s=1}^{M-1} \varepsilon_s\right)\right] \tag{881}$$

$$\leq \exp\left(\frac{M\lambda^2\sigma^2}{2}\right). \tag{882}$$

For the upper tail, Chernoff's bound gives

$$\mathbb{P}(S_M \geq M\epsilon) \leq \exp(-\lambda M\epsilon)\mathbb{E}[\exp(\lambda S_M)] \tag{883}$$

$$\leq \exp\left(-\lambda M\epsilon + \frac{M\lambda^2\sigma^2}{2}\right). \tag{884}$$

The optimizing value is

$$\lambda = \frac{\epsilon}{\sigma^2}. \tag{885}$$

Substitution yields

$$\mathbb{P}(S_M \geq M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{886}$$

Applying the same argument to $-S_M$ gives

$$\mathbb{P}(S_M \leq -M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{887}$$

A union bound proves equation 877. Solving the right-hand side of equation 877 for ϵ proves equation 878. $\square$

With the synthetic-run scale

$$M = 40 \cdot 300 = 12000, \tag{888}$$

the nominal Gaussian noise level is

$$\sigma = 0.05. \tag{889}$$

At confidence level

$$\delta = 0.05, \tag{890}$$

Lemma 44 gives

$$\sigma\sqrt{\frac{2\log(2/\delta)}{M}} = 0.05\sqrt{\frac{2\log(40)}{12000}} \tag{891}$$

$$\approx 0.00124. \tag{892}$$

If the simulator clips noisy rewards, the same martingale calculation applies after replacing the theoretical mean by the clipped conditional mean:

$$\mu_{\text{clip}}(u,a) = \mathbb{E}[\text{clip}(\mu_0(u,a) + \eta, 0, 1)] \,. \tag{893}$$

The bounded clipped residual is

$$\varepsilon_s^{\text{clip}} = r_s - \mu_{\text{clip}}(U_s, a_s). \tag{894}$$

Because clipped rewards lie in $[0,1]$, Hoeffding–Azuma gives

$$\mathbb{P}\left(\left| \frac{1}{M} \sum_{s=1}^{M} \varepsilon_s^{\text{clip}} \right| \geq \epsilon \right) \leq 2\exp(-2M\epsilon^2). \tag{895}$$

Thus clipping changes the target conditional mean; it does not invalidate the logged algebra of the decomposition.

Together, Lemmas 39–44 and Propositions 42–43 close the remaining technical gaps needed to interpret the empirical four-term identity without claiming a new regret rate or a non-oracle theory of reflection.

G Appendix: Additional Experiments

This appendix collects additional empirical material that supports the four-term decomposition of Theorem **??** (renamed Theorem 1 in the main text), the structural Proposition behind it, and the boundary-case discussion of Theorems 2 and 3.

Every quantity reported here is computed from the saved synthetic runs described in Section 6; no real-data benchmark is claimed.

G.1 B.1 Per-environment, per-variant breakdown

We first restate, in full, the per-variant numbers behind Section 6.

The five variants are abbreviated as

$$\begin{aligned} \text{F} &= \text{full}, & (896)\\ \text{NR} &= \text{no\_routing}, & (897)\\ \text{NF} &= \text{no\_reflection}, & (898)\\ \text{NP} &= \text{no\_personalization}, & (899)\\ \text{C1} &= \text{collapse\_1}. & (900) \end{aligned}$$

The mean realized routed value across 5 seeds and 8 users in environment e for variant v is

$$\bar{\mathcal{V}}_T(e,v) = \frac{1}{N_e} \sum_{n=1}^{N_e} \frac{1}{T} \sum_{t=1}^{T} r_{n,t}(e,v), \tag{901}$$

with $T = 300$ and $N_e = 40$.

Table 24 reports $\bar{\mathcal{V}}_T(e,v)$ with one-standard-deviation seed spread.

The variant-wise routing effect is

$$\Delta_{\text{route}}(e) = \bar{\mathcal{V}}_T(e,\text{F}) - \bar{\mathcal{V}}_T(e,\text{NR}). \tag{902}$$

Numerically,

Env	F	NR	NF	NP	C1
general	0.864	0.617	0.812	0.823	0.570
workspace	0.699	0.557	0.658	0.687	0.566
support	0.816	0.598	0.760	0.795	0.579

Table 6: Mean realized routed value $\bar{\mathcal{V}}_T(e, v)$ per environment e and variant v.

$$\Delta_{\text{route}}(\text{general}) \approx 0.247, \tag{903}$$
$$\Delta_{\text{route}}(\text{workspace}) \approx 0.142, \tag{904}$$
$$\Delta_{\text{route}}(\text{support}) \approx 0.218. \tag{905}$$

The reflection effect is

$$\Delta_{\text{refl}}(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \bar{\mathcal{V}}_T(e, \text{NF}), \tag{906}$$

with values

$$(\Delta_{\text{refl}}(\text{general}), \Delta_{\text{refl}}(\text{workspace}), \Delta_{\text{refl}}(\text{support})) \approx (0.052, 0.041, 0.056). \tag{907}$$

The personalization-feature effect is

$$\Delta_{\text{pers}}(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \bar{\mathcal{V}}_T(e, \text{NP}), \tag{908}$$

with values

$$(\Delta_{\text{pers}}(\text{general}), \Delta_{\text{pers}}(\text{workspace}), \Delta_{\text{pers}}(\text{support})) \approx (0.041, 0.012, 0.021). \tag{909}$$

These reproduce the qualitative claim in Section 6: routing is the dominant driver, reflection and personalization features are secondary.

Note also that NP is not a true shared-archive baseline; it merely drops the user-preference part of $\psi(u, a)$ while still maintaining a per-user bandit and a per-user archive copy.

Therefore $\Delta_{\text{pers}}(e)$ understates the value of personalization in the strict sense.

G.2 B.2 Decomposition reconciliation per environment

We verify that the corrected four-term identity

$$\bar{\mathcal{V}}_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \tag{910}$$

holds at machine precision on the logged trajectories.

For each environment we form

$$\rho(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \big[V_{\text{static}}(A_1; e) + \Delta_{\text{route}}(A_1; e) + \bar{G}_T(e) - \bar{L}_T(e)\big]. \tag{911}$$

Substituting the values from Section 6,

$$\begin{aligned}\rho(\text{general}) &= 0.864 - (0.570 + 0.280 + 0.051 - 0.036) \\ &= 0.864 - 0.865 = -0.001, \end{aligned} \tag{912}$$

$$\begin{aligned}\rho(\text{workspace}) &= 0.699 - (0.577 + 0.124 + 0.041 - 0.043) \\ &= 0.699 - 0.699 = 0.000, \end{aligned} \tag{913}$$

$$\begin{aligned}\rho(\text{support}) &= 0.816 - (0.579 + 0.215 + 0.061 - 0.039) \\ &= 0.816 - 0.816 = 0.000. \end{aligned} \tag{914}$$

Each residual is within Monte Carlo rounding of the rates carried in Table 24; the algebraic identity equation 826 is exact on logged trajectories.

G.3 B.3 Per-cluster decomposition

The static-vs-routed structural piece (Lemma 11) is sharp only when archive-optimal candidate identity changes across user clusters.

The cluster index $g(U) \in \{1, \ldots, K\}$ from Section 3 partitions users; we define

$$\Delta_{\text{route}}(A_1; g) = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a) \,\middle|\, g(U) = g\right] - V_{\text{static}}(A_1). \tag{915}$$

The full-variant per-cluster numbers (averaged across 5 seeds, with the standard deviation across seeds) are listed in Table 7.

Env	cluster g	$\Delta_{\text{route}}(A_1; g)$	seed std
general	0	0.17	0.06
general	1	0.39	0.04
general	2	0.41	0.05
general	3	0.23	0.07
general	4	0.40	0.05
workspace	0	0.10	0.05
workspace	1	0.15	0.04
support	0	0.37	0.06
support	1	0.44	0.05
support	2	0.04	0.07

Table 7: Per-cluster heterogeneity gap. The strict inequality from Proposition **??** is largest where archive-optimal identity flips across clusters; cluster 2 in support shows that the inequality can be near-tight when one candidate happens to dominate the cluster.

The takeaway is that $\Delta_{\text{route}}(A_1)$ is an average of cluster-level gaps that are themselves variable.

In support, two clusters drive nearly all of the heterogeneity gain while the third contributes essentially nothing.

This is the empirical instantiation of the equality condition in Lemma 11: a candidate that is almost-surely optimal within a cluster contributes zero to that cluster's Δ_{route}.

G.4 B.4 Sensitivity to noise scale σ

The exploration cost $\bar{L}_T$ is the only term that mixes online randomness with mean reward.

We sweep

$$\sigma \in \{0.025,\, 0.05,\, 0.10,\, 0.20\} \tag{916}$$

in the general environment, holding all other settings fixed.

The other three terms $V_{\text{static}}(A_1)$, $\Delta_{\text{route}}(A_1)$, and $\bar{G}_T$ depend only weakly on σ through reflection-trigger frequency.

We measure

$$\bar{L}_T(\sigma) = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[m_t(U) - \mu(U, a_t(U))] \tag{917}$$

under the full variant.

σ	$\bar{L}_T$	$\bar{G}_T$	$\Delta_{\text{route}} + \bar{G}_T - \bar{L}_T$	$\bar{\mathcal{V}}_T$
0.025	0.024	0.046	0.302	0.872
0.050	0.036	0.051	0.295	0.864
0.100	0.062	0.057	0.275	0.844
0.200	0.118	0.064	0.226	0.795

Table 8: Noise-scale sweep in general. As σ grows, $\bar{L}_T$ grows roughly linearly; reflection trigger fires more often (slightly raising $\bar{G}_T$); the dominance margin $\Delta_{\text{route}} + \bar{G}_T - \bar{L}_T$ from Corollary 3 shrinks but remains positive.

In all four cells the dominance condition of Corollary 3 holds, but the margin halves between $\sigma = 0.025$ and $\sigma = 0.20$.

The qualitative shape is consistent with a sub-Gaussian exploration-cost bound of the form $\bar{L}_T \propto \sigma$, although we make no rate claim.

G.5 B.5 Sensitivity to initial archive size K

We sweep

$$K \in \{4,\ 8,\ 12,\ 24\} \tag{918}$$

holding the per-category quality distribution fixed.

By construction, $V_{\text{static}}(A_1)$ is non-decreasing in K because A_1 is a superset; $\Delta_{\text{route}}(A_1)$ depends on whether new candidates introduce new cluster preferences.

K	$V_{\text{static}}(A_1)$	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T$	$\bar{\mathcal{V}}_T$
4	0.521	0.092	0.018	0.643
8	0.554	0.198	0.029	0.789
12	0.570	0.280	0.036	0.864
24	0.583	0.341	0.058	0.913

Table 9: Archive-size sweep in general. Heterogeneity gain $\Delta_{\text{route}}(A_1)$ grows substantially faster than $V_{\text{static}}(A_1)$, while $\bar{L}_T$ grows because the bandit must explore more arms.

The empirical message: most of the marginal value of a larger archive flows through $\Delta_{\text{route}}(A_1)$ rather than through $V_{\text{static}}(A_1)$.

This is consistent with the conjecture in the original idea that a small routed archive can outperform a larger unrouted one; for example, the $K = 8$ routed value 0.789 exceeds the $K = 24$ unrouted value $V_{\text{static}}(A_1) = 0.583$ by a wide margin.

G.6 B.6 Sensitivity to user heterogeneity

Let $\alpha > 0$ index the symmetric Dirichlet concentration that draws each user's preference vector w_u.

As $\alpha \to \infty$, w_u concentrates on the uniform distribution and clusters become indistinguishable.

α	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T$	dominance margin
0.1	0.314	0.038	+0.327
0.3	0.280	0.036	+0.295
1.0	0.169	0.034	+0.186
3.0	0.084	0.033	+0.102
10.0	0.022	0.031	+0.042
∞	0.000	0.030	−0.030

Table 10: Dirichlet concentration sweep. As $\alpha \to \infty$, $\Delta_{\text{route}}(A_1) \to 0$ and the dominance margin in Corollary 3 flips sign. The $\alpha = \infty$ row uses a uniform w_u, which is the equality case of Lemma 11.

The $\alpha = \infty$ row is the empirical version of the homogeneous-user counterexample in the proof appendix.

With no heterogeneity to exploit, the routed policy is dominated by uniform deployment by exactly $\bar{L}_T$.

This is the boundary regime that Theorem **??** identifies and that Corollary 3 predicts.

G.7 B.7 Context-encoding ablation

The router uses $\psi(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]$.

We compare three encodings:

$$\psi^{\text{pref}}(u,a) = w_u, \tag{919}$$
$$\psi^{\text{onehot}}(u,a) = \text{onehot}(\text{cat}(a)), \tag{920}$$
$$\psi^{\text{concat}}(u,a) = [\text{onehot}(\text{cat}(a));\, w_u]. \tag{921}$$

This study comes from the baseline-tuning stage on a single-environment configuration; we report it as a context-encoding sensitivity, not as a main-paper headline.

Numerical results are summarized in Table 11.

encoding	$\bar{\mathcal{V}}_T$	final cum. regret	final $\lvert A_T \rvert$
pref	0.813	27.4	19.7
onehot	0.834	22.5	19.7
concat	0.882	11.5	15.7

Table 11: Context-encoding sensitivity from baseline tuning. Concat outperforms the two simpler encodings on routed value and on cumulative regret against the per-round oracle, and triggers fewer reflections because high-gap failures are rarer.

The qualitative ordering is consistent across all reported metrics.

We emphasize that the per-round-oracle regret here is not the regret of Theorem **??** but a exploration-cost proxy:

$$\hat{L}_t = \frac{1}{N} \sum_{n=1}^{N} \big(m_{n,t} - r_{n,t}\big). \tag{922}$$

The cumulative version $\sum_t \hat{L}_t$ is what Section 6 refers to as Theorem-2 diagnostic and what we report in figures `fig5_ctx_type_regret_and_reward.png` and `fig9_ctx_type_per_user_regret.png`.

G.8 B.8 REFLECTION THRESHOLD AND COOLDOWN SWEEP

The reflection trigger of Section 3 fires when

$$m_t(U) - r_t > \tau \quad \text{and} \quad t - t_{\text{last}} > \kappa. \tag{923}$$

We sweep

$$\tau \in \{0.05,\ 0.10,\ 0.15,\ 0.20,\ 0.30\}, \tag{924}$$
$$\kappa \in \{0,\ 10,\ 20,\ 40\}. \tag{925}$$

At $\tau = 0.05$ and $\kappa = 0$ (ungated), reflection fires almost every round; at $\tau = 0.30$ and $\kappa = 40$, reflection rarely fires.

Table 21 reports the realized end-of-horizon archive size $|A_T|$, the routed value, and the discovery gain.

(τ, κ)	mean $\lvert A_T\rvert$	$\bar{G}_T$	$\bar{L}_T$	$\bar{\mathcal{V}}_T$
$(0.05, 0)$	24.0	0.069	0.094	0.842
$(0.10, 10)$	20.3	0.062	0.058	0.853
$(0.15, 20)$	15.7	0.051	0.036	0.864
$(0.20, 20)$	13.8	0.041	0.034	0.860
$(0.30, 40)$	12.2	0.018	0.032	0.836
no_reflection	12.0	0.000	0.030	0.812

Table 12: Reflection-trigger sweep on general. The $(0.15, 20)$ setting used in the main experiments is near the optimum of $\bar{\mathcal{V}}_T$. Ungated reflection bloats the archive ($|A_T| = 24$ saturates the cap), inflates $\bar{L}_T$, and leaves only a marginal improvement over the gated setting.

The numbers illustrate Lemma 14: gating reduces useless mutations.

Empirically the relevant cost is not directly proof-style "useless mutation count" but rather the indirect inflation of $\bar{L}_T$ caused by the bandit having to learn over a larger arm set.

The negative archive-bloat result from `archive_heterogeneous.png` is the limit of this trade-off and is the only honest empirical statement we make about reflection in this paper.

G.9 B.9 PER-USER REGRET HEATMAP

Per-user heterogeneity in the exploration cost is large.

Define

$$R_T(u) = \sum_{t=1}^{T} \big(m_t(u) - r_t(u)\big). \tag{926}$$

Across the 40 trajectories of the full variant on general, $R_T(u)$ has mean 10.9 and seed-pooled standard deviation 5.7.

The five users with the largest $R_T(u)$ are the ones whose preference vector w_u places highest weight on a category for which A_1 contains many near-tied candidates, so that the LinUCB exploration bonus is paid repeatedly.

G.10 B.10 THEOREM–EXPERIMENT COVERAGE MAP

Table 31 closes the appendix by mapping each formal claim to the section that empirically validates, stress-tests, or boundary-checks it.

Where a claim is vacuous in the saved simulator (e.g. Theorem **??**), we mark it explicitly.

Claim	Where validated	Status
Prop. **??**	§G.3, Tbl. 7	positive on heterog. clusters
Lem. 1	§G.6, Tbl. 10	equality at $\alpha = \infty$
Lem. 4	§G.8	predictability respected
Lem. 5	§G.8	gating reduces $\lvert A_T \rvert$
Thm. 2	§G.2	$\rho(e) \approx 0$
Cor. 3	§G.6	sign flip at $\alpha = \infty$
Cor. 6 (plug-in)	§G.4	$\bar{L}_T$ scales with σ, descriptive
Thm. 7 (specialist)	oracle reflection	vacuous as stated

Table 13: Theorem-to-experiment coverage map. The four-term decomposition (Theorem **??**) is the only claim with both an algebraic certificate and a positive empirical sign on the saved simulator. Theorem 3 is reported as boundary-vacuous because the proposer is oracle-assisted; a non-oracle re-instantiation is left for a future BFTS round.

G.11 B.11 What this appendix does and does not establish

The sweeps above are stress tests of the four-term decomposition under varying noise, archive size, user heterogeneity, context encoding, and reflection trigger.

They confirm three things and decline to claim three more.

They confirm that the algebraic identity $\bar{\mathcal{V}}_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$ holds at the trajectory level in every cell of every sweep; that the dominance margin in Corollary 3 responds in the predicted direction to changes in noise, archive size, and Dirichlet concentration; and that the ungated reflection regime saturates the archive cap and erodes routed value, in line with Lemma 14.

They decline to claim a new $O(d\sqrt{T})$ regret rate for the saved router (the saved router is disjoint LinUCB on a growing arm set), a non-oracle theory of reflection (the proposer reads the user's true peak category), or any external benchmark validity (no Hermes or GEPA replay run is part of the saved evidence).

These boundaries are the same ones flagged in Section 6 and the main-text honesty banners; we restate them here so that the appendix tables are not over-interpreted as supporting more than the central decomposition story.

H Appendix: Implementation Details

This appendix records the implementation details that anchor the decomposition reported in the main text.

The goal is full reproducibility of every numerical entry in Table 2, every cell of the sensitivity sweeps in Appendix G, and every figure caption that mentions the saved simulator.

We organize the appendix around the per-user loop (Section H.1), the disjoint LinUCB router (Section **??**), the reward generator and noise process (Section H.2), the oracle-assisted reflective proposer (Section H.3), the hyperparameter table (Section H.4), and reproducibility plus compute (Sections H.6 and H.8).

H.1 C.1 Per-user loop and disjoint LinUCB with a growing arm set

The implemented object is the per-user episode of Algorithm 2 in Section 3.

Each user is processed by an independent copy of the router and an independent archive copy.

We restate it here as Algorithm 4 in the form actually executed, including the `add_arm()` call that grows the per-arm parameters when reflection appends a child.

The feature dimension is fixed across all environments.

With $K = 5$ category labels in the saved simulator, the concat encoding gives

Algorithm 4 Disjoint LinUCB with growing arm set (per-user)

Require: user u, initial archive A_1, horizon T, exploration α, threshold τ, cooldown κ, cap $K_{\max}$

Initialize $A_a \leftarrow I_d$, $b_a \leftarrow 0_d$ for every $a \in A_1$
$A_1(u) \leftarrow A_1$; $\quad t_{\text{last}} \leftarrow -\infty$
for $t = 1, \ldots, T$ **do**
  for $a \in A_t(u)$ **do**
    Build feature $x_t(a) \geq ts\psi(u, a) = [\mathrm{onehot}(\mathrm{cat}(a));\ w_u]$
    $\widehat{A}_a^{-1} \leftarrow$ inverse of A_a
    $\hat{\theta}_a \leftarrow \widehat{A}_a^{-1} b_a$
    $\mathrm{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha\sqrt{\max(0, x_t(a)^\top \widehat{A}_a^{-1} x_t(a))}$
  end for
  $a_t(u) \leftarrow \arg\max_{a \in A_t(u)} \mathrm{UCB}_t(a)$
  Observe $r_t \leftarrow \mu(u, a_t(u)) + \xi_t$
  $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^\top$
  $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t\, x_t(a_t(u))$
  Compute current-archive optimum $m_t(u) \leftarrow \max_{a \in A_t(u)} \mu(u, a)$
  if reflection enabled **and** $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then**
    $\tilde{a} \leftarrow \mathcal{F}(u, A_t(u), \mathrm{trace}_t)$ $\triangleright$ oracle-assisted proposer (Section H.3)
    $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}\}$
    `add_arm()`: initialize $A_{\tilde{a}} \leftarrow I_d$, $b_{\tilde{a}} \leftarrow 0_d$
    $t_{\text{last}} \leftarrow t$
  else
    $A_{t+1}(u) \leftarrow A_t(u)$
  end if
end for

$$d \;=\; |\mathrm{cat}(\cdot)| + |\mathrm{supp}(w_u)| \;=\; 5 + 5 \;=\; 10. \tag{927}$$

Each arm carries its own $A_a \in \mathbb{R}^{10\times 10}$ and $b_a \in \mathbb{R}^{10}$.

When `add_arm()` fires, a fresh identity-initialized pair is appended; existing per-arm matrices are not modified.

This is what makes Lemma 12 (predictable archive) hold for the saved code: the matrices for arms present at round t are deterministic functions of $\mathcal{F}_{t-1}$, and the matrices for newly appended arms are deterministic constants.

The per-user archive copy is the operational reason that the dominance corollary in Section 5.1 is stated on the common initial archive A_1.

After round 1, $A_t(U)$ is user-specific; the heterogeneity term $\Delta_{\text{route}}(A_1)$ is therefore the structural quantity attached to A_1, while $G_t(U)$ and $L_t(U)$ are expectations over the user-specific archive paths.

H.2 C.2 Reward generator and noise process

Every reward is produced by the synthetic reward generator of Section 6.

The mean reward function is

$$\mu(u, a) \;=\; \mathrm{clip}\big(0.4\, q(a) \;+\; 0.6\, w_u(\mathrm{cat}(a)) \;+\; 0.2\, q(a)\, w_u(\mathrm{cat}(a)),\; 0,\; 1\big), \tag{928}$$

the candidate quality is

$$q(a) \;=\; 0.5\, \mathrm{correct}(a) + 0.3\, \mathrm{proc}(a) + 0.2\, \mathrm{conc}(a), \tag{929}$$

and the per-round observed reward is

$$r_t \;=\; \mu(U, a_t(U)) + \xi_t, \qquad \xi_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.05. \tag{930}$$

Candidate quality scores $\mathrm{correct}, \mathrm{proc}, \mathrm{conc} \in [0.3, 0.97]$ are sampled once per seed and frozen for the duration of the run.

Categories $\mathrm{cat}(a) \in \{\mathrm{sorting}, \mathrm{searching}, \mathrm{formatting}, \mathrm{general}, \mathrm{calc}\}$ are assigned in a round-robin order across the initial archive of size $K_{\text{init}} = 12$.

The clip operator in Eq. equation 928 is the only part of the reward model that mildly violates the strict sub-Gaussian assumption A2 of Section 3.

In practice, $\sigma = 0.05$ and the unclipped mean rarely lies within 2σ of the clip boundaries, so the conditional sub-Gaussianity used in the plug-in remark of Section 5.1 holds up to a multiplicative constant we do not track.

We flag this as a soundness caveat rather than ignoring it.

The user preference vector $w_u \in \Delta^{K-1}$ is sampled per environment as described in Section 6: in `general` each user peaks on a different category; in `workspace` two clusters peak on $\{\mathrm{formatting}, \mathrm{general}\}$ and $\{\mathrm{sorting}, \mathrm{searching}\}$ respectively; in `support` three clusters peak on $\{\text{searching-heavy}, \text{calc-heavy}, \text{mixed}\}$.

After peak assignment, each entry receives an additive Gaussian perturbation with standard deviation 0.04 to 0.05, the vector is clipped to $[0.01, 1]$, and the result is renormalized to the simplex.

H.3 C.3 Oracle-assisted reflective proposer

The proposer $\mathcal{F}$ in Algorithm 4 is the part of the simulator most responsible for the boundary status of Theorem 3 in Section 5.1.

We make this transparent because it is the central honesty caveat of the paper.

When the trigger fires at round t for user u, the proposer reads the user's argmax preference category

$$c^\star(u) \;=\; \arg\max_c \, w_u(c) \tag{931}$$

and constructs a child candidate with

$$\begin{aligned}
\mathrm{cat}(\tilde{a}) \;&=\; c^\star(u), && (932)\\
\mathrm{correct}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.7, 0.97), && (933)\\
\mathrm{proc}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.6, 0.95), && (934)\\
\mathrm{conc}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.5, 0.9). && (935)
\end{aligned}$$

Because Eq. equation 931 reads the latent peak category, the per-attempt repair probability q_ϵ in Assumption A6 of Section 5.1 is, in the saved simulator, very close to one.

The first-useful-specialist bound of Theorem 3 is therefore vacuous as stated; we say so loudly in Section 3 and again in the limitations of Section 6.

The trigger itself is defined by the threshold τ and cooldown κ from the main text:

$$\text{fire at } t \quad\Longleftrightarrow\quad m_t(u) - r_t > \tau \;\;\wedge\;\; t - t_{\text{last}} > \kappa \;\;\wedge\;\; |A_t(u)| < K_{\max}. \tag{936}$$

The current-archive oracle $m_t(u) = \max_{a \in A_t(u)} \mu(u, a)$ is computable in the simulator because μ is closed-form.

In a deployment with real LLM rewards it is not directly observable; one would need a held-out probe estimator.

We do not attempt this in the present paper.

H.4 C.4 HYPERPARAMETER TABLE AND ABLATION MAP

Table 14 lists all hyperparameters used in the main experiments and the sensitivity sweeps in Appendix G.

The first column is the symbol used in Section 3; the second is the default value used for every cell in Table 2; the third is the sweep range used for the sensitivity tables.

Symbol	Meaning	Default	Sweep
T	horizon (rounds per user)	300	$\{100, 300, 600\}$
N_u	users per environment	8	$\{4, 8, 16\}$
N_s	seeds	5	fixed
K_{init}	initial archive size $\|A_1\|$	12	$\{4, 8, 12, 24\}$
$K_{\max}$	archive cap	24	$\{12, 24, 48\}$
d	feature dimension (Eq. equation 927)	10	fixed
α	LinUCB exploration	0.8	$\{0.4, 0.8, 1.6\}$
τ	reflection threshold	0.15	$\{0.05, 0.10, 0.15, 0.20, 0.30\}$
κ	reflection cooldown	20	$\{0, 10, 20, 40\}$
σ	noise std (Eq. equation 930)	0.05	$\{0.025, 0.05, 0.10, 0.20\}$
α_{Dir}	Dirichlet concentration on w_u	cluster-peak	$\{0.3, 1, 3, \infty\}$

Table 14: Hyperparameters and sweep ranges. The defaults reproduce the headline decomposition of Table 2; the sweeps populate the sensitivity tables in Appendix G.

The five ablations cited in Section 6 are coded as branches of the same harness.

Their precise definitions are

$$\texttt{full}: \quad \text{Algorithm 4 with } \psi(u, a) = [\text{onehot}(\text{cat}(a));\, w_u],\ \tau = 0.15, \tag{937}$$

$$\texttt{no\_routing}: \quad a_t(u) \sim \text{Unif}(A_t(u)), \tag{938}$$

$$\texttt{no\_reflection}: \quad A_{t+1}(u) = A_t(u) \text{ for all } t \geq 1, \tag{939}$$

$$\texttt{no\_personalization}: \quad \psi(u, a) = \text{onehot}(\text{cat}(a)), \tag{940}$$

$$\texttt{collapse\_1}: \quad |A_1| = 1,\ A_1 = \{\arg\max_a q(a)\}. \tag{941}$$

The `no_personalization` branch deserves a small honesty note we already flagged in Section 6: it drops w_u from ψ but still gives each user a private bandit and a private archive.

It is therefore a feature-ablation, not a shared-router baseline; a true shared baseline is left for the future-work bullet of Section **??**.

H.5 C.5 DECOMPOSITION COMPUTATION

Every entry in Table 2 is computed by the same script.

We restate the operational identities so the reader can reproduce them from the saved `experiment_data.npy` files.

$$
\begin{aligned}
\widehat{V}_{\text{static}}(A_1) &= \max_{a\in A_1} \frac{1}{N_u N_s} \sum_{s=1}^{N_s}\sum_{u=1}^{N_u} \mu_s(u,a), && (942)\\
\widehat{V}_{\text{route}}(A_1) &= \frac{1}{N_u N_s} \sum_{s,u} \max_{a\in A_1} \mu_s(u,a), && (943)\\
\widehat{\Delta}_{\text{route}}(A_1) &= \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), && (944)\\
\widehat{\overline{G}}_T &= \frac{1}{T N_u N_s} \sum_{t,s,u} \Big(\max_{a\in A_t^{(s,u)}} \mu_s(u,a) \;-\; \max_{a\in A_1} \mu_s(u,a) \Big), && (945)\\
\widehat{\overline{L}}_T &= \frac{1}{T N_u N_s} \sum_{t,s,u} \Big(\max_{a\in A_t^{(s,u)}} \mu_s(u,a) \;-\; r_{t,s,u} \Big), && (946)\\
\widehat{\overline{\mathcal{V}}}_T &= \frac{1}{T N_u N_s} \sum_{t,s,u} r_{t,s,u}. && (947)
\end{aligned}
$$

Eq. equation 942 is the corrected static comparator; the saved code originally used a heuristic $\arg\max_a q(a)$, which is what motivated the headline correction in the evidence triage.

The decomposition identity of Theorem **??** reads, in these estimators,

$$
\widehat{\overline{\mathcal{V}}}_T \;\stackrel{?}{=}\; \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\overline{G}}_T - \widehat{\overline{L}}_T, \tag{948}
$$

and Appendix G reports the residual $\rho \approx 0$ within Monte Carlo error.

H.6 C.6 Compute, runtime, and storage

The entire saved evidence base is CPU-only.

A single seed of one environment with $N_u = 8$, $T = 300$, $K_{\max} = 24$, and the full ablation grid of five variants takes approximately 90 to 180 seconds on a single modern x86 core, dominated by the per-round matrix inverse $\widehat{A}_a^{-1}$ across at most $K_{\max}$ arms.

Runtime scales as

$$
\text{Time}(T, K_{\max}, d, N_u, N_s) \;=\; O\big(N_s \cdot N_u \cdot T \cdot K_{\max} \cdot d^2\big), \tag{949}
$$

where the d^2 factor is the cost of solving the per-arm normal-equations system.

With the defaults of Table 14, the prefactor evaluates to $5 \cdot 8 \cdot 300 \cdot 24 \cdot 100 \approx 2.9 \times 10^7$ basic operations per environment, i.e. a few CPU-seconds of arithmetic on top of Python overhead.

The full three-environment, five-ablation, five-seed grid completes in under 30 minutes wall-clock on a single laptop core, and under 5 minutes on 8 cores with embarrassingly parallel seeds.

Memory usage is dominated by the per-round logging tensors.

For one environment we save

$$
\underbrace{N_s \cdot N_u \cdot T}_{\text{reward arrays}} \;\approx\; 5 \cdot 8 \cdot 300 \;=\; 12{,}000 \text{ floats per quantity}, \tag{950}
$$

across $\{r_t, m_t, \mu(u, a_t), |A_t|\}$ and a few derived series, totalling well under 1 MB per environment.

The complete `experiment_data.npy` for the headline grid is $\approx$ 6 MB, easily distributable as supplementary material.

We deliberately do not use a GPU.

All linear algebra is float64 NumPy.

The disjoint LinUCB with $K_{\max} \leq 24$ arms and $d = 10$ is too small to benefit from GPU offload, and using float64 makes the algebraic identity in Eq. equation 948 hold to machine precision modulo the Monte Carlo residual ρ in Appendix G.

H.7 C.7 RUNTIME COMPARISON ACROSS ABLATIONS

Table 15 reports per-seed wall-clock for each of the five ablation branches at the default hyperparameters of Table 14, averaged across the three environments.

The ranking is what one would expect: `collapse_1` is fastest because it trivially has $K = 1$ and never inverts; `no_routing` is fast because the $\arg\max$ is replaced by a uniform draw and the LinUCB linear system is never solved; `full` sits between `no_reflection` and the worst case because gating keeps the average archive size below $K_{\max}$.

Variant	wall-clock (s/seed)	mean $\lvert A_T \rvert$	inverses solved
`collapse_1`	1.4	1.0	0
`no_routing`	2.1	19.7	0
`no_reflection`	7.6	12.0	$\sim N_u T K_{\text{init}}$
`no_personalization`	8.4	19.7	$\sim N_u T \bar{K}$
`full`	9.2	15.7	$\sim N_u T \bar{K}$

Table 15: Per-seed wall-clock and arithmetic intensity for each ablation, averaged over the three environments. "Inverses solved" is the count of per-round matrix inverse calls $\widehat{A}_a^{-1}$ across all arms.

H.8 C.8 REPRODUCIBILITY HARNESS

The minimal reproduction harness consists of three Python scripts, each of which reads a YAML config and writes a single `.npy` file plus the figures used in Section 6 and Appendix G.

$$\texttt{run\_decomposition.py} \;\rightarrow\; \text{Table 2 and Eq. equation 948,} \tag{951}$$

$$\texttt{run\_fixed\_archive.py} \;\rightarrow\; \text{Lemma 11 sanity check; } \widehat{\Delta}_{\text{route}}(A_1) \text{ vs. } \alpha_{\text{Dir}}, \tag{952}$$

$$\texttt{run\_reflection\_limits.py} \;\rightarrow\; \text{ungated-reflection negative result; Table 21.} \tag{953}$$

Each script is parameterised by a tuple (env, variant, seed) and uses

$$\text{rng}(s, u) \;=\; \text{np.random.default\_rng}(s + 1000 + u) \tag{954}$$

for the per-user noise stream, and an independent

$$\text{rng}_{\text{arch}}(s) \;=\; \text{np.random.default\_rng}(s + 1) \tag{955}$$

for the initial-archive draw.

With these two seeds fixed, every entry of Table 2 reproduces bitwise on a single machine, and to within float64 round-off across machines.

The codebase bridge document referenced in the writeup packet specifies which existing experiment files were lifted into `src/` and which were rewritten.

The most important rewritten module is the metrics layer, where the original "global best" label is replaced by the corrected $\widehat{V}_{\text{static}}(A_1)$ from Eq. equation 942, and "`no_routing`" is renamed in-place to make the comparator unambiguous against the decomposition of Section 3.

Reflection code, feature builders, and the LinUCB inner loop are reused from the saved `19d7...` experiment, including the `add_arm()` primitive of Algorithm 4.

H.9 C.9 What is intentionally not in the harness

Three things are absent by design.

First, there is no LLM call: $\mathcal{F}$ is the closed-form proposer of Section H.3, not a language model.

A non-oracle reflection harness is the headline open direction in Section **??**.

Second, there is no real-data benchmark: the configuration is synthetic-only, matching the evidence triage memo.

Third, there is no shared-archive variant: each user gets a private archive copy after round 1, which is the operational reason that Δ_{route} in the main paper is attached to the common initial archive A_1 rather than to a population-level archive.

Adding a shared-archive sibling experiment is straightforward in this harness but would change the comparator and is therefore deferred.

These three absences are what keep the paper a decomposition paper rather than a system paper.

Every entry in Table 2 and every cell in Appendix G is generated by the harness above; nothing in the main text or appendix is taken from a different code path.

I Appendix: Full Tables

This appendix consolidates every numerical artifact referenced in the main paper.

All numbers are computed from the harness of Appendix H.8 on the corrected static comparator of Eq. equation 942.

Throughout, we use $N_s = 5$ seeds, $N_u = 8$ users per environment, $T = 300$, $K_{\max} = 24$, and feature dimension $d = 2K = 10$.

The five ablation variants are

$$\mathcal{V} \;=\; \{\texttt{full},\ \texttt{no\_routing},\ \texttt{no\_reflection},\ \texttt{no\_personalization},\ \texttt{collapse\_1}\}, \tag{956}$$

and the three environments are

$$\mathcal{E} \;=\; \{\texttt{general},\ \texttt{workspace},\ \texttt{support}\}. \tag{957}$$

For every aggregate statistic $\widehat{\theta}$ reported below we form a paired-seed 95% confidence interval as

$$\mathrm{CI}_{95}(\widehat{\theta}) \;=\; \widehat{\theta} \pm 1.96 \cdot \frac{\widehat{\sigma}_s}{\sqrt{N_s}}, \tag{958}$$

where $\widehat{\sigma}_s$ is the across-seed standard deviation.

Bootstrapped CIs in Section I.8 use $B = 10{,}000$ resamples drawn with replacement from the seed index set.

I.1 D.1 Full decomposition table

Table 16 reports the four decomposition terms of Theorem **??** for each environment and each variant, together with the realized routed value $\widehat{\widehat{\mathcal{V}}}_T$ and the residual

$$\rho \;=\; \widehat{\widehat{\mathcal{V}}}_T - \big[\widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T\big]. \tag{959}$$

By Theorem **??** the population residual is exactly zero; the empirical $|\rho|$ is bounded by the Monte Carlo error from finite (N_s, N_u, T).

Env	Variant	$\widehat{V}_{\text{static}}(A_1)$	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\widehat{G}}_T$	$\widehat{\widehat{L}}_T$	$\widehat{\widehat{\mathcal{V}}}_T$	$\lvert\rho\rvert$
general	`full`	0.570	0.280	0.051	0.036	0.864	0.001
	`no_routing`	0.570	0.280	0.051	0.281	0.621	0.001
	`no_reflection`	0.570	0.280	0.000	0.038	0.812	0.000
	`no_personalization`	0.570	0.280	0.052	0.080	0.821	0.001
	`collapse_1`	0.570	0.000	0.000	0.069	0.501	0.000
workspace	`full`	0.577	0.124	0.041	0.043	0.699	0.001
	`no_routing`	0.577	0.124	0.041	0.183	0.559	0.001
	`no_reflection`	0.577	0.124	0.000	0.044	0.657	0.000
	`no_personalization`	0.577	0.124	0.040	0.061	0.680	0.000
	`collapse_1`	0.577	0.000	0.000	0.073	0.504	0.000
support	`full`	0.579	0.215	0.061	0.039	0.816	0.001
	`no_routing`	0.579	0.215	0.061	0.255	0.600	0.000
	`no_reflection`	0.579	0.215	0.000	0.041	0.753	0.000
	`no_personalization`	0.579	0.215	0.060	0.078	0.776	0.000
	`collapse_1`	0.579	0.000	0.000	0.060	0.519	0.000

Table 16: Full decomposition table across environments and ablations. Residual $|\rho|$ is bounded by Monte Carlo error and is at most 1.5×10^{-3} across all 15 cells.

I.2 D.2 Per-seed breakdown of the `full` variant

Table 17 reports per-seed values of the four decomposition terms and the realized routed value for the `full` variant, demonstrating that across-seed variability in $\widehat{\Delta}_{\text{route}}(A_1)$ is dominated by the initial-archive draw $\text{rng}_{\text{arch}}(s)$ of Eq. equation 955, while across-seed variability in $\widehat{\widehat{L}}_T$ is dominated by the noise stream of Eq. equation 954.

Env	Seed	$\widehat{V}_{\text{static}}$	$\widehat{\Delta}_{\text{route}}$	$\widehat{\widehat{G}}_T$	$\widehat{\widehat{L}}_T$	$\widehat{\widehat{\mathcal{V}}}_T$
general	0	0.568	0.282	0.052	0.034	0.868
	1	0.572	0.276	0.049	0.037	0.860
	2	0.569	0.283	0.053	0.036	0.869
	3	0.571	0.279	0.050	0.038	0.862
	4	0.570	0.280	0.051	0.035	0.866
workspace	0	0.575	0.126	0.040	0.044	0.697
	1	0.578	0.123	0.042	0.042	0.701
	2	0.577	0.125	0.041	0.043	0.700
	3	0.579	0.122	0.040	0.045	0.696
	4	0.576	0.124	0.042	0.041	0.701
support	0	0.580	0.213	0.062	0.038	0.817
	1	0.578	0.217	0.060	0.040	0.815
	2	0.579	0.214	0.063	0.039	0.817
	3	0.581	0.216	0.060	0.041	0.816
	4	0.577	0.215	0.061	0.037	0.816

Table 17: Per-seed decomposition for the `full` variant. Across-seed standard deviations are at most 4×10^{-3} for every column.

I.3 D.3 Per-cluster heterogeneity decomposition

The fixed-archive heterogeneity term $\widehat{\Delta}_{\text{route}}(A_1)$ is averaged over the user marginal Π.

Stratifying by the cluster index $g(U)$ of Eq. equation 957 reveals which clusters carry the gain.

Define

$$\widehat{\Delta}^{(g)}_{\text{route}}(A_1) \;=\; \mathbb{E}\big[m_1(U) \mid g(U) = g\big] - \max_{a \in A_1} \mathbb{E}\big[\mu(U,a) \mid g(U) = g\big], \tag{960}$$

which is the within-cluster routed-minus-static gap.

Table 18 reports $\widehat{\Delta}^{(g)}_{\text{route}}$ for each cluster in each environment under the `full` variant.

Env	Cluster g	$\widehat{\Delta}^{(g)}_{\text{route}}(A_1)$	95% CI
`general`	0	0.17	$[0.11, 0.23]$
	1	0.39	$[0.34, 0.44]$
	2	0.41	$[0.36, 0.46]$
	3	0.23	$[0.17, 0.29]$
	4	0.40	$[0.35, 0.45]$
`workspace`	0	0.10	$[0.06, 0.14]$
	1	0.15	$[0.10, 0.20]$
`support`	0	0.37	$[0.32, 0.42]$
	1	0.44	$[0.39, 0.49]$
	2	0.03	$[-0.02, 0.08]$

Table 18: Per-cluster heterogeneity gaps for the `full` variant, with paired-seed 95% CIs from Eq. equation 958. The cluster-2 row of `support` confirms the strict-inequality clause of Proposition 10: when within-cluster rankings agree, $\widehat{\Delta}^{(g)}_{\text{route}} \approx 0$.

I.4 D.4 Per-seed final cumulative regret

Per-seed final cumulative regret of the `full` variant against the current-archive oracle is reported in Table 19, where

$$\text{CumReg}_T = \sum_{t=1}^{T} \big(m_t(U) - r_t\big), \tag{961}$$

averaged across users.

By Theorem **??** this quantity equals $T \cdot \widehat{\widehat{L}}_T$ in expectation.

Env	seed 0	seed 1	seed 2	seed 3	seed 4	mean ± std
`general`	10.21	11.10	10.84	11.46	10.87	10.90 ± 0.42
`workspace`	13.21	12.62	12.95	13.49	12.91	13.04 ± 0.30
`support`	11.41	11.99	11.71	12.35	10.92	11.68 ± 0.49

Table 19: Per-seed final cumulative regret of the `full` variant against the current-archive oracle. The implied $T \cdot \widehat{\widehat{L}}_T$ is $300 \cdot 0.036 = 10.8$, $300 \cdot 0.043 = 12.9$, and $300 \cdot 0.039 = 11.7$ for the three environments, matching the table to within Monte Carlo error.

I.5 D.5 Final regret across ablations

Table 20 extends Table 19 to all five variants.

The `no_routing` variant has cumulative regret roughly seven times that of `full`, consistent with $\widehat{L}_T^{\texttt{no\_routing}} \approx 7 \cdot \widehat{L}_T^{\texttt{full}}$ in Table 16.

I.6 D.6 Reflection diagnostics

Table 21 summarises the reflection bookkeeping under the `full` variant: mean number of reflection events per user, fraction of children that strictly improve the per-user archive optimum ("beneficial-child fraction"), final archive size, and time-to-first-specialist (TTFS).

Env	Variant	seed 0	seed 1	seed 2	seed 3	seed 4
general	full	10.21	11.10	10.84	11.46	10.87
	no_routing	84.30	85.12	83.71	84.95	84.42
	no_reflection	11.18	11.92	11.41	11.95	11.55
	no_personalization	23.71	24.92	24.10	24.85	24.04
	collapse_1	20.40	21.12	20.61	21.32	20.85
workspace	full	13.21	12.62	12.95	13.49	12.91
	no_routing	54.70	55.21	54.95	55.42	54.81
	no_reflection	13.30	12.95	13.21	13.65	13.04
	no_personalization	18.21	18.95	18.41	18.81	18.32
	collapse_1	21.40	22.10	21.81	22.30	21.95
support	full	11.41	11.99	11.71	12.35	10.92
	no_routing	76.30	77.12	76.41	77.85	76.21
	no_reflection	12.30	12.85	12.51	12.95	12.20
	no_personalization	23.30	23.95	23.61	24.10	23.45
	collapse_1	17.81	18.45	18.10	18.62	18.02

Table 20: Per-seed final cumulative regret across ablations. Each entry is averaged across the $N_u = 8$ users for that seed.

Env	mean reflections	beneficial-child frac.	mean $\lvert A_T \rvert$	TTFS (rounds)
general	7.6 ± 0.4	0.42	15.7 ± 0.6	8.98 ± 1.2
workspace	4.0 ± 0.3	0.22	11.9 ± 0.5	2.75 ± 0.7
support	5.1 ± 0.3	0.35	13.4 ± 0.5	0.35 ± 0.4

Table 21: Reflection diagnostics for the `full` variant. The TTFS column is the empirical instance of the bound in Theorem 7, with the caveat from Section H.9 that $q_\epsilon \approx 1$ in this simulator.

I.7 D.7 Ungated reflection diagnostics

Removing the gating clause of Eq. equation 948 produces the negative result of Section 6.

Table 22 reports the gated/ungated comparison, showing that the archive grows to the cap $K_{\max}$ and reward erodes by 0.024 to 0.041 depending on environment.

Env	Setting	mean $\lvert A_T \rvert$	$\widehat{\mathcal{V}}_T$	$\widehat{L}_T$
general	gated	15.7	0.864	0.036
	ungated	19.7	0.823	0.077
workspace	gated	11.9	0.699	0.043
	ungated	19.7	0.675	0.067
support	gated	13.4	0.816	0.039
	ungated	19.7	0.781	0.074

Table 22: Gated vs. ungated reflection. Ungated reflection inflates the per-arm sample requirement and the exploration cost dominates the modest extra discovery.

I.8 D.8 Ablation effect sizes with bootstrapped CIs

For each pair $(\texttt{full}, \texttt{ablation})$ we form the seed-paired difference

$$\widehat{\Delta}^{(\texttt{abl})} = \widehat{\mathcal{V}}_T^{(\texttt{full})} - \widehat{\mathcal{V}}_T^{(\texttt{abl})}, \tag{962}$$

and resample seeds with replacement $B = 10{,}000$ times to form a bootstrap distribution.

Table 23 reports the mean effect, the bootstrap 95% CI, and the bootstrap two-sided p-value against $\widehat{\Delta}^{(\mathtt{abl})} = 0$.

Env	Comparison	$\widehat{\Delta}$	95% CI (boot.)	p
`general`	vs. `no_routing`	+0.243	[0.235, 0.251]	$< 10^{-4}$
	vs. `no_reflection`	+0.052	[0.046, 0.057]	$< 10^{-4}$
	vs. `no_personalization`	+0.043	[0.038, 0.049]	$< 10^{-4}$
	vs. `collapse_1`	+0.363	[0.354, 0.371]	$< 10^{-4}$
`workspace`	vs. `no_routing`	+0.140	[0.133, 0.147]	$< 10^{-4}$
	vs. `no_reflection`	+0.042	[0.037, 0.047]	$< 10^{-4}$
	vs. `no_personalization`	+0.019	[0.012, 0.026]	$< 10^{-4}$
	vs. `collapse_1`	+0.195	[0.187, 0.203]	$< 10^{-4}$
`support`	vs. `no_routing`	+0.216	[0.208, 0.224]	$< 10^{-4}$
	vs. `no_reflection`	+0.063	[0.057, 0.069]	$< 10^{-4}$
	vs. `no_personalization`	+0.040	[0.034, 0.046]	$< 10^{-4}$
	vs. `collapse_1`	+0.297	[0.288, 0.306]	$< 10^{-4}$

Table 23: Bootstrapped effect sizes for the four ablations. Routing and archive collapse are the dominant effects; reflection and personalization features each contribute a small but significant increment.

I.9 D.9 Average reward by variant

Table 24 reports the realized routed value $\widehat{\widehat{\mathcal{V}}}_T$ for every (env, variant) pair with 95% CIs from Eq. equation 958.

The values match Table 16 but are repeated here in a flat layout for ease of citation.

Env	`full`	`no_routing`	`no_reflection`	`no_pers.`	`collapse_1`
`general`	0.864 ± 0.003	0.621 ± 0.005	0.812 ± 0.003	0.821 ± 0.004	0.501 ± 0.004
`workspace`	0.699 ± 0.004	0.559 ± 0.005	0.657 ± 0.004	0.680 ± 0.004	0.504 ± 0.004
`support`	0.816 ± 0.003	0.600 ± 0.005	0.753 ± 0.004	0.776 ± 0.004	0.519 ± 0.004

Table 24: Mean routed value $\widehat{\widehat{\mathcal{V}}}_T$ (seed mean ± paired-seed 95% CI half-width) by environment and ablation.

I.10 D.10 Per-user breakdown of routed value

Table 25 reports per-user $\widehat{\widehat{\mathcal{V}}}_T$ for the `full` variant in the `general` environment; analogous tables for `workspace` and `support` are omitted for space and follow the same pattern.

Seed	u0	u1	u2	u3	u4	u5	u6	u7
0	0.881	0.870	0.862	0.852	0.873	0.880	0.864	0.862
1	0.876	0.861	0.870	0.844	0.866	0.875	0.860	0.858
2	0.880	0.872	0.871	0.851	0.872	0.879	0.863	0.864
3	0.871	0.863	0.866	0.846	0.870	0.877	0.861	0.860
4	0.882	0.866	0.872	0.853	0.875	0.881	0.866	0.862

Table 25: Per-user, per-seed $\widehat{\widehat{\mathcal{V}}}_T$ in the `general` environment under the `full` variant. Per-user variance is dominated by the cluster index $g(u)$.

I.11 D.11 Context-type tuning sweep

Table 26 restates the baseline-tuning context-type comparison from Appendix G.

Three encodings are compared: `pref` (user preference vector only), `onehot` (candidate one-hot only), and `concat` (concatenation, used in the `full` variant of all main-paper experiments).

ctx type	$\widehat{\mathcal{V}}_T$	$\widehat{\bar{L}}_T$	final $\sum_t (m_t - r_t)$	mean $\lvert A_T \rvert$
`pref`	0.813	0.091	27.41	19.67
`onehot`	0.834	0.075	22.49	19.67
`concat`	0.882	0.038	11.53	15.67

Table 26: Context-type ablation in the `general` environment, single seed, $T = 300$. Values reproduced from the baseline-tuning stage; `concat` is selected for the main-paper experiments.

I.12 D.12 Sensitivity to noise scale

Table 27 sweeps the noise scale σ in Eq. equation 942 for the `full` variant in the `general` environment.

The exploration cost $\widehat{\bar{L}}_T$ scales approximately linearly with σ, while the structural term $\widehat{\Delta}_{\text{route}}(A_1)$ is invariant.

σ	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\bar{G}}_T$	$\widehat{\bar{L}}_T$	$\widehat{\mathcal{V}}_T$
0.025	0.280	0.052	0.020	0.882
0.050	0.280	0.051	0.036	0.864
0.100	0.280	0.049	0.064	0.836
0.200	0.280	0.044	0.121	0.776

Table 27: Noise-scale sensitivity. The dominance condition of Corollary 3 continues to hold at all four noise levels because $\widehat{\Delta}_{\text{route}} + \widehat{\bar{G}}_T \in [0.324, 0.332]$ exceeds $\widehat{\bar{L}}_T \in [0.020, 0.121]$.

I.13 D.13 Sensitivity to archive size

Table 28 sweeps the initial archive size $K = |A_1|$ in $\{4, 8, 12, 24\}$ in the `general` environment.

Larger archives raise both $\widehat{\Delta}_{\text{route}}(A_1)$ and $\widehat{\bar{L}}_T$, illustrating the tradeoff foreshadowed by Theorem **??**.

K	$\widehat{V}_{\text{static}}(A_1)$	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\bar{L}}_T$	$\widehat{\mathcal{V}}_T$
4	0.534	0.171	0.022	0.683
8	0.561	0.244	0.030	0.775
12	0.570	0.280	0.036	0.864
24	0.578	0.318	0.058	0.838

Table 28: Archive-size sensitivity in the `general` environment. The dominance margin $\widehat{\Delta}_{\text{route}} + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T$ peaks near $K = 12$.

I.14 D.14 Sensitivity to user heterogeneity

Let α_{Dir} be the Dirichlet concentration parameter governing per-user preference vectors.

As $\alpha_{\text{Dir}} \to \infty$ the user marginal collapses to a Dirac and $\widehat{\Delta}_{\text{route}}(A_1) \to 0$, recovering the homogeneous counterexample C1 of Section J.

I.15 D.15 Reflection threshold and cooldown sweep

Table 30 varies the reflection trigger τ and cooldown κ in Eq. equation 948.

Lower τ or κ approach the ungated regime of Table 22.

α_{Dir}	$\widehat{\Delta}_{\mathrm{route}}(A_1)$	$\widehat{\widehat{L}}_T$	$\widehat{\widehat{\mathcal{V}}}_T - \widehat{V}_{\mathrm{static}}(A_1)$
0.1	0.341	0.041	+0.347
0.3	0.280	0.036	+0.293
1.0	0.182	0.034	+0.198
3.0	0.071	0.030	+0.085
10.0	0.018	0.028	+0.030
∞	0.000	0.026	−0.026

Table 29: Heterogeneity sensitivity in the `general` environment. Personalized routing beats the static comparator until α_{Dir} pushes $\widehat{\Delta}_{\mathrm{route}}$ below $\widehat{\widehat{L}}_T - \widehat{\widehat{G}}_T$.

τ	κ	mean $\lvert A_T \rvert$	$\widehat{\widehat{G}}_T$	$\widehat{\widehat{\mathcal{V}}}_T$
0.05	5	19.7	0.060	0.823
0.10	10	18.4	0.057	0.842
0.15	20	15.7	0.051	0.864
0.20	30	12.9	0.030	0.851
0.30	50	9.6	0.011	0.829

Table 30: Reflection (τ, κ) sweep in the `general` environment. The default $(\tau, \kappa) = (0.15, 20)$ used in the main paper is on the Pareto frontier of $(\widehat{\widehat{G}}_T, \widehat{\widehat{\mathcal{V}}}_T)$.

I.16 D.16 THEOREM-TO-EXPERIMENT COVERAGE MAP

Table 31 cross-references each theory result to the empirical row that validates it.

Result	Statement	Empirical row	Status
Prop. 10	$\Delta_{\mathrm{route}}(A) \geq 0$, strict iff disagreement	Tab. 16, 18, 29	confirmed
Lem. 11	predictable archive preserves concentration	Algo. 1; Tab. 16 residual $\lvert\rho\rvert$	confirmed
Lem. 12	failure-prioritisation Markov bound	Tab. 30	confirmed
Thm. **??**	exact 4-term identity	Tab. 16 ($\lvert\rho\rvert \leq 10^{-3}$)	confirmed
Thm. **??**	conditional plug-in regret bound	Tab. 19, 27	descriptive
Thm. 7	first-useful-specialist time	Tab. 21 TTFS	vacuous (oracle proposer)
Cor. 3	$\Delta_{\mathrm{route}} + \bar{G}_T > \bar{L}_T$ dominance	Tab. 24, 29	confirmed

Table 31: Coverage map. "Descriptive" indicates we use Corollary 6 as a exploration-cost proxy rather than as a tight finite-sample bound; "vacuous" indicates the oracle-assisted reflection of Section H.9.

I.17 D.17 HEADLINE SUMMARY

Table 32 aggregates the four decomposition terms across environments for the `full` variant in a single row, with 95% CIs from Eq. equation 958, and reports the headline cross-environment dominance margin

$$\widehat{M} = \widehat{\Delta}_{\mathrm{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T. \tag{963}$$

This concludes the table appendix.

All numerical entries reproduce bitwise from the harness of Appendix H.8 on a single CPU core in under 30 minutes.

Quantity	`general`	`workspace`	`support`	cross-env mean
$\widehat{V}_{\text{static}}(A_1)$	0.570 ± 0.003	0.577 ± 0.003	0.579 ± 0.002	0.575
$\widehat{\Delta}_{\text{route}}(A_1)$	0.280 ± 0.004	0.124 ± 0.003	0.215 ± 0.004	0.206
$\widehat{\widehat{G}}_T$	0.051 ± 0.002	0.041 ± 0.002	0.061 ± 0.002	0.051
$\widehat{\widehat{L}}_T$	0.036 ± 0.002	0.043 ± 0.002	0.039 ± 0.001	0.039
$\widehat{\widehat{\mathcal{V}}}_T$	0.864 ± 0.003	0.699 ± 0.004	0.816 ± 0.003	0.793
$\widehat{M}$	$+0.295$	$+0.122$	$+0.237$	$+0.218$

Table 32: Cross-environment summary for the `full` variant. The dominance margin $\widehat{M}$ of Eq. equation 963 is positive in all three environments, validating Corollary 3.

J HYPER-RIDGE EXPLORATION-COST ANALYSIS

This appendix block analyzes the new Hyper-Ridge router. The reward accounting is inherited unchanged from Theorem 2 and Corollary 3; we do not re-prove the four-term decomposition. The only question addressed here is how replacing ordinary ridge geometry by a shared archive covariance changes the exploration-cost term.

J.1 GENERALIZED-RIDGE OBJECTS

Fix one user trajectory and one candidate in the current archive. The user-candidate feature at round s is

$$x_s(a) \in \mathbb{R}^d. \tag{964}$$

The linear reward model for candidate a is

$$\mu(U_s, a) = x_s(a)^\top \theta_a. \tag{965}$$

The observed reward on a selected candidate is

$$r_s = x_s(a_s)^\top \theta_{a_s} + \xi_s. \tag{966}$$

The new modeling assumption is that candidate parameters share a hyper-covariance:

$$\theta_a \sim (0, \Omega). \tag{967}$$

Here the notation means

$$\mathbb{E}[\theta_a] = 0, \tag{968}$$

and

$$\mathrm{Cov}(\theta_a) = \Omega. \tag{969}$$

The covariance is required to be positive definite:

$$\Omega \succ 0. \tag{970}$$

For candidate a, the pre-t selected-round index set is

$$\mathcal{T}_{a,t} = \{\, s < t : \; a_s = a \,\}. \tag{971}$$

The candidate-specific design matrix stacks the selected features:

$$X_{a,t} = \left[x_s(a)^\top\right]_{s \in \mathcal{T}_{a,t}}. \tag{972}$$

The corresponding reward vector is

$$y_{a,t} = [r_s]_{s \in \mathcal{T}_{a,t}}. \tag{973}$$

The hyper-covariance estimate available before round t is

$$\widehat{\Omega}_t \succ 0. \tag{974}$$

The generalized-ridge estimator used by Hyper-Ridge PersonalEvo is

$$\widehat{\theta}_{a,t} = \arg\min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda\, \theta^\top \widehat{\Omega}_t^{-1} \theta \right\}. \tag{975}$$

Equivalently,

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1} \right)^{-1} X_{a,t}^\top y_{a,t}. \tag{976}$$

The generalized design matrix is

$$V_{a,t}^{\mathrm{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{977}$$

The Hyper-Ridge UCB score is

$$\mathrm{UCB}_t^{\mathrm{hyper}}(U_t, a) = x_t(a)^\top \widehat{\theta}_{a,t} + \alpha_t \sqrt{x_t(a)^\top \left(V_{a,t}^{\mathrm{hyper}}\right)^{-1} x_t(a)}. \tag{978}$$

The selected candidate is

$$a_t(U_t) \in \arg\max_{a \in A_t(U_t)} \mathrm{UCB}_t^{\mathrm{hyper}}(U_t, a). \tag{979}$$

For a generic design matrix X, the effective dimension induced by Ω is

$$d_{\mathrm{eff}}(\Omega; X) = \mathrm{tr}\left[X^\top X \left(X^\top X + \lambda \Omega^{-1}\right)^{-1}\right]. \tag{980}$$

When the design is clear from context, we write

$$d_{\mathrm{eff}}(\Omega) = d_{\mathrm{eff}}(\Omega; X). \tag{981}$$

A small calculation illustrates why Ω can reduce the statistical dimension. Take

$$\Omega = \begin{bmatrix} 1 & 0 \\ 0 & 0.01 \end{bmatrix}. \tag{982}$$

Let

$$X^\top X = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}. \tag{983}$$

With $\lambda = 1$, the effective dimension is

$$\begin{aligned} d_{\mathrm{eff}}(\Omega; X) &= \mathrm{tr}\left[\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix}\right)^{-1}\right] && (984) \\ &= \frac{9}{10} + \frac{1}{101} && (985) \\ &\approx 0.910. && (986) \end{aligned}$$

By contrast, identity ridge gives

$$\begin{aligned} d_{\mathrm{eff}}(I; X) &= \frac{9}{10} + \frac{1}{2} && (987) \\ &= 1.400. && (988) \end{aligned}$$

Thus the same data matrix has smaller exploration geometry when the archive covariance says the second coordinate is unlikely to matter.

For the estimator itself, consider

$$X = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}. \tag{989}$$

Let the observed rewards be

$$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{990}$$

Using the same Ω and $\lambda = 1$, the generalized design is

$$\begin{aligned} V^{\mathrm{hyper}} &= X^\top X + \Omega^{-1} && (991) \\ &= \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} && (992) \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}. && (993) \end{aligned}$$

Since

$$X^\top y = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \tag{994}$$

the estimate is

$$\widehat{\theta} = \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 0 \end{bmatrix} \tag{995}$$

$$= \begin{bmatrix} 5/6 \\ 0 \end{bmatrix}. \tag{996}$$

The confidence width for $x = (1, 0)^\top$ is

$$\sqrt{x^\top (V^{\mathrm{hyper}})^{-1} x} = \sqrt{1/6}. \tag{997}$$

J.2 SHARED-STRUCTURE ASSUMPTION

We now restate the shared-structure condition used by the Hyper-Ridge analysis. This is a regime assumption, not an empirical finding; the planned low-rank archive-geometry sweep in Section 6 is designed to stress-test it once new-method artifacts are produced.

Assumption 17 (Shared archive geometry)**.** *There is a positive definite hyper-covariance matrix*

$$\Omega \in \mathbb{R}^{d \times d}. \tag{998}$$

Each candidate parameter follows the random-effects model

$$\theta_a \sim (0, \Omega). \tag{999}$$

For the high-probability regret analysis, the realized candidate parameters obey the ellipsoid bound

$$\|\theta_a\|_{\Omega^{-1}} \le S_\Omega \qquad \textit{for every } a \in A_t(U_t) \textit{ and } t \le T. \tag{1000}$$

The features are predictable:

$$x_t(a) \in \mathcal{F}_{t-1} \qquad \textit{for every } a \in A_t(U_t). \tag{1001}$$

The rewards satisfy the linear model

$$r_t = x_t(a_t)^\top \theta_{a_t} + \xi_t. \tag{1002}$$

The noise is conditionally σ-sub-Gaussian:

$$\mathbb{E}[\exp(\eta \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \qquad \textit{for every } \eta \in \mathbb{R}. \tag{1003}$$

The archive covariance gives a bounded feature radius:

$$x_t(a)^\top \Omega x_t(a) \le L_\Omega^2 \qquad \textit{for every } a \in A_t(U_t). \tag{1004}$$

The effective dimension is small in the target regime:

$$d_{\mathrm{eff}}(\Omega; X) \ll d. \tag{1005}$$

Within Assumption 17, the predictability condition equation 1001 and the sub-Gaussian noise condition equation 1003 restate Assumptions 3 and 2 of Section 3 in the hyper-ridge notation, and the Ω-weighted feature radius equation 1004 is the Ω-geometry analogue of Assumption 4. The genuinely new hypotheses are the random-effects model, the ellipsoid bound S_Ω, and the low-effective-dimension regime; the latter matches Assumption 7 in the main text. The last display is not required for validity of the concentration lemmas. It is the structural regime in which the resulting bound improves over ordinary d-dimensional LinUCB.

J.3 Known-Ω concentration

The first lemma is the generalized-ridge analogue of the standard OFUL confidence bound (Abbasi-Yadkori et al., 2011; Chu et al., 2011). It is stated for known Ω, which is the clean oracle-covariance case. The estimated-covariance case adds a perturbation term and is handled separately.

Lemma 45 (Self-normalized concentration in the Ω geometry)**.** *Fix a candidate a. Assume $\widehat{\Omega}_t = \Omega$. Define the known-covariance design matrix by*

$$V_{a,t}^{\Omega} = \lambda\Omega^{-1} + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top. \tag{1006}$$

Define the known-covariance estimator by

$$\widehat{\theta}_{a,t}^{\Omega} = \left(V_{a,t}^{\Omega}\right)^{-1} \sum_{s\in\mathcal{T}_{a,t}} x_s(a)r_s. \tag{1007}$$

Define the confidence radius

$$\beta_{a,t}^{\Omega}(\delta) = \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}\delta}\right)} + \sqrt{\lambda}\,S_\Omega. \tag{1008}$$

Then, under Assumption 17, with probability at least $1-\delta$, uniformly over t,

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \le \beta_{a,t}^{\Omega}(\delta). \tag{1009}$$

Consequently, for every predictable test feature $x\in\mathbb{R}^d$,

$$\left|x^\top\left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| \le \beta_{a,t}^{\Omega}(\delta)\sqrt{x^\top\left(V_{a,t}^{\Omega}\right)^{-1}x}. \tag{1010}$$

Proof. For this fixed candidate, define the regularizer

$$A = \lambda\Omega^{-1}. \tag{1011}$$

Define the noise-weighted feature martingale

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s. \tag{1012}$$

Using the reward model, the estimator error satisfies

$$\widehat{\theta}_{a,t}^{\Omega} - \theta_a = \left(V_{a,t}^{\Omega}\right)^{-1}\sum_{s\in\mathcal{T}_{a,t}} x_s(a)\left(x_s(a)^\top\theta_a + \xi_s\right) - \theta_a \tag{1013}$$

$$= \left(V_{a,t}^{\Omega}\right)^{-1}\left[\sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top\theta_a + S_{a,t}\right] - \theta_a \tag{1014}$$

$$= \left(V_{a,t}^{\Omega}\right)^{-1}\left[\left(V_{a,t}^{\Omega} - A\right)\theta_a + S_{a,t}\right] - \theta_a \tag{1015}$$

$$= \left(V_{a,t}^{\Omega}\right)^{-1}S_{a,t} - \left(V_{a,t}^{\Omega}\right)^{-1}A\theta_a. \tag{1016}$$

Taking the $V_{a,t}^{\Omega}$-norm and applying the triangle inequality gives

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \le \|S_{a,t}\|_{(V_{a,t}^{\Omega})^{-1}} + \|A\theta_a\|_{(V_{a,t}^{\Omega})^{-1}}. \tag{1017}$$

Since $V_{a,t}^{\Omega} \succeq A$, we have

$$\|A\theta_a\|_{(V_{a,t}^{\Omega})^{-1}}^2 = \theta_a^\top A\left(V_{a,t}^{\Omega}\right)^{-1}A\theta_a \tag{1018}$$

$$\le \theta_a^\top A\theta_a \tag{1019}$$

$$= \lambda\theta_a^\top\Omega^{-1}\theta_a \tag{1020}$$

$$\le \lambda S_\Omega^2. \tag{1021}$$

It remains to control the martingale term. For any fixed vector $z \in \mathbb{R}^d$, conditional sub-Gaussianity implies

$$\mathbb{E}\left[\exp\left(z^\top S_{a,t} - \frac{\sigma^2}{2} z^\top \left(V_{a,t}^\Omega - A\right) z\right)\right] \le 1. \tag{1022}$$

Integrating the preceding display over a Gaussian density with precision A/σ^2 yields the standard mixture supermartingale identity

$$\mathbb{E}\left[\frac{\det(A)^{1/2}}{\det(V_{a,t}^\Omega)^{1/2}} \exp\left(\frac{\|S_{a,t}\|^2_{(V_{a,t}^\Omega)^{-1}}}{2\sigma^2}\right)\right] \le 1. \tag{1023}$$

By Markov's inequality,

$$\Pr\left(\|S_{a,t}\|_{(V_{a,t}^\Omega)^{-1}} > \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(A)^{1/2}\delta}\right)}\right) \le \delta. \tag{1024}$$

The standard stopping-time version of the same mixture argument gives the same event uniformly over t. Substituting $A = \lambda\Omega^{-1}$ gives

$$\|S_{a,t}\|_{(V_{a,t}^\Omega)^{-1}} \le \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}\delta}\right)}. \tag{1025}$$

Combining equation 1017, equation 1018, and equation 1025 proves

$$\left\|\widehat{\theta}_{a,t}^\Omega - \theta_a\right\|_{V_{a,t}^\Omega} \le \beta_{a,t}^\Omega(\delta). \tag{1026}$$

Finally, Cauchy–Schwarz in the $V_{a,t}^\Omega$ inner product gives

$$\begin{aligned}
\left|x^\top\left(\widehat{\theta}_{a,t}^\Omega - \theta_a\right)\right| &= \left|\left[\left(V_{a,t}^\Omega\right)^{-1/2} x\right]^\top \left[\left(V_{a,t}^\Omega\right)^{1/2}\left(\widehat{\theta}_{a,t}^\Omega - \theta_a\right)\right]\right| && (1027)\\
&\le \sqrt{x^\top\left(V_{a,t}^\Omega\right)^{-1} x}\left\|\widehat{\theta}_{a,t}^\Omega - \theta_a\right\|_{V_{a,t}^\Omega} && (1028)\\
&\le \beta_{a,t}^\Omega(\delta)\sqrt{x^\top\left(V_{a,t}^\Omega\right)^{-1} x}. && (1029)
\end{aligned}$$

□

A concrete confidence-bound computation is useful. Suppose

$$\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}} = 10. \tag{1030}$$

Let

$$\sigma = 0.05. \tag{1031}$$

Let

$$\delta = 0.05. \tag{1032}$$

Let the regularization bias term be

$$\sqrt{\lambda}S_\Omega = 0.20. \tag{1033}$$

Then the confidence radius is

$$\begin{aligned}
\beta_{a,t}^\Omega(\delta) &= 0.05\sqrt{2\log(10/0.05)} + 0.20 && (1034)\\
&\approx 0.05\sqrt{10.596} + 0.20 && (1035)\\
&\approx 0.363. && (1036)
\end{aligned}$$

If the uncertainty width at a candidate is

$$\sqrt{x^\top (V^{\Omega}_{a,t})^{-1} x} = 0.10, \tag{1037}$$

then Lemma 45 gives the prediction error bound

$$\left| x^\top \left(\widehat{\theta}^{\Omega}_{a,t} - \theta_a \right) \right| \leq 0.363 \cdot 0.10 \tag{1038}$$

$$= 0.0363. \tag{1039}$$

J.4 Elliptical potential in the Ω geometry

The next lemma is the step that converts per-round confidence widths into a dimension-dependent exploration cost. Its role is the same as the elliptical-potential lemma in LinUCB, except that the determinant is measured relative to $\lambda\Omega^{-1}$, and the resulting log-determinant is controlled by $d_{\text{eff}}(\Omega; X)$ up to a logarithmic factor.

Lemma 46 (Ω-elliptical potential and effective dimension)**.** *Fix one candidate a, and write its selected features in chronological order as*

$$x_1, \ldots, x_n. \tag{1040}$$

Define the recursive generalized-ridge matrices by

$$V_0 = \lambda\Omega^{-1}, \tag{1041}$$

and

$$V_s = V_{s-1} + x_s x_s^\top. \tag{1042}$$

Define the uncertainty increment by

$$q_s = x_s^\top V_{s-1}^{-1} x_s. \tag{1043}$$

Let the full design be

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix}. \tag{1044}$$

Define the covariance-scaled Gram matrix by

$$M = \lambda^{-1}\Omega^{1/2} X^\top X \Omega^{1/2}. \tag{1045}$$

Let its largest eigenvalue be

$$R = \lambda_{\max}(M). \tag{1046}$$

Define

$$c(R) = \begin{cases} 1, & R = 0, \\ \dfrac{1+R}{R}\log(1+R), & R > 0. \end{cases} \tag{1047}$$

Then

$$\sum_{s=1}^{n} \min\{1, q_s\} \leq 2\, c(R)\, d_{\text{eff}}(\Omega; X). \tag{1048}$$

If additionally $x_s^\top \Omega x_s \leq L_\Omega^2$, then

$$R \leq \frac{nL_\Omega^2}{\lambda}, \tag{1049}$$

and hence

$$\sum_{s=1}^{n} \min\{1, q_s\} \leq 2\left[1 + \log\left(1 + \frac{nL_\Omega^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega; X). \tag{1050}$$

Proof. The matrix determinant lemma gives

$$\det(V_s) = \det(V_{s-1})\left(1 + x_s^\top V_{s-1}^{-1} x_s\right) \tag{1051}$$
$$= \det(V_{s-1})(1 + q_s). \tag{1052}$$

Multiplying over $s = 1, \ldots, n$ gives

$$\frac{\det(V_n)}{\det(V_0)} = \prod_{s=1}^{n} (1 + q_s). \tag{1053}$$

Taking logarithms yields

$$\log \frac{\det(V_n)}{\det(V_0)} = \sum_{s=1}^{n} \log(1 + q_s). \tag{1054}$$

For every $q \geq 0$, the scalar inequality

$$\min\{1, q\} \leq 2\log(1 + q) \tag{1055}$$

holds. Therefore

$$\sum_{s=1}^{n} \min\{1, q_s\} \leq 2 \sum_{s=1}^{n} \log(1 + q_s) \tag{1056}$$
$$= 2 \log \frac{\det(V_n)}{\det(V_0)}. \tag{1057}$$

The determinant ratio can be rewritten in the covariance-scaled coordinates:

$$\log \frac{\det(V_n)}{\det(V_0)} = \log\det\left(I + \lambda^{-1}\Omega^{1/2} X^\top X \Omega^{1/2}\right) \tag{1058}$$
$$= \log\det(I + M). \tag{1059}$$

Let the eigenvalues of M be

$$\nu_1, \ldots, \nu_d. \tag{1060}$$

Then

$$0 \leq \nu_i \leq R. \tag{1061}$$

The log-determinant is

$$\log\det(I + M) = \sum_{i=1}^{d} \log(1 + \nu_i). \tag{1062}$$

The effective dimension is

$$d_{\text{eff}}(\Omega; X) = \operatorname{tr}\left[X^\top X \left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\right] \tag{1063}$$
$$= \operatorname{tr}\left[\Omega^{1/2} X^\top X \Omega^{1/2} \left(\Omega^{1/2} X^\top X \Omega^{1/2} + \lambda I\right)^{-1}\right] \tag{1064}$$
$$= \sum_{i=1}^{d} \frac{\nu_i}{1 + \nu_i}. \tag{1065}$$

For $z > 0$, define

$$g(z) = \frac{1 + z}{z} \log(1 + z). \tag{1066}$$

Its derivative is

$$g'(z) = -\frac{\log(1+z)}{z^2} + \frac{1+1/z}{1+z} \tag{1067}$$

$$= \frac{z - \log(1+z)}{z^2} \tag{1068}$$

$$\geq 0. \tag{1069}$$

Thus $g(z) \leq g(R)$ for $0 < z \leq R$. Equivalently,

$$\log(1+z) \leq c(R)\frac{z}{1+z} \qquad \text{for } 0 \leq z \leq R. \tag{1070}$$

Applying this scalar bound eigenvalue by eigenvalue gives

$$\log\det(I+M) = \sum_{i=1}^{d} \log(1+\nu_i) \tag{1071}$$

$$\leq c(R)\sum_{i=1}^{d} \frac{\nu_i}{1+\nu_i} \tag{1072}$$

$$= c(R)d_{\text{eff}}(\Omega; X). \tag{1073}$$

Combining equation 1056 and equation 1071 proves

$$\sum_{s=1}^{n} \min\{1, q_s\} \leq 2c(R)d_{\text{eff}}(\Omega; X). \tag{1074}$$

It remains to bound R. Since $M \succeq 0$,

$$R \leq \operatorname{tr}(M) \tag{1075}$$

$$= \lambda^{-1} \operatorname{tr}\left(\Omega^{1/2} X^\top X \Omega^{1/2}\right) \tag{1076}$$

$$= \lambda^{-1} \sum_{s=1}^{n} x_s^\top \Omega x_s \tag{1077}$$

$$\leq \frac{nL_\Omega^2}{\lambda}. \tag{1078}$$

Finally, for $R \geq 0$,

$$c(R) \leq 1 + \log(1+R). \tag{1079}$$

Substituting equation 1075 into equation 1079 yields

$$c(R) \leq 1 + \log\left(1 + \frac{nL_\Omega^2}{\lambda}\right). \tag{1080}$$

The claimed logarithmic bound follows. □

K ORACLE HYPER-COVARIANCE: PROOF OF THE EFFECTIVE-DIMENSION COST

We now combine Lemma 45 and Lemma 46. The result is an oracle statement: the router is allowed to use the true hyper-covariance Ω. The estimated-Ω case is treated afterward as a perturbation of this oracle bound.

Assumption 18 (Oracle covariance and calibrated confidence)**.** *The hyper-covariance used by the router is the true matrix:*

$$\widehat{\Omega}_t = \Omega. \tag{1081}$$

The matrix is positive definite:

$$\Omega \succ 0. \tag{1082}$$

The archive size is uniformly bounded:

$$|A_t(U)| \le K_{\max}. \tag{1083}$$

The Ω-scaled feature norm is uniformly bounded:

$$x_t(a)^\top \Omega x_t(a) \le L_\Omega^2. \tag{1084}$$

There is an event $\mathcal{E}_\Omega$ on which every available arm has valid hyper-ridge confidence intervals:

$$\mathbb{P}(\mathcal{E}_\Omega) \ge 1 - \delta. \tag{1085}$$

On this event, the confidence multiplier is bounded by a deterministic radius:

$$\alpha_T \ge 1. \tag{1086}$$

The radius satisfies the logarithmic calibration condition:

$$\alpha_T^2 \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] \le C_\Omega \log\left(\frac{eT}{\delta}\right). \tag{1087}$$

The effective dimension used in the bound is the largest realized per-candidate effective dimension. For candidate a, let its realized design through time T be

$$X_{a,T} = \left[x_s(a_s)^\top\right]_{s \le T:\, a_s = a}. \tag{1088}$$

Define the candidate-wise effective dimension by

$$d_{\mathrm{eff}}(\Omega; X_{a,T}) = \mathrm{tr}\left[X_{a,T}^\top X_{a,T} \left(X_{a,T}^\top X_{a,T} + \lambda \Omega^{-1}\right)^{-1}\right]. \tag{1089}$$

The oracle archive effective dimension is

$$d_{\mathrm{eff}}(\Omega) = \max_{a \in A_T(U)} d_{\mathrm{eff}}(\Omega; X_{a,T}). \tag{1090}$$

Proposition 47 (Known-Ω effective-dimension exploration cost)**.** *Under Assumption 18, the oracle hyper-ridge router satisfies the pathwise bound on $\mathcal{E}_\Omega$:*

$$\sum_{t=1}^T L_t(U) \le 2\alpha_T \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)}. \tag{1091}$$

Consequently, its expected exploration cost satisfies

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \le 2\sqrt{2C_\Omega K_{\max} d_{\mathrm{eff}}(\Omega) T \log\left(\frac{eT}{\delta}\right)} + \delta T. \tag{1092}$$

In big-O notation, for fixed δ, fixed L_Ω, fixed λ, and calibrated α_T,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] = O\left(\sqrt{K_{\max} d_{\mathrm{eff}}(\Omega) T \log T}\right). \tag{1093}$$

Proof. Let the current-archive oracle action at round t be

$$a_t^\star \in \arg\max_{a \in A_t(U)} x_t(a)^\top \theta_a. \tag{1094}$$

The exploration cost is therefore

$$L_t(U) = x_t(a_t^\star)^\top \theta_{a_t^\star} - x_t(a_t)^\top \theta_{a_t}. \tag{1095}$$

On $\mathcal{E}_\Omega$, Lemma 45 gives the upper confidence bound for the oracle action:

$$x_t(a_t^\star)^\top \theta_{a_t^\star} \le x_t(a_t^\star)^\top \widehat{\theta}^\Omega_{a_t^\star,t} + \alpha_T \sqrt{x_t(a_t^\star)^\top \left(V^\Omega_{a_t^\star,t}\right)^{-1} x_t(a_t^\star)}. \tag{1096}$$

Since the router chooses the largest oracle-Ω UCB score,

$$x_t(a_t^\star)^\top \widehat{\theta}^{\Omega}_{a_t^\star,t} + \alpha_T \sqrt{x_t(a_t^\star)^\top \left(V^{\Omega}_{a_t^\star,t}\right)^{-1} x_t(a_t^\star)} \le x_t(a_t)^\top \widehat{\theta}^{\Omega}_{a_t,t} + \alpha_T \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t}\right)^{-1} x_t(a_t)}. \tag{1097}$$

Again by Lemma 45, the selected arm has the lower confidence bound

$$x_t(a_t)^\top \widehat{\theta}^{\Omega}_{a_t,t} \le x_t(a_t)^\top \theta_{a_t} + \alpha_T \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t}\right)^{-1} x_t(a_t)}. \tag{1098}$$

Combining equation 1095–equation 1098 yields

$$L_t(U) \le 2\alpha_T \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t}\right)^{-1} x_t(a_t)}. \tag{1099}$$

The rewards are bounded, so the same gap also satisfies

$$0 \le L_t(U) \le 1. \tag{1100}$$

Define the realized uncertainty at the selected arm by

$$q_t = x_t(a_t)^\top \left(V^{\Omega}_{a_t,t}\right)^{-1} x_t(a_t). \tag{1101}$$

Using $\alpha_T \ge 1$, equation 1099 and equation 1100 imply

$$\begin{aligned} L_t(U) &\le \min\{1, 2\alpha_T \sqrt{q_t}\} && (1102)\\ &\le 2\alpha_T \min\{1, \sqrt{q_t}\}. && (1103) \end{aligned}$$

Summing over time gives

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_T \sum_{t=1}^{T} \min\{1, \sqrt{q_t}\}. \tag{1104}$$

Partition the selected rounds by candidate. Let n_a be the number of times candidate a is selected:

$$n_a = \sum_{t=1}^{T} \mathbf{1}\{a_t = a\}. \tag{1105}$$

Let the within-candidate uncertainties be

$$q_{a,s} = x_{a,s}^\top \left(V^{\Omega}_{a,s-1}\right)^{-1} x_{a,s}. \tag{1106}$$

Then the total width sum is

$$\sum_{t=1}^{T} \min\{1, \sqrt{q_t}\} = \sum_{a \in A_T(U)} \sum_{s=1}^{n_a} \min\{1, \sqrt{q_{a,s}}\}. \tag{1107}$$

Cauchy–Schwarz gives

$$\begin{aligned} \sum_{a \in A_T(U)} \sum_{s=1}^{n_a} \min\{1, \sqrt{q_{a,s}}\} &\le \sqrt{\left(\sum_{a \in A_T(U)} n_a\right)\left(\sum_{a \in A_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\}\right)} && (1108)\\ &= \sqrt{T \sum_{a \in A_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\}}. && (1109) \end{aligned}$$

Lemma 46 applied separately to each candidate yields

$$\begin{aligned} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\} &\le 2\left[1 + \log\left(1 + \frac{n_a L_\Omega^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega; X_{a,T}) && (1110)\\ &\le 2\left[1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega). && (1111) \end{aligned}$$

Since at most $K_{\max}$ candidates are available,

$$\sum_{a\in A_T(U)}\sum_{s=1}^{n_a}\min\{1,q_{a,s}\}\le 2K_{\max}\left[1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)\right]d_{\mathrm{eff}}(\Omega). \tag{1112}$$

Substituting equation 1112 into equation 1108 and then into equation 1104 proves

$$\sum_{t=1}^{T}L_t(U)\le 2\alpha_T\sqrt{2K_{\max}T\left[1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)\right]d_{\mathrm{eff}}(\Omega)}. \tag{1113}$$

This is equation 1091. Taking expectations and splitting on $\mathcal{E}_\Omega$ gives

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)]=\mathbb{E}\left[\mathbf{1}_{\mathcal{E}_\Omega}\sum_{t=1}^{T}L_t(U)\right]+\mathbb{E}\left[\mathbf{1}_{\mathcal{E}_\Omega^c}\sum_{t=1}^{T}L_t(U)\right] \tag{1114}$$

$$\le 2\alpha_T\sqrt{2K_{\max}T\left[1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)\right]d_{\mathrm{eff}}(\Omega)}+\delta T. \tag{1115}$$

Using equation 1087 gives

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)]\le 2\sqrt{2C_\Omega K_{\max}d_{\mathrm{eff}}(\Omega)T\log\left(\frac{eT}{\delta}\right)}+\delta T. \tag{1116}$$

This is equation 1092. The asymptotic statement equation 1093 is the same inequality with fixed constants absorbed. □

For scale, suppose

$$K_{\max}=10. \tag{1117}$$

Suppose the learned archive geometry has effective dimension

$$d_{\mathrm{eff}}(\Omega)=4. \tag{1118}$$

Let the horizon be

$$T=1000. \tag{1119}$$

Ignoring constants and using $\log T\approx 6.9$, the oracle cost scale is

$$\sqrt{K_{\max}d_{\mathrm{eff}}(\Omega)T\log T}=\sqrt{10\cdot 4\cdot 1000\cdot 6.9} \tag{1120}$$

$$\approx 525. \tag{1121}$$

The corresponding ordinary dimension-d scale with $d=40$ is

$$\sqrt{K_{\max}dT\log T}=\sqrt{10\cdot 40\cdot 1000\cdot 6.9} \tag{1122}$$

$$\approx 1661. \tag{1123}$$

Thus the proof isolates the statistical benefit of low archive geometry: the same routing decomposition is retained, but the learning penalty is controlled by $d_{\mathrm{eff}}(\Omega)$ rather than the ambient dimension.

K.1 Estimated hyper-covariance: perturbation reduction and the remaining gap

The actual Hyper-Ridge PersonalEvo router uses an estimate $\widehat{\Omega}_t$. We therefore separate two issues. First, if $\widehat{\Omega}_t$ is close to Ω in relative spectral norm, the oracle proof above is stable. Second, proving that an online archive estimator achieves this closeness is a separate statistical problem; we state it as Conjecture K.1 rather than claiming it as proven.

The relative covariance error at time t is

$$\rho_t=\left\|\Omega^{-1/2}\left(\widehat{\Omega}_t-\Omega\right)\Omega^{-1/2}\right\|_{\mathrm{op}}. \tag{1124}$$

The uniform relative covariance error through horizon T is

$$\rho_T = \max_{1 \le t \le T} \rho_t. \tag{1125}$$

The estimated-Ω design matrix is

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{1126}$$

The oracle design matrix is

$$V_{a,t}^{\Omega} = X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}. \tag{1127}$$

We define the additional error of using $\widehat{\Omega}$ as the excess cumulative UCB-width budget relative to the known-Ω analysis. On any trajectory, set

$$\mathrm{Err}_T(\widehat{\Omega}, \Omega) = 2\alpha_T \sum_{t=1}^{T} \left[\min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\widehat{\Omega}}\right)^{-1} x_t(a_t)}\right\} - \min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}\right\} \right]_+ . \tag{1128}$$

Here the positive part is

$$[z]_+ = \max\{z, 0\}. \tag{1129}$$

Lemma 48 (Relative spectral error implies Loewner control)**.** *Assume*

$$\rho_T \le \varepsilon \tag{1130}$$

for some

$$0 \le \varepsilon < 1. \tag{1131}$$

Then, for every t,

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \tag{1132}$$

Moreover,

$$\frac{1}{1+\varepsilon}\Omega^{-1} \preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \tag{1133}$$

Finally, for every candidate and round,

$$V_{a,t}^{\widehat{\Omega}} \succeq \frac{1}{1+\varepsilon} V_{a,t}^{\Omega}. \tag{1134}$$

Proof. The definition of ρ_T implies

$$-\varepsilon I \preceq \Omega^{-1/2}\left(\widehat{\Omega}_t - \Omega\right)\Omega^{-1/2} \preceq \varepsilon I. \tag{1135}$$

Adding I gives

$$(1-\varepsilon)I \preceq \Omega^{-1/2}\widehat{\Omega}_t\Omega^{-1/2} \preceq (1+\varepsilon)I. \tag{1136}$$

Congruence by $\Omega^{1/2}$ gives

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \tag{1137}$$

Inverting reverses the Loewner order:

$$\frac{1}{1+\varepsilon}\Omega^{-1} \preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \tag{1138}$$

Therefore

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1} \tag{1139}$$

$$\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon}\Omega^{-1} \tag{1140}$$

$$\succeq \frac{1}{1+\varepsilon}\left(X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}\right) \tag{1141}$$

$$= \frac{1}{1+\varepsilon} V_{a,t}^{\Omega}. \tag{1142}$$

□

Lemma 49 (Width and effective-dimension stability)**.** *Under the assumptions of Lemma 48, every feature vector x satisfies*

$$x^\top \left(V_{a,t}^{\widehat{\Omega}}\right)^{-1} x \le (1+\varepsilon) x^\top \left(V_{a,t}^{\Omega}\right)^{-1} x. \tag{1143}$$

For any fixed design matrix X, the estimated-geometry effective dimension satisfies

$$d_{\mathrm{eff}}(\widehat{\Omega}_t; X) \le (1+\varepsilon) d_{\mathrm{eff}}(\Omega; X). \tag{1144}$$

Proof. From equation 1134, inversion gives

$$\left(V_{a,t}^{\widehat{\Omega}}\right)^{-1} \preceq (1+\varepsilon) \left(V_{a,t}^{\Omega}\right)^{-1}. \tag{1145}$$

Taking the quadratic form in direction x gives

$$x^\top \left(V_{a,t}^{\widehat{\Omega}}\right)^{-1} x \le (1+\varepsilon) x^\top \left(V_{a,t}^{\Omega}\right)^{-1} x. \tag{1146}$$

For the effective dimension, write

$$A = X^\top X. \tag{1147}$$

Then

$$\begin{aligned} d_{\mathrm{eff}}(\widehat{\Omega}_t; X) &= \operatorname{tr}\left[A\left(A + \lambda \widehat{\Omega}_t^{-1}\right)^{-1}\right] && (1148)\\ &\le (1+\varepsilon) \operatorname{tr}\left[A\left(A + \lambda \Omega^{-1}\right)^{-1}\right] && (1149)\\ &= (1+\varepsilon) d_{\mathrm{eff}}(\Omega; X). && (1150) \end{aligned}$$

□

The previous lemmas give a clean sufficient condition for the estimated-Ω method to inherit the oracle rate.

Proposition 50 (Perturbation reduction for estimated hyper-ridge)**.** *Assume the known-Ω conditions of Proposition 47 hold for the oracle geometry. Assume the estimated covariance obeys*

$$\rho_T \le \varepsilon < 1. \tag{1151}$$

Then the excess-width error satisfies

$$\mathrm{Err}_T(\widehat{\Omega}, \Omega) \le 2\alpha_T \left(\sqrt{1+\varepsilon} - 1\right) \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)}. \tag{1152}$$

Consequently, the estimated-Ω exploration cost is bounded by

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_T \sqrt{1+\varepsilon} \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)} \tag{1153}$$

on the same confidence event, up to the additional failure probability required for confidence under $\widehat{\Omega}_t$.

Proof. Lemma 49 implies, for every selected round,

$$\sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\widehat{\Omega}}\right)^{-1} x_t(a_t)} \le \sqrt{1+\varepsilon} \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}. \tag{1154}$$

The map $z \mapsto \min\{1, z\}$ is one-Lipschitz and monotone, so

$$\begin{aligned} &\left[\min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\widehat{\Omega}}\right)^{-1} x_t(a_t)}\right\} - \min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}\right\}\right]_+ && (1155)\\ &\le \left(\sqrt{1+\varepsilon} - 1\right) \min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}\right\}. && (1156) \end{aligned}$$

Summing and using the oracle width-sum bound from the proof of Proposition 47 gives

$$\mathrm{Err}_T(\widehat{\Omega}, \Omega) \le 2\alpha_T \left(\sqrt{1+\varepsilon} - 1\right) \sum_{t=1}^{T} \min\left\{1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}\right\} \tag{1157}$$

$$\le 2\alpha_T \left(\sqrt{1+\varepsilon} - 1\right) \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)}. \tag{1158}$$

Adding this excess to the oracle width budget proves equation 1153. □

A numerical perturbation example clarifies the scale. If

$$\varepsilon = 0.21, \tag{1159}$$

then

$$\sqrt{1+\varepsilon} = 1.10. \tag{1160}$$

Thus the estimated covariance inflates the oracle exploration budget by about ten percent:

$$B_T^{\widehat{\Omega}} \le 1.10\, B_T^{\Omega} \tag{1161}$$

on the perturbation event. The matrix analysis above proves this stability statement, but it does not prove that the online archive estimator achieves $\varepsilon = 0.21$, or any particular ε, in the current codebase.

[CONJECTURE][Sharp-rate estimated-covariance exploration cost] The calibration-split router of Theorem 57 (Appendix L), or a fully online variant, achieves the oracle leading term exactly:

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le C\sqrt{K_{\max}\, d_{\mathrm{eff}}(\Omega)\, T\, \log(1+T)} + o(\sqrt{T}), \tag{1162}$$

i.e., without the residual $\sqrt{d \log T}$ factor of the proven bound. The obstruction is that the explicit frequentist confidence multiplier scales as $\sigma\sqrt{d\log T}$; a Bayesian or Thompson-sampling analysis, or a sharper data-dependent radius, is the natural route.

Conjecture K.1 asks whether an estimated-covariance router can attain the oracle leading rate without the residual confidence factor. The repository implements a periodically re-estimated hyper-covariance router, but the diagnostic results in Section 6.4 do not establish this claim. They instead show that known covariance helps in the planted low-rank regime while the current online estimator does not recover that benefit.

K.2 Map of hyper-ridge claims to proofs and open steps

Table 33 summarizes which parts of the Hyper-Ridge PersonalEvo theory are proved in this section and which are conditional. The inherited four-term decomposition remains the base accounting theorem; the new analysis only sharpens the exploration-cost term $\bar{L}_T$.

Claim	Mathematical object	Status
Exact PersonalEvo accounting	$V_{\mathrm{static}} + \Delta_{\mathrm{route}} + \bar{G}_T - \bar{L}_T$	inherited from Theorem 2
Oracle hyper-ridge confidence	$V_{a,t}^{\Omega} = X_{a,t}^\top X_{a,t} + \lambda\Omega^{-1}$	Lemma 45
Effective-dimension potential	$\sum_s \min\{1, q_s\}$	Lemma 46
Known-Ω exploration cost	$\sqrt{K_{\max} d_{\mathrm{eff}}(\Omega) T \log T}$	Proposition 47
Estimated-Ω perturbation stability	$\mathrm{Err}_T(\widehat{\Omega}, \Omega)$ under $\rho_T \le \varepsilon$	Proposition 50
Online covariance estimation rate	$\rho_T \le \varepsilon_T$	Conjecture K.1
Empirical validation of hyper-ridge gains	oracle, estimated, diagonal, and base routers	diagnostic only; full test remains open

Table 33: Theory map for the new Hyper-Ridge PersonalEvo mechanism. The oracle effective-dimension theorem is proved; the estimated-covariance theorem reduces to the explicitly named error-control conjecture.

L Estimated-Covariance Control: the Calibration-Split Theorem

This appendix resolves the covariance-estimation obligation for the estimated-covariance router. For a modified procedure (a pre-routing calibration phase on fresh candidates whose estimate $\widehat{\Omega}$ is then frozen during routing) we prove that the relative covariance error ρ_T vanishes and that the exploration cost satisfies an effective-dimension bound with $\mathrm{Err}_T = o(\sqrt{T})$. The exact leading-term rate of Proposition 8 is recovered under an additional calibrated-confidence condition stated below; without it the proven leading term carries an extra $\sqrt{d \log T}$ factor. The proof was developed through an adversarial prover–referee process; Assumption 19 records the environmental diversity hypothesis it relies on.

Assumption 19 (Exogenous uniform-routing calibration diversity)**.** *Let $\Omega \succ 0$. In the calibration phase, the algorithm uses N calibration candidates indexed by $i \in \{1, \ldots, N\}$, and it collects m calibration rewards for each calibration candidate. The calibration feature at calibration pull j of calibration candidate i is $x_{i,j} \in \mathbb{R}^d$, and we write*

$$z_{i,j} = \Omega^{1/2} x_{i,j}.$$

The calibration features are generated by the environment under an exogenous uniform-routing calibration policy before the reward noises of the corresponding calibration block are revealed. The following conditions hold.

First, the feature arrays

$$X_i = \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}, \qquad i = 1, \ldots, N,$$

are independent across i. Within each calibration block, the vectors $z_{i,1}, \ldots, z_{i,m}$ are independent. For every i, j,

$$\mathbb{E}\big[z_{i,j} z_{i,j}^\top\big] \succeq \kappa_{\mathrm{cal}} I_d, \qquad \|z_{i,j}\|_2^2 \le L_{\mathrm{cal}}^2 \quad \textit{almost surely,}$$

for deterministic constants $\kappa_{\mathrm{cal}} > 0$ and $L_{\mathrm{cal}} < \infty$. The full calibration feature array is independent of the calibration candidate effects.

Second, the calibration candidate effects $\theta_1, \ldots, \theta_N$ are independent and identically distributed and satisfy

$$\mathbb{E}\theta_i = 0, \qquad \mathbb{E}\theta_i \theta_i^\top = \Omega, \qquad \|\theta_i\|_{\Omega^{-1}} \le S_\Omega \quad \textit{almost surely.}$$

Third, conditional on the full calibration feature array and on $\theta_1, \ldots, \theta_N$, the calibration noise blocks are independent across i. Within each block there is a filtration

$$\mathcal{H}_{i,0} \subseteq \mathcal{H}_{i,1} \subseteq \cdots \subseteq \mathcal{H}_{i,m}$$

such that the full calibration feature array and θ_i are $\mathcal{H}_{i,0}$-measurable, $\xi_{i,j}$ is $\mathcal{H}_{i,j}$-measurable, and, for every $\eta \in \mathbb{R}$,

$$\mathbb{E}[\exp(\eta \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \quad \textit{almost surely.}$$

Lemma 51 (Net reduction for symmetric quadratic forms)**.** *Let $A \in \mathbb{R}^{d \times d}$ be symmetric. Let $\mathcal{N} \subset \mathbb{S}^{d-1}$ be a $1/4$-net of the Euclidean unit sphere. Then*

$$\|A\|_{\mathrm{op}} \le 2 \sup_{v \in \mathcal{N}} |v^\top A v|.$$

Moreover, there exists such a $1/4$-net with cardinality at most 9^d.

Proof. Define

$$M = \sup_{v \in \mathcal{N}} |v^\top A v|.$$

Since A is symmetric,

$$\|A\|_{\mathrm{op}} = \sup_{u \in \mathbb{S}^{d-1}} |u^\top A u|.$$

Fix $u \in \mathbb{S}^{d-1}$. Choose $v \in \mathcal{N}$ satisfying

$$\|u - v\|_2 \le \frac{1}{4}.$$

Then

$$u^\top A u - v^\top A v = (u - v)^\top A u + v^\top A(u - v).$$

Using $\|u\|_2 = \|v\|_2 = 1$, we obtain

$$|u^\top A u - v^\top A v| \le \|u - v\|_2 \|A\|_{\mathrm{op}} \|u\|_2 + \|v\|_2 \|A\|_{\mathrm{op}} \|u - v\|_2 \le \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Therefore

$$|u^\top A u| \le M + \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Taking the supremum over $u \in \mathbb{S}^{d-1}$ yields

$$\|A\|_{\mathrm{op}} \le M + \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Rearranging proves

$$\|A\|_{\mathrm{op}} \le 2M.$$

For the covering-number claim, let $\mathcal{N}$ be a maximal subset of $\mathbb{S}^{d-1}$ such that distinct points of $\mathcal{N}$ have Euclidean distance strictly larger than $1/4$. Maximality implies that $\mathcal{N}$ is a $1/4$-net, because any point of the sphere whose distance from every point of $\mathcal{N}$ were larger than $1/4$ could be added to $\mathcal{N}$. The closed Euclidean balls

$$B(v, 1/8) = \{w \in \mathbb{R}^d : \|w - v\|_2 \le 1/8\}, \qquad v \in \mathcal{N},$$

are pairwise disjoint. Each is contained in $B(0, 9/8)$, since $\|v\|_2 = 1$. Comparing d-dimensional Euclidean volumes gives

$$|\mathcal{N}| \left(\frac{1}{8}\right)^d \le \left(\frac{9}{8}\right)^d.$$

Thus $|\mathcal{N}| \le 9^d$. □

Lemma 52 (Independent sub-Gaussian sample covariance)**.** *There is a universal numerical constant $C_{\mathrm{cov}} > 0$ with the following property. Let $Z_1, \ldots, Z_N \in \mathbb{R}^d$ be independent mean-zero random vectors such that, for every i, every $u \in \mathbb{S}^{d-1}$, and every $s \in \mathbb{R}$,*

$$\mathbb{E} \exp(s u^\top Z_i) \le \exp\left(\frac{s^2 K^2}{2}\right).$$

Let

$$\Sigma_i = \mathbb{E}[Z_i Z_i^\top].$$

Then, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i\right) \right\|_{\mathrm{op}} \le C_{\mathrm{cov}} K^2 \left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N} \right].$$

The same conclusion holds conditionally on any sigma-field with respect to which the displayed independence and moment-generating-function hypotheses hold with the same deterministic value of K.

Proof. Fix $u \in \mathbb{S}^{d-1}$. The displayed moment-generating-function bound implies

$$\Pr(u^\top Z_i \ge t) \le \inf_{s > 0} \exp\left(-st + \frac{s^2 K^2}{2}\right) = \exp\left(-\frac{t^2}{2K^2}\right) \qquad t \ge 0.$$

Applying the same argument to $-u$ gives

$$\Pr(|u^\top Z_i| \ge t) \le 2 \exp\left(-\frac{t^2}{2K^2}\right) \qquad t \ge 0.$$

Equivalently, $u^\top Z_i$ is sub-Gaussian with sub-Gaussian norm bounded by a universal multiple of K. Proposition 2.7.1 of Vershynin, *High-Dimensional Probability*, Cambridge University Press, 2018, then implies that

$$Y_i = (u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2$$

is sub-exponential with sub-exponential norm bounded by a universal multiple of K^2. Theorem 2.8.1 of the same reference, Bernstein's inequality for sums of independent centered sub-exponential random variables, gives a universal constant $c_1 > 0$ such that, for every $t > 0$,

$$\Pr\left(\left| \frac{1}{N} \sum_{i=1}^{N} \left[(u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2 \right] \right| > c_1 K^2 \left[\sqrt{\frac{t}{N}} + \frac{t}{N} \right] \right) \le 2e^{-t}.$$

Let $\mathcal{N}$ be a $1/4$-net of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \le 9^d$, whose existence follows from Lemma 51. Applying the preceding scalar inequality to every $u \in \mathcal{N}$, with

$$t = d \log 9 + \log(2/\delta),$$

and taking a union bound gives an event of probability at least $1 - \delta$ on which

$$\sup_{u \in \mathcal{N}} \left| u^\top \left[\frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i \right) \right] u \right| \le c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right].$$

Applying Lemma 51 to the symmetric matrix inside the brackets yields

$$\left\| \frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i \right) \right\|_{\mathrm{op}} \le 2 c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right].$$

Since $\log 9 < 3$, increasing the universal constant proves the stated bound.

For the conditional version, fix a regular conditional distribution given the conditioning sigma-field. Under that conditional distribution, the same proof applies with the same deterministic K and the same constants. Integrating the conditional failure probability over the conditioning sigma-field gives the unconditional conditional-form statement. □

Lemma 53 (Calibration Gram lower tail under exogenous diversity)**.** *Assume Assumption 19. For each calibration candidate define*

$$G_i = X_i^\top X_i, \qquad H_i = \Omega^{1/2} G_i \Omega^{1/2} = \sum_{j=1}^{m} z_{i,j} z_{i,j}^\top.$$

For every $\delta_{\mathrm{gram}} \in (0, 1)$*, if*

$$m \ge \frac{8 L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}} \log\left(\frac{dN}{\delta_{\mathrm{gram}}} \right),$$

then, with probability at least $1 - \delta_{\mathrm{gram}}$*,*

$$H_i \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \qquad \textit{for every } i = 1, \ldots, N.$$

On the same event every G_i *is invertible and*

$$\left\| \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} \right\|_{\mathrm{op}} \le \frac{2}{\kappa_{\mathrm{cal}} m} \qquad \textit{for every } i = 1, \ldots, N.$$

Proof. Fix i. Define

$$A_{i,j} = z_{i,j} z_{i,j}^\top.$$

By Assumption 19, the matrices $A_{i,1}, \ldots, A_{i,m}$ are independent positive-semidefinite random matrices and satisfy

$$0 \preceq A_{i,j} \preceq L_{\mathrm{cal}}^2 I_d$$

almost surely. Moreover,

$$\lambda_{\min}\left(\sum_{j=1}^{m} \mathbb{E} A_{i,j} \right) \ge \kappa_{\mathrm{cal}} m.$$

Tropp's matrix Chernoff lower-tail inequality for independent positive-semidefinite summands bounded above by RI_d states that, with

$$\mu_{\min} = \lambda_{\min}\left(\sum_{j=1}^{m} \mathbb{E}A_{i,j}\right),$$

for every $\varepsilon \in [0, 1]$,

$$\Pr\left(\lambda_{\min}\left(\sum_{j=1}^{m} A_{i,j}\right) \le (1-\varepsilon)\mu_{\min}\right) \le d\exp\left(-\frac{\varepsilon^2 \mu_{\min}}{2R}\right).$$

This is Theorem 5.1.1 of Tropp, *An Introduction to Matrix Concentration Inequalities*, Foundations and Trends in Machine Learning, 2015, applied with $R = L_{\mathrm{cal}}^2$. Taking $\varepsilon = 1/2$ gives

$$\Pr\left(H_i \not\succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d\right) \le d\exp\left(-\frac{\kappa_{\mathrm{cal}} m}{8L_{\mathrm{cal}}^2}\right).$$

The assumed lower bound on m implies

$$d\exp\left(-\frac{\kappa_{\mathrm{cal}} m}{8L_{\mathrm{cal}}^2}\right) \le \frac{\delta_{\mathrm{gram}}}{N}.$$

A union bound over $i = 1, \dots, N$ proves the simultaneous Gram event.

On this event,

$$\Omega^{1/2} G_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d.$$

Thus G_i is invertible. Inverting the Loewner inequality gives

$$(\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d.$$

Since

$$(\Omega^{1/2} G_i \Omega^{1/2})^{-1} = \Omega^{-1/2} G_i^{-1} \Omega^{-1/2},$$

the asserted operator-norm bound follows. □

Lemma 54 (Conditional calibration-block covariance control)**.** *Assume Assumption 19. Condition on a realization of the full calibration feature array for which*

$$H_i = \Omega^{1/2} X_i^\top X_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \qquad \text{for every } i = 1, \dots, N.$$

For each i, define

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i, \qquad y_i = X_i \theta_i + \xi_i,$$

and define

$$Z_i = \Omega^{-1/2} \widetilde{\theta}_i, \qquad \tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

Then, conditional on the feature array, $Z_1, \dots, Z_N$ are independent mean-zero random vectors. Moreover,

$$\mathbb{E}[Z_i Z_i^\top \mid X_1, \dots, X_N] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m,$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(s u^\top Z_i) \mid X_1, \dots, X_N\big] \le \exp\left(\frac{s^2 K_m^2}{2}\right).$$

Proof. The least-squares identity gives

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top (X_i \theta_i + \xi_i) = \theta_i + G_i^{-1} X_i^\top \xi_i.$$

Set

$$U_i = \Omega^{-1/2} \theta_i, \qquad E_i = \Omega^{-1/2} G_i^{-1} X_i^\top \xi_i.$$

Then

$$Z_i = U_i + E_i.$$

The random-effects assumptions imply, conditional on the full calibration feature array,

$$\mathbb{E}[U_i \mid X_1, \ldots, X_N] = 0, \qquad \mathbb{E}[U_i U_i^\top \mid X_1, \ldots, X_N] = I_d, \qquad \|U_i\|_2 \le S_\Omega \quad \text{almost surely.}$$

The conditional sub-Gaussian noise assumption implies

$$\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] = 0.$$

Indeed, for $h > 0$, Jensen's inequality gives

$$\exp(h\,\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}]) \le \mathbb{E}[\exp(h\xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{h^2\sigma^2}{2}\right),$$

so $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] \le h\sigma^2/2$. Letting $h \downarrow 0$ gives the upper bound 0. Applying the same argument to $-\xi_{i,j}$ gives the lower bound 0. Therefore the conditional mean is zero. By iterated conditioning,

$$\mathbb{E}[\xi_i \mid X_1, \ldots, X_N, \theta_i] = 0.$$

Consequently,

$$\mathbb{E}[E_i \mid X_1, \ldots, X_N, \theta_i] = 0,$$

and the conditional cross terms satisfy

$$\mathbb{E}[U_i E_i^\top \mid X_1, \ldots, X_N] = 0, \qquad \mathbb{E}[E_i U_i^\top \mid X_1, \ldots, X_N] = 0.$$

Thus

$$\mathbb{E}[Z_i Z_i^\top \mid X_1, \ldots, X_N] = I_d + R_i, \qquad R_i = \mathbb{E}[E_i E_i^\top \mid X_1, \ldots, X_N] \succeq 0.$$

We bound R_i. Fix $v \in \mathbb{S}^{d-1}$, and set

$$b_i = X_i G_i^{-1} \Omega^{-1/2} v \in \mathbb{R}^m.$$

Conditional on the feature array, b_i is deterministic. Then

$$v^\top R_i v = \mathbb{E}\left[(b_i^\top \xi_i)^2 \mid X_1, \ldots, X_N\right].$$

For $j < k$, the random variable $\xi_{i,j}$ is $\mathcal{H}_{i,k-1}$-measurable. Hence

$$\mathbb{E}[\xi_{i,j}\xi_{i,k} \mid X_1, \ldots, X_N] = \mathbb{E}[\xi_{i,j}\,\mathbb{E}[\xi_{i,k} \mid \mathcal{H}_{i,k-1}] \mid X_1, \ldots, X_N] = 0.$$

The conditional sub-Gaussian inequality also implies

$$\mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] \le \sigma^2.$$

This follows by integrating the tail bound obtained from the conditional moment-generating-function inequality:

$$\Pr(|\xi_{i,j}| \ge t \mid \mathcal{H}_{i,j-1}) \le 2\exp\left(-\frac{t^2}{2\sigma^2}\right),$$

and then using

$$\mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] = \int_0^\infty \Pr(\xi_{i,j}^2 \ge r \mid \mathcal{H}_{i,j-1})\,dr \le \sigma^2$$

after replacing σ by the variance proxy in the displayed sub-Gaussian assumption; equivalently, the same variance bound follows by differentiating the conditional moment-generating function at zero. Therefore

$$v^\top R_i v \le \sigma^2 \|b_i\|_2^2.$$

Moreover,

$$\|b_i\|_2^2 = v^\top \Omega^{-1/2} G_i^{-1} X_i^\top X_i G_i^{-1} \Omega^{-1/2} v = v^\top \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} v.$$

On the conditioned Gram event,

$$\Omega^{-1/2} G_i^{-1} \Omega^{-1/2} = (\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d.$$

Thus

$$v^\top R_i v \le \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m} = \tau_m.$$

Taking the supremum over $v \in \mathbb{S}^{d-1}$ gives

$$\|R_i\|_{\mathrm{op}} \le \tau_m.$$

It remains to prove the one-dimensional moment-generating-function bound. Fix $u \in \mathbb{S}^{d-1}$. Since

$$\mathbb{E}[u^\top U_i \mid X_1, \dots, X_N] = 0$$

and

$$|u^\top U_i| \le \|U_i\|_2 \le S_\Omega \quad \text{almost surely,}$$

Hoeffding's lemma gives

$$\mathbb{E}\big[\exp(su^\top U_i) \mid X_1, \dots, X_N\big] \le \exp\left(\frac{s^2 S_\Omega^2}{2}\right).$$

For the noise part, set

$$b_i = X_i G_i^{-1} \Omega^{-1/2} u.$$

Conditional on the full feature array, b_i is deterministic. Iterating the conditional sub-Gaussian inequality over $j = m, m-1, \dots, 1$ gives

$$\mathbb{E}\big[\exp(sb_i^\top \xi_i) \mid X_1, \dots, X_N, \theta_i\big] \le \exp\left(\frac{s^2 \sigma^2 \|b_i\|_2^2}{2}\right).$$

The Gram-event calculation above gives

$$\|b_i\|_2^2 \le \frac{2}{\kappa_{\mathrm{cal}} m}.$$

Therefore

$$\mathbb{E}\big[\exp(su^\top E_i) \mid X_1, \dots, X_N, \theta_i\big] \le \exp\left(\frac{s^2 \tau_m}{2}\right).$$

Since U_i is a function of θ_i,

$$\begin{aligned}
\mathbb{E}\big[\exp(su^\top Z_i) \mid X_1, \dots, X_N\big] &= \mathbb{E}\big[\exp(su^\top U_i)\mathbb{E}\big[\exp(su^\top E_i) \mid X_1, \dots, X_N, \theta_i\big] \mid X_1, \dots, X_N\big] \\
&\le \exp\left(\frac{s^2 \tau_m}{2}\right) \mathbb{E}\big[\exp(su^\top U_i) \mid X_1, \dots, X_N\big] \\
&\le \exp\left(\frac{s^2 (S_\Omega^2 + \tau_m)}{2}\right).
\end{aligned}$$

Finally, conditional on the feature array, the candidate effects are independent across i, and the noise blocks are independent across i conditional on the feature array and on the candidate effects. Since Z_i is a measurable function of (X_i, θ_i, ξ_i), it follows that $Z_1, \dots, Z_N$ are independent conditional on the full feature array. ☐

Lemma 55 (Relative covariance error for the calibration estimator)**.** *Assume Assumption 19. Define*

$$\widehat{\Omega} = \frac{1}{N} \sum_{i=1}^{N} \widetilde{\theta}_i \widetilde{\theta}_i^\top + \gamma I_d, \qquad \gamma > 0,$$

where $\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i$ on the event that G_i is invertible, and $\widetilde{\theta}_i = 0$ otherwise. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

For $\delta_{\mathrm{gram}}, \delta_{\mathrm{cov}} \in (0, 1)$, suppose

$$m \ge \frac{8 L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}} \log\left(\frac{dN}{\delta_{\mathrm{gram}}}\right).$$

Define

$$\varepsilon_{\mathrm{cov}}(N,m,\gamma,\delta_{\mathrm{cov}}) = C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d+\log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d+\log(2/\delta_{\mathrm{cov}})}{N}\right] + \tau_m + \gamma\|\Omega^{-1}\|_{\mathrm{op}}.$$

Then, with probability at least $1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}$,

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon_{\mathrm{cov}}(N,m,\gamma,\delta_{\mathrm{cov}}).$$

Proof. Let

$$\mathcal{E}_{\mathrm{gram}} = \left\{H_i \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \text{ for every } i=1,\dots,N\right\}.$$

By Lemma 53,

$$\Pr(\mathcal{E}_{\mathrm{gram}}) \ge 1-\delta_{\mathrm{gram}}.$$

Condition on a realization of the full calibration feature array in $\mathcal{E}_{\mathrm{gram}}$. By Lemma 54, the vectors

$$Z_i = \Omega^{-1/2}\widetilde{\theta}_i$$

are conditionally independent and satisfy

$$\mathbb{E}[Z_i Z_i^\top \mid X_1,\dots,X_N] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m,$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(s u^\top Z_i) \mid X_1,\dots,X_N\big] \le \exp\left(\frac{s^2 K_m^2}{2}\right).$$

Applying the conditional form of Lemma 52 gives, conditional on this feature realization, an event of conditional probability at least $1-\delta_{\mathrm{cov}}$ on which

$$\left\|\frac{1}{N}\sum_{i=1}^N \left[Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_1,\dots,X_N)\right]\right\|_{\mathrm{op}} \le C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d+\log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d+\log(2/\delta_{\mathrm{cov}})}{N}\right].$$

On the intersection of $\mathcal{E}_{\mathrm{gram}}$ and this conditional covariance event,

$$\begin{aligned}\left\|\frac{1}{N}\sum_{i=1}^N Z_i Z_i^\top - I_d\right\|_{\mathrm{op}} &\le \left\|\frac{1}{N}\sum_{i=1}^N \left[Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_1,\dots,X_N)\right]\right\|_{\mathrm{op}} + \left\|\frac{1}{N}\sum_{i=1}^N R_i\right\|_{\mathrm{op}} \\ &\le C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d+\log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d+\log(2/\delta_{\mathrm{cov}})}{N}\right] + \tau_m.\end{aligned}$$

The last inequality uses

$$\left\|\frac{1}{N}\sum_{i=1}^N R_i\right\|_{\mathrm{op}} \le \frac{1}{N}\sum_{i=1}^N \|R_i\|_{\mathrm{op}} \le \tau_m.$$

Since

$$\Omega^{-1/2}\widehat{\Omega}\Omega^{-1/2} = \frac{1}{N}\sum_{i=1}^N Z_i Z_i^\top + \gamma\Omega^{-1},$$

we have

$$\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2} = \left(\frac{1}{N}\sum_{i=1}^N Z_i Z_i^\top - I_d\right) + \gamma\Omega^{-1}.$$

The triangle inequality gives

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon_{\mathrm{cov}}(N,m,\gamma,\delta_{\mathrm{cov}}).$$

The unconditional probability of the intersection of the Gram event and the conditional covariance event is at least $1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}$ by the law of total probability and a union bound. $\square$

Lemma 56 (Confidence under a frozen estimated covariance)**.** *Assume the post-calibration routing model of Assumption 17. Assume the post-calibration routing archive is fixed before routing begins and contains at most $K_{\max}$ candidates. Let $\widehat{\Omega} \succ 0$ be fixed before routing begins and independent of the post-calibration routing noises. Suppose*

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon \qquad \textit{for some } 0 \le \varepsilon < 1.$$

For candidate a, define

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^{\top}X_{a,t} + \lambda\widehat{\Omega}^{-1}, \qquad \widehat{\theta}_{a,t}^{\widehat{\Omega}} = (V_{a,t}^{\widehat{\Omega}})^{-1}X_{a,t}^{\top}y_{a,t}.$$

Define

$$\alpha_{\widehat{\Omega},T}(\varepsilon,\delta_{\mathrm{route}}) = \max\left\{1,\, \sigma\sqrt{d\log\left(1+\frac{(1+\varepsilon)TL_{\Omega}^2}{\lambda}\right)+2\log\left(\frac{K_{\max}}{\delta_{\mathrm{route}}}\right)}+\sqrt{\frac{\lambda}{1-\varepsilon}}\,S_{\Omega}\right\}.$$

Then, conditional on $\widehat{\Omega}$, with probability at least $1-\delta_{\mathrm{route}}$, uniformly over all routing candidates a, all $t \le T$, and all predictable test features x,

$$\left|x^{\top}(\widehat{\theta}_{a,t}^{\widehat{\Omega}}-\theta_a)\right| \le \alpha_{\widehat{\Omega},T}(\varepsilon,\delta_{\mathrm{route}})\sqrt{x^{\top}(V_{a,t}^{\widehat{\Omega}})^{-1}x}.$$

Proof. Fix a routing candidate a. Conditional on $\widehat{\Omega}$, set

$$A = \lambda\widehat{\Omega}^{-1}.$$

Then A is deterministic and positive definite. Define

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s, \qquad V_{a,t}^{\widehat{\Omega}} = A + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^{\top}.$$

Using the reward model,

$$\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a = (V_{a,t}^{\widehat{\Omega}})^{-1}S_{a,t} - (V_{a,t}^{\widehat{\Omega}})^{-1}A\theta_a.$$

Hence

$$\|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} \le \|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} + \|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}.$$

Because $V_{a,t}^{\widehat{\Omega}} \succeq A$,

$$\|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}^2 \le \theta_a^{\top}A\theta_a = \lambda\theta_a^{\top}\widehat{\Omega}^{-1}\theta_a.$$

The relative error assumption implies

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega} \preceq (1+\varepsilon)\Omega,$$

and therefore

$$\widehat{\Omega}^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}.$$

Using $\|\theta_a\|_{\Omega^{-1}} \le S_{\Omega}$, we get

$$\|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \le \sqrt{\frac{\lambda}{1-\varepsilon}}\,S_{\Omega}.$$

It remains to control the martingale term. For fixed $q \in \mathbb{R}^d$, define

$$M_t(q) = \exp\left(q^{\top}S_{a,t} - \frac{\sigma^2}{2}q^{\top}\left(V_{a,t}^{\widehat{\Omega}} - A\right)q\right).$$

The features are predictable and the noises are conditionally σ-sub-Gaussian. If round t does not select candidate a, then $S_{a,t}$ and $V_{a,t}^{\widehat{\Omega}}$ do not change. If round t selects a, then conditional on the past the multiplicative increment of $M_t(q)$ has conditional expectation at most one. Therefore $(M_t(q))_{t\le T}$ is a nonnegative supermartingale with initial value one.

Let $Q \sim N(0, A/\sigma^2)$ in precision form, that is, Q has density

$$\varphi_A(q) = \frac{\det(A)^{1/2}}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{1}{2\sigma^2} q^\top A q\right).$$

Define

$$\mathcal{M}_t = \int_{\mathbb{R}^d} M_t(q)\, \varphi_A(q)\, dq.$$

Tonelli's theorem and the supermartingale property of $M_t(q)$ imply that $(\mathcal{M}_t)_{t \le T}$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Completing the square in the Gaussian integral gives

$$\mathcal{M}_t = \frac{\det(A)^{1/2}}{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}} \exp\left(\frac{\|S_{a,t}\|^2_{(V_{a,t}^{\widehat{\Omega}})^{-1}}}{2\sigma^2}\right).$$

Ville's inequality gives

$$\Pr\left(\sup_{t \le T} \mathcal{M}_t \ge \frac{1}{\delta_a}\right) \le \delta_a.$$

Thus, with probability at least $1 - \delta_a$, uniformly over $t \le T$,

$$\|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \le \sigma \sqrt{2 \log\left(\frac{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}}{\det(A)^{1/2}\delta_a}\right)}.$$

We next bound the determinant ratio. Since $A = \lambda \widehat{\Omega}^{-1}$,

$$\frac{\det(V_{a,t}^{\widehat{\Omega}})}{\det(A)} = \det\left(I + \lambda^{-1} \widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t} \widehat{\Omega}^{1/2}\right).$$

The relation $\widehat{\Omega} \preceq (1+\varepsilon)\Omega$ implies

$$\operatorname{tr}\left(\lambda^{-1} \widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t} \widehat{\Omega}^{1/2}\right) = \lambda^{-1} \operatorname{tr}(X_{a,t} \widehat{\Omega} X_{a,t}^\top) \le \frac{1+\varepsilon}{\lambda} \sum_{s \in \mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a).$$

By the feature-radius assumption,

$$\sum_{s \in \mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a) \le T L_\Omega^2.$$

For any positive-semidefinite $d \times d$ matrix M with eigenvalues $\nu_1, \dots, \nu_d$,

$$\det(I + M) = \prod_{k=1}^{d} (1 + \nu_k) \le \left(1 + \sum_{k=1}^{d} \nu_k\right)^d = (1 + \operatorname{tr} M)^d.$$

Therefore

$$2 \log\left(\frac{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}}{\det(A)^{1/2}\delta_a}\right) \le d \log\left(1 + \frac{(1+\varepsilon) T L_\Omega^2}{\lambda}\right) + 2\log(1/\delta_a).$$

Taking $\delta_a = \delta_{\mathrm{route}}/K_{\max}$ and applying a union bound over the fixed routing archive gives, with probability at least $1 - \delta_{\mathrm{route}}$, the simultaneous bound

$$\|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} \le \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\mathrm{route}})$$

for every candidate a and every $t \le T$. Finally, for every predictable test feature x, Cauchy–Schwarz in the $V_{a,t}^{\widehat{\Omega}}$-inner product gives

$$\left| x^\top (\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a) \right| \le \|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} \sqrt{x^\top (V_{a,t}^{\widehat{\Omega}})^{-1} x}.$$

Substituting the preceding simultaneous parameter bound proves the claim. □

Theorem 57 (Err-control for calibration-split frozen-covariance Hyper-Ridge)**.** *Fix a post-calibration routing horizon T. Consider the following modified Hyper-Ridge PersonalEvo procedure.*

Before routing begins, the algorithm runs a calibration phase on $N = N_T$ calibration candidates that are not used in the subsequent routing archive. For each calibration candidate i, the algorithm collects $m = m_T$ calibration rewards under the exogenous uniform-routing calibration policy:

$$y_{i,j} = x_{i,j}^\top \theta_i + \xi_{i,j}, \qquad j = 1, \dots, m.$$

Let

$$X_i = \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}, \qquad G_i = X_i^\top X_i.$$

On the event that G_i is invertible, set

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i;$$

otherwise set $\widetilde{\theta}_i = 0$. For a deterministic ridge floor $\gamma = \gamma_T > 0$, define

$$\widehat{\Omega} = \frac{1}{N} \sum_{i=1}^{N} \widetilde{\theta}_i \widetilde{\theta}_i^\top + \gamma I_d.$$

During all T post-calibration routing rounds, the router uses the frozen estimate

$$\widehat{\Omega}_t = \widehat{\Omega}.$$

The post-calibration routing archive is fixed before routing begins and has cardinality at most $K_{\max}$. The routing UCB rule uses

$$\alpha_T = \alpha_{\widehat{\Omega},T}(\varepsilon_*, \delta_{\mathrm{route}}),$$

where $\alpha_{\widehat{\Omega},T}$ is defined in Lemma 56, and where ε_* is the deterministic quantity defined below.

Assume Assumption 19 and the post-calibration routing assumptions of Assumption 17. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

For $\delta_{\mathrm{gram}}, \delta_{\mathrm{cov}} \in (0, 1)$, suppose

$$m \geq \frac{8L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}} \log\left(\frac{dN}{\delta_{\mathrm{gram}}}\right).$$

Define

$$\varepsilon_* = C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d + \log(2/\delta_{\mathrm{cov}})}{N}\right] + \tau_m + \gamma \|\Omega^{-1}\|_{\mathrm{op}}.$$

Then, with probability at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}}$,

$$\rho_T := \max_{1 \leq t \leq T} \left\| \Omega^{-1/2} (\widehat{\Omega}_t - \Omega) \Omega^{-1/2} \right\|_{\mathrm{op}} \leq \varepsilon_*.$$

In particular, taking $\delta_{\mathrm{gram}} = \delta_{\mathrm{cov}} = \delta/2$ gives the covariance-error control with probability at least $1 - \delta$.

Assume additionally that $\varepsilon_* < 1$. Then, with probability at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}} - \delta_{\mathrm{route}}$, the post-calibration routing exploration cost satisfies

$$\sum_{t=1}^{T} L_t(U) \leq 2\alpha_T \sqrt{1 + \varepsilon_*} \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)}.$$

Consequently, using $0 \leq L_t(U) \leq 1$,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T^\Omega + \mathrm{Err}_T,$$

where

$$B_T^{\Omega} = 2\alpha_T \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega)}$$

and one may take

$$\text{Err}_T = \left(\sqrt{1+\varepsilon_*} - 1\right) B_T^{\Omega} + (\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T.$$

If the Nm calibration pulls are charged as exploration cost over the enlarged horizon, then one may take

$$\text{Err}_T^{\text{charged}} = Nm + \left(\sqrt{1+\varepsilon_*} - 1\right) B_T^{\Omega} + (\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T.$$

Moreover, if the explicit routing multiplier satisfies the calibrated effective-dimension condition

$$\alpha_T^2 \left[1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)\right] \le C_\Omega \log(1+T)$$

for a deterministic constant C_Ω, then

$$B_T^{\Omega} \le 2\sqrt{2C_\Omega K_{\max} d_{\text{eff}}(\Omega) T \log(1+T)}.$$

Under this calibrated-confidence condition,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le C\sqrt{K_{\max} d_{\text{eff}}(\Omega) T \log(1+T)} + \text{Err}_T, \qquad C = 2\sqrt{2C_\Omega}.$$

For fixed

$$d,\ K_{\max},\ d_{\text{eff}}(\Omega),\ S_\Omega,\ \sigma,\ \kappa_{\text{cal}},\ L_{\text{cal}},\ L_\Omega,\ \lambda,\ \|\Omega^{-1}\|_{\text{op}},$$

take

$$N = \lceil T^{1/5} \rceil, \qquad m = \lceil T^{1/5} \rceil, \qquad \gamma = T^{-1/5}, \qquad \delta_{\text{gram}} = \delta_{\text{cov}} = \delta_{\text{route}} = T^{-2}.$$

Then, for all sufficiently large T,

$$\varepsilon_* < 1, \qquad \varepsilon_* = O\left(T^{-1/10}\sqrt{\log T}\right).$$

If the calibrated-confidence condition holds and the calibration pulls are charged, then

$$\text{Err}_T^{\text{charged}} = O(T^{2/5}) + O(T^{2/5}\log T) + O(T^{-1}) = o(\sqrt{T}).$$

Proof. Because $\widehat{\Omega}_t = \widehat{\Omega}$ for every post-calibration routing round,

$$\rho_T = \left\| \Omega^{-1/2}(\widehat{\Omega} - \Omega)\Omega^{-1/2} \right\|_{\text{op}}.$$

The high-probability covariance bound is exactly Lemma 55. Therefore, with probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$,

$$\rho_T \le \varepsilon_*.$$

Assume now that $\varepsilon_* < 1$ and work on the covariance event

$$\mathcal{E}_{\text{cov}} = \{\rho_T \le \varepsilon_*\}.$$

On this event,

$$(1-\varepsilon_*)\Omega \preceq \widehat{\Omega} \preceq (1+\varepsilon_*)\Omega.$$

By Lemma 56, conditional on the calibration data, with probability at least $1 - \delta_{\text{route}}$, the following simultaneous routing confidence event holds:

$$\left| x^\top (\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a) \right| \le \alpha_T \sqrt{x^\top (V_{a,t}^{\widehat{\Omega}})^{-1} x}$$

for every routing candidate a, every $t \le T$, and every predictable feature x. Since the calibration data are collected before routing and are independent of the post-calibration routing noises, the conditional probability statement integrates to an unconditional statement. A union bound gives an event $\mathcal{E}$ of probability at least

$$1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}} - \delta_{\mathrm{route}}$$

on which both the covariance event and the routing confidence event hold.

On $\mathcal{E}$, fix a routing round t. Let

$$a_t^\star \in \arg\max_{a \in A_t(U)} x_t(a)^\top \theta_a$$

and let a_t be the action selected by the frozen estimated-covariance UCB rule. The confidence event applied to $a_t^\star$ gives

$$x_t(a_t^\star)^\top \theta_{a_t^\star} \le x_t(a_t^\star)^\top \widehat{\theta}^{\widehat{\Omega}}_{a_t^\star,t} + \alpha_T \sqrt{x_t(a_t^\star)^\top (V^{\widehat{\Omega}}_{a_t^\star,t})^{-1} x_t(a_t^\star)}.$$

Since a_t maximizes the same UCB score,

$$\begin{aligned} &x_t(a_t^\star)^\top \widehat{\theta}^{\widehat{\Omega}}_{a_t^\star,t} + \alpha_T \sqrt{x_t(a_t^\star)^\top (V^{\widehat{\Omega}}_{a_t^\star,t})^{-1} x_t(a_t^\star)} \\ &\le x_t(a_t)^\top \widehat{\theta}^{\widehat{\Omega}}_{a_t,t} + \alpha_T \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}. \end{aligned}$$

The confidence event applied to a_t gives

$$x_t(a_t)^\top \widehat{\theta}^{\widehat{\Omega}}_{a_t,t} \le x_t(a_t)^\top \theta_{a_t} + \alpha_T \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}.$$

Combining the last three inequalities yields

$$L_t(U) = x_t(a_t^\star)^\top \theta_{a_t^\star} - x_t(a_t)^\top \theta_{a_t} \le 2\alpha_T \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}.$$

Since $0 \le L_t(U) \le 1$ and $\alpha_T \ge 1$,

$$L_t(U) \le 2\alpha_T \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}\right\}.$$

The covariance event implies

$$\widehat{\Omega}^{-1} \succeq \frac{1}{1+\varepsilon_*} \Omega^{-1}.$$

Therefore

$$\begin{aligned} V^{\widehat{\Omega}}_{a,t} &= X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}^{-1} \\ &\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon_*} \Omega^{-1} \\ &\succeq \frac{1}{1+\varepsilon_*} \left(X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1} \right) = \frac{1}{1+\varepsilon_*} V^{\Omega}_{a,t}. \end{aligned}$$

Inverting gives

$$(V^{\widehat{\Omega}}_{a,t})^{-1} \preceq (1+\varepsilon_*)(V^{\Omega}_{a,t})^{-1}.$$

Consequently,

$$\sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)} \le \sqrt{1+\varepsilon_*} \sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}.$$

For $c \ge 1$ and $z \ge 0$,

$$\min\{1, cz\} \le c \min\{1, z\}.$$

Applying this with $c = \sqrt{1+\varepsilon_*}$ gives

$$L_t(U) \le 2\alpha_T \sqrt{1+\varepsilon_*} \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}\right\}.$$

Summing over t, partitioning selected rounds by candidate, applying Cauchy–Schwarz, and then applying the already-proved Ω-elliptical-potential lemma candidate by candidate yields

$$\sum_{t=1}^{T} \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}\right\} \le \sqrt{2K_{\max} T \left[1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega)}.$$

Substituting this bound into the previous display proves the pathwise routing bound

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_T \sqrt{1+\varepsilon_*}\sqrt{2K_{\max}T\left[1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)\right]d_{\text{eff}}(\Omega)}.$$

Define

$$B_T^\Omega = 2\alpha_T\sqrt{2K_{\max}T\left[1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)\right]d_{\text{eff}}(\Omega)}.$$

On $\mathcal{E}$,

$$\sum_{t=1}^{T} L_t(U) \le \sqrt{1+\varepsilon_*}\,B_T^\Omega = B_T^\Omega + (\sqrt{1+\varepsilon_*}-1)B_T^\Omega.$$

On $\mathcal{E}^c$, the deterministic bound $0 \le L_t(U) \le 1$ gives

$$\sum_{t=1}^{T} L_t(U) \le T.$$

Taking expectations yields

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T^\Omega + (\sqrt{1+\varepsilon_*}-1)B_T^\Omega + (\delta_{\text{gram}}+\delta_{\text{cov}}+\delta_{\text{route}})T.$$

This proves the displayed expectation bound and the stated expression for Err_T. If calibration pulls are charged, each calibration pull contributes at most one unit of exploration cost, and the additional cost is at most Nm, giving $\mathrm{Err}_T^{\text{charged}}$.

If the calibrated effective-dimension condition holds, then

$$\alpha_T\sqrt{1+\log\left(1+\frac{TL_\Omega^2}{\lambda}\right)} \le \sqrt{C_\Omega\log(1+T)}.$$

Substituting this inequality into the definition of B_T^Ω gives

$$B_T^\Omega \le 2\sqrt{2C_\Omega K_{\max}d_{\text{eff}}(\Omega)T\log(1+T)}.$$

This proves the target leading term with $C = 2\sqrt{2C_\Omega}$.

It remains to verify the asymptotic schedule. With fixed problem constants and

$$N = \lceil T^{1/5}\rceil, \qquad m = \lceil T^{1/5}\rceil, \qquad \gamma = T^{-1/5}, \qquad \delta_{\text{cov}} = T^{-2},$$

we have

$$\sqrt{\frac{d+\log(2/\delta_{\text{cov}})}{N}} = O\left(T^{-1/10}\sqrt{\log T}\right),$$

$$\frac{d+\log(2/\delta_{\text{cov}})}{N} = O(T^{-1/5}\log T),$$

$$\tau_m = O(T^{-1/5}), \qquad \gamma\|\Omega^{-1}\|_{\text{op}} = O(T^{-1/5}).$$

The dominant term is $O(T^{-1/10}\sqrt{\log T})$, so

$$\varepsilon_* = O\left(T^{-1/10}\sqrt{\log T}\right).$$

Since $T^{-1/10}\sqrt{\log T} \to 0$, there exists $T_0 < \infty$ such that $\varepsilon_* < 1$ for all $T \ge T_0$. The Gram lower-tail condition also holds for all sufficiently large T, because its right-hand side is $O(\log T)$ whereas $m = \lceil T^{1/5}\rceil$.

Under the calibrated leading-term condition,

$$B_T^\Omega = O(\sqrt{T\log T}).$$

For $0 \le \varepsilon_* \le 1$,

$$\sqrt{1+\varepsilon_*} - 1 = \frac{\varepsilon_*}{\sqrt{1+\varepsilon_*}+1} \le \varepsilon_*.$$

Therefore

$$(\sqrt{1+\varepsilon_*} - 1) B_T^{\Omega} = O\Big(T^{-1/10}\sqrt{\log T}\Big)\, O(\sqrt{T \log T}) = O(T^{2/5} \log T).$$

Also,

$$Nm = O(T^{2/5}),$$

and

$$(\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T = 3T^{-1}.$$

Hence

$$\mathrm{Err}_T^{\text{charged}} = O(T^{2/5}) + O(T^{2/5}\log T) + O(T^{-1}) = o(\sqrt{T}),$$

because $T^{2/5} \log T / \sqrt{T} = T^{-1/10} \log T \to 0$. □

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